UNIVERSITY OF CHINESE ACADEMY OF SCIENCES

DOCTORAL THESIS

The measurement of Higgs production cross section in the four-lepton final state at CMS

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Abstract

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The measurement of Higgs production cross section in the four-lepton final state at CMS

by Tahir JAVAID

The measurement of properties of Higgs boson is presented in its four-lepton decay channel. The fiducial cross sections (integrated and differential) of the Higgs boson production via $H \to 4\ell$ decays ($\ell = \mathrm{e}$, μ) are measured using the data collected with CMS detector in pp collisions at $\sqrt{s} = 13$ TeV . The measurements use Higgs mass $m_H = 125.09$ GeV and correspond to an integrated luminosity of 137.1 fb^{-1}at 13 TeV(Full Run 2 data). These measurements provide a direct test of perturbative QCD calculations in the Higgs sector of Standard Model at the TeV energy scale.

In order to make all the measurements independent of how Higgs is produced, all the cross sections are extracted within a fiducial phase space defined by the selections on lepton kinematics and event topology. The integrated fiducial cross sections are measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})\ \mathrm{fb}$ at 13 TeV. The differential fiducial cross sections are presented as a function of several kinematic observables which are sensitive to the Higgs-boson production mechanisms: rapidity and transverse momentum of the four-lepton system, associated jet multiplicity and transverse momentum of the leading jet. These measurements are important to test the Standard Model predictions in the Higgs sector and can be used to constrain phase spaces of many theories beyond the Standard Model.

All the integrated and differential measurements are compared with the Standard Model based theoretical predictions and found to be consistent within their uncertainties.

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Chapter 1

Introduction

After a quest, ordinary matter and underlying forces or interactions (electromagnetic, weak and strong interactions and not the gravity) which bind it in a structure are successfully explained by the Standard Model (SM) of the particle physics. In SM, description of these interactions is made by symmetry laws of mathematics which could only explain the characteristic of particles with zero mass. Higgs mechanism was then introduced to make the theory realistic enough to encorporate mass problem. Under this mechanism, electroweak symmetry breaks and as a result particles gain the mass. Higgs boson is the direct manifestation of this Higgs mechanism [1, 2, 3, 4, 5, 6].

Precision measurements on the high energy physics experiments like Tevatron and LEP have proven the authenticity of SM; however, Higgs observation could not happen. In 2010, Large Hadron Collider (LHC), the largest experiment ever of its kind in physics world, was then set to put more tests on SM and it endorsed the measurements on SM from earlier experiments. In 2012, the “hidden" Higgs boson, was discovered in shape of heavy scalar boson as jointly reported by the CMS and ATLAS experiments [7, 8]. With this happening, physicists were all set to measure other properties of Higgs for further tests on SM predictions. Mass of Higgs boson was only free parameter in Higgs sector of SM until measured to be $m_H = 125.09 \pm 0.21(stat.) \pm 0.11(syst.)$ GeV [9].

Model independent production cross section for a particular process is an important property of H boson, where the cross section gives the possibility of the occurrence of that process. In this dissertation, measurements of the integrated or inclusive and differential fiducial cross sections for H boson production in $H \to 4\ell$ decay channel ($\ell = e, \mu$) using the full Run 2 CMS data recorded in proton-proton collisions at $\sqrt{s} = 13$ TeV are presented. Included Higgs differential measurements are based on Higgs kinematics and motivated from several aspects. For instance, Higgs trasverse momentum can be used to test the pQCD calculations (especially when measured with exclusive multiplicities of jets). Additionally this variable is sensitive to Lagrangian structure of Higgs. Rapidity of Higgs is sensitive to parton distribution functions (PDFs) of the protons involved in the collisions. Jet related observables which are sensitive to theoretical modelling, boosted quark and also to the gluon emission are also covered in the dissertation. Current measurements are consistent with SM prediciton; however, in such measurements, any deviation from SM expectations provides a direct hint to BSM contributions. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different decay cahannels of Higgs which can been studied in Ref. [**Khachatryan:2015yvw**, **CMS:2015hja**, **CMS:2016rqf**, **Sirunyan:2017exp**, **CMS:2018hhg**, **ATLAS-CONF-2019-029**, **ATLAS-CONF-2019-025** , **ATLAS:2018uso**, **ATLAS-CONF-2019-032**, **ATLAS:2017myr**, **CMS:2019chr**, 10, 11]. More precise Higgs boson properties measurements are needed to spot any possible deviation from SM predictions which can be a doorway to new physics.

This thesis is structured as followed:

- **Chapter 2:** An overview of theoretical background i.e. the SM of particle physics, production and decays of H boson are described.
- **Chapter 3:** Overview of LHC experiment and its operation are described.
- **Chapter 4:** CMS detector, its sub-detectors and gradual upgrades are described.
- **Chapter 5:** The calibrations and reconstructions of the physics objects within the CMS detector are described.
- **Chapter 6:** Presents the procedure of measurements of inclusive and differential fiducial cross sections of H boson production at $\sqrt{s} = 13$ TeV.
- **Chapter 7:** Finally, the conclusions and outlook of the dissertation are summarized.

Chapter 2

Theoretical Background

In this chapter, theoretical grounds upon which this dissertation is built, are briefly discussed. Under the umbrella of “elementary particle physics”, we have been able to get answers to long lasting questions about the visible universe end even beyond. We have been able to know about the smallest structures of which the uinverse is made and forces which govern them to bind together in the form or ordinary matter. Standard Model of the particle physics is a theoretical framework capable to explain the dynamics of these subatomic world of particles in its beautiful mathematical formulations which are based on symmetries. Higgs boson is manifestation of arguments from core components of Standard Model which was first discovered at CERN,LHC facility in proton proton collisions. In next Sections, details on Standard Model, its Higgs sector are described. Additionally, underlying theoretical concepts on collision experiment (i.e. proton-proton) are discussed.

2.1 The Standard Model

The Standard Model (SM) of particle physics, is the theory which has provided modern concept of understanding the dynamics of the constituents of the matter and underlying principles through which they interact with each other. It is capable to give unified explanation of electromagnetic (EM), weak and strong forces (except the gravity) [12].

2.1.1 From the History

SM is a quantum field theory (QFT) belonging to $\mathrm{SU(3)_C \otimes SU(2)_L \otimes U(1)_\gamma}$ gauge symmetry group. It was formulated from a collaborative work spanned throughout the 20^{th} century. In 1961, there was a proposal from Glashow that electromagnetic and weak intractions are similar substrctures of electroweak intraction and presented their unified interpretation with $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ symmetry group. Brout, Englert, Higgs *et al.* proposed a mechanism with which fundamental particles may acquire mass through the idea of spontaneous symmetry breaking and associated new elementary particle called Higgs boson. Salam and Weinberg then annexed this idea to the Glashow's model ($\mathrm{SU(2)_L \otimes U(1)_\gamma}$) and completed the EW sector of SM.

In 1964, Gell-Mann and Zweig proposed the quark model independently and at the same time Nambu *et al.*, in 1965, gave the idea of color charge an important property of a quark. In 1973, formulation of strong force as a QFT (Quantum Chromodynamics, QCD), was done with $\mathrm{SU(3)_C}$ by Gell-Mann and Fritzsh. Earlier in 1972, Veltnam and Hooft proved SM to be a renormalizable theory making perturbation theory applicable for detailed calculations and resulting predictions be tested in the experiments.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [13]. Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [14, 15, 16, 17]. After EW was widely accepted, the Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification.

Meanwhile, QCD part of the SM was also being devolped. There were evidences of the quarks in different scattering experiment facilities in late 60's. In 1974, existence of the quarks was further verified with the discovery of J/ψ meson. Friedman *et al.* and Richter with Ting were nominated for the Nobel prizes for the discoveries of quark and charmonium respectively. Later on, discoveries of particle (a 3^{rd} generation lepton), bottom and anti-bottom quarks were discovered in 1976, 1977 and 1995 respectively. Top quark was finally discovered at Fermilab in 1995. All the discoveries in this era, provided a consistent test of SM predictions. Higgs boson, however, still remained unobserved untill July 2012 when CMS and ATLAS collaborations of CERN announced its discovery [7, 8]. Over this extraordinary experimental success, Higgs and Englert were awarded with Physics Nobel Prize in 2013 for their theoretical predictions about this gauge boson.Despite of an incredibly successful theory of particle world, the SM does fail to fully describe the known universe.

2.2 Fundamental Particles and underlying Interactions

2.2.1 Fundamental Particles

The elementary particles [1] of SM have certain trends and can be listed in terms of a table like a periodic table of elements in Chemistry. Fundamental particles can be categorized into fermions (characterized by Fermi-Dirac statistics) and bosons (Bose-Einstein statistics). The spin-statistics theorem classifies the particles with integral spin as bosons, while particles with half-integral spin as fermions. Fermions includes all leptons, quarks and baryons (composite particles three quarks). Ordinary matter is composed of 1^{st} generation of fermions, i.e. leptons (e.g. electron that revolves around the nucleus) and quarks (which bind to give protons and neutrons of which nucleus is composed) which are categorized in three "generations" as illustrated in Table 2.1.

Leptons, in each doublet or generation, there is a lepton and corresponding neutrino (ν) as neutral particle. Leptons only interact via electromagnetic and weak interaction, since they have no color charge. Neutrinos are electrically neutral and carry no color charges, so interact through weak force only. Charged leptons masses are listed in order of increasing mass $m_e = 0.511$ MeV, $m_\mu = 105.7$ MeV and $\mathrm{m}_\tau = 1.78$ GeV. Charged leptons carry -1 and their antiparticle carry $+1$ electric charge. In addition, lepton number can be assigned to the particles which is "1" for leptons and "-1" for anti-leptons. Lepton number is conserved in SM processes.

In all generations, quarks carry only fractional charges. First generation includes an "up" type quark which carries electric charge $+\frac{2}{3}$ and a "down" type

[1] Particle having unknown structure is called fundamental particle.

quark carrying electric charge $-\frac{1}{3}$. Quarks also carry color charge (red, green, or blue) which is responsible for the strong interaction. The types of the quarks in order of increasing mass are "up", "down", "strange", "charm", "bottom" and "top". The whole visible world is only made of light type of quarks,"up" and "down". The heavy quarks(second and third generation quarks) being unstable can be produced only in high energy collisions (cosmic rays, particle accelerators) and quickly decays into "up" and "down" quarks. All quarks have a property which can be regarded as color charge which is conserved in all SM processes. Each flavor of quark can have three different colors; red, green and blue (R,G,B).

Quarks interact through all three Standard Model forces, weak, strong, and electromagnetic. According to the Quantum Chromodynamics (QCD), two quarks (quark antiquark pair) or three quarks can combine as mesons ($q_1\bar{q}_2$) or baryons ($q_1q_2q_3$) respectively to make color-neutral hadrons. Out of hadrons, protons and neutrons are most widely recognized making nuclei of matter which are formed of uud and udd quarks respectively.

TABLE 2.1: Summary of different Standard Model particles, and values of parameters related to them. Masses of the particles are taken from **[masses]** (In SM neutrino masses are taken as zero.).

Family (generation)	Name	Symbol	Mass	Charge[e]	Interactions
leptons ($\mathbf{1^{st}}$)	electron-neutrino	ν_e	<2 eV	0	weak
leptons ($\mathbf{1^{st}}$)	electron	e	0.511 MeV	-1	weak, EM
leptons ($\mathbf{2^{nd}}$)	muon-neutrino	ν_μ	<0.19 MeV	0	weak
leptons ($\mathbf{2^{nd}}$)	muon	μ	105.658 MeV	-1	weak, EM
leptons ($\mathbf{3^{rd}}$)	tau-neutrino	ν_τ	<18.2 MeV	0	weak
leptons ($\mathbf{3^{rd}}$)	tau	τ	1776.82 MeV	-1	weak, EM
quarks ($\mathbf{1^{st}}$)	up	u	2.3 MeV	$+\frac{2}{3}$	EM, weak, strong
quarks ($\mathbf{1^{st}}$)	down	d	4.8 MeV	$-\frac{1}{3}$	EM, weak, strong
quarks ($\mathbf{2^{nd}}$)	charm	c	1.275 GeV	$+\frac{2}{3}$	EM, weak, strong
quarks ($\mathbf{2^{nd}}$)	strange	s	95 MeV	$-\frac{1}{3}$	EM, weak, strong
quarks ($\mathbf{3^{rd}}$)	top	t	173.07 GeV	$+\frac{2}{3}$	EM, weak, strong
quarks ($\mathbf{3^{rd}}$)	bottom	b	4.18 GeV	$-\frac{1}{3}$	EM, weak, strong

Carriers of fundamental forces are bosons, more details would be described in Section 2.2.2. Electromagnetic force is mediated by photon which is massless, neutral and of spin-1. On the other hand, weak force is carried by $W^{\pm}$ (80.4 GeV) and Z (91.2 GeV) bosons having electric charges ± 1 and 0 respectively. At last, strong forces is carried by massless, neutral spin-1 gluons. Gluon is the only gauge

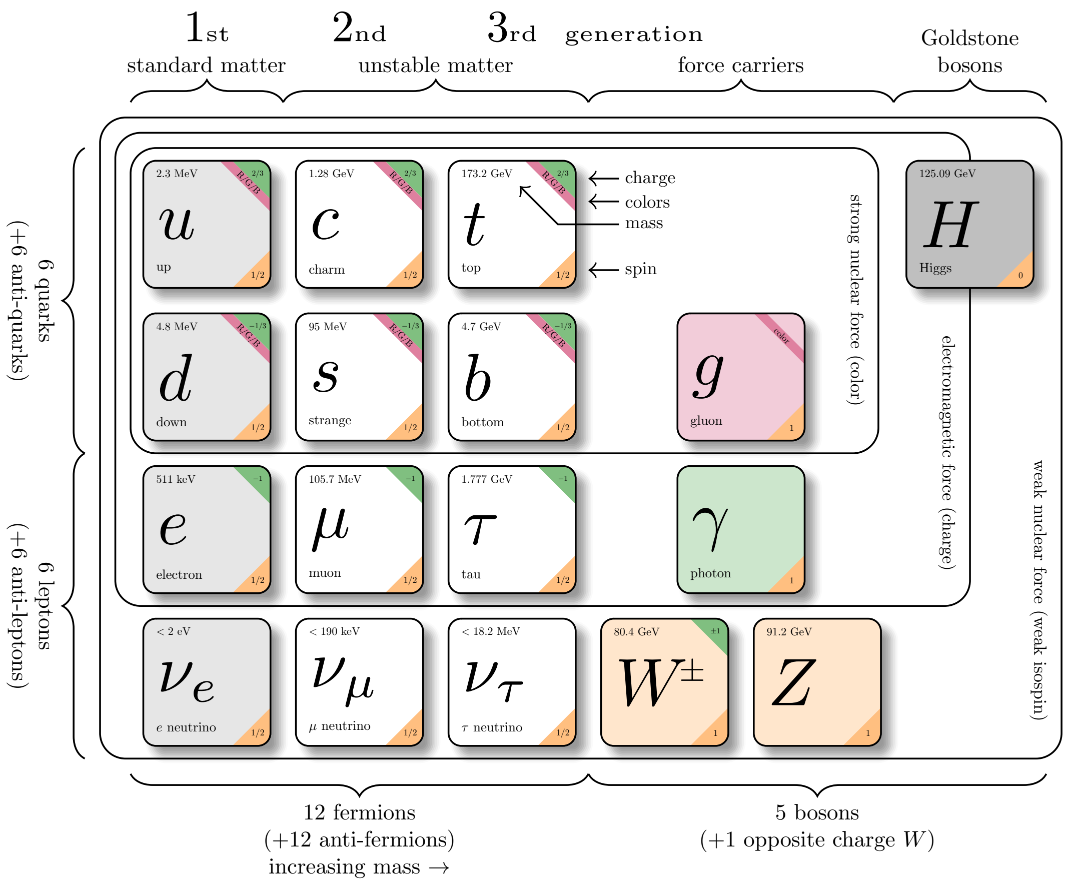


FIGURE 2.1: The elementary particles given in three generations of matter, the gauge bosons and the Higgs boson.

boson which carries color and anti-color charge, thus it can interact with other gluons.

TABLE 2.2: Gauge bosons and their masses, coupling constants and interaction ranges for the three forces of the SM

Force	Weak	Electromagnetic	Strong
Mediator boson	$W^{\pm}$ and Z	photon (γ)	Gluon (g)
Mass(GeV)	80.4, 91.2 GeV	0	0
Coupling Constant	$G_F \approx 1.2 \times 10^{-5}$ GeV^{-2}	$\alpha = \frac{1}{137}$	$\alpha_s \approx 0.1$
Interaction Range	10^{-16} cm	∞	10^{-13} cm

2.2.2 Fundamental Interactions

The three fundamental interactions (electromagnetic, strong and weak) are explained three dedicated theoretical descriptions Quantum Electrodynamics (QED), Quantum Chromodynamics (QCD) and Quantum Flavourdynamics (QFD) which are encoded in SM.

Weak interaction, one of the four fundamental forces, was first revealed by the understanding of beta decay in 1933 by Enrico Fermi called Fermi's interaction [13]. Weak force is mediated by $W^{\pm}$ and Z bosons. In terms of a field theory, it is represented by gauge theory with symmetry group $SU(2)$, having three gauge bosons (W^{+}, W^{-}, Z). Weak force, rarely known from quantum flavordynamics (QFD), is the only force which is proficient to change the flavor of quarks. This is very short range force because exchange bosons ($W^{\pm}$, Z) are massive. All fermions and the Higgs boson can interact via weak force. In 1970s, electromagnetic and weak forces were unified as electroweak force above unification energy (on the order of 100 GeV). The Physics Nobel Prize was awarded to Sheldon Glashow, Abdus Salam, and Steven Weinberg for their role in electroweak unification. In the field theory representation, Electroweak unification is settled under the gauge group $\mathrm{SU(2)_L \otimes U(1)_\gamma}$. Experimental observation of existence of electroweak interaction was accomplished in two phases, the first was disclosure of existence of neutral current in scattering of neutrino in 1973 by Gargamelle collaboration and then the discovery of W and Z gauge bosons in proton-antiproton collisions by both the UA1 and the UA2 collaborations in 1983 [14, 15, 16, 17].

Electroweak unification

As discussed above, Glashow proved that EM and weak interactions can be experessed by $\mathrm{SU(2)_L \otimes U(1)_\gamma}$ gauge group, where "L" represents left handed chiral states of fermionic fields. "Y" is the weak hyper charge and can be related to the weak-isospin (T) and electric charge (Q) as follows:

$$W = T_3 + \frac{Y}{2} \tag{2.1}$$

This is known as Gell-Mann-Nishijima formula and applies to left-handed and right-handed fermions.

Left handed states form weak-isospin doublets i.e. $\chi_L = (\nu_\ell, \ell)^T_L$, whereas right-handed components from weak-isospin singlet $\psi_R = \ell_R$.

To establish the local gauge invariance in the proposed theory, three vector fields (W_μ) couple to weak isospin currents with strength "g". A neutral field denoted by B_μ couples with strength $\frac{g^{'}}{2}$ to the weak hypercharge. To construct the Lagrangian, the covariant derivative can be written as:

$$D_\mu = \partial_\mu + ig\mathbf{T}.\mathbf{W}_\mu + \frac{ig'}{2}YB_\mu \tag{2.2}$$

This covariant derivative can be operated on the left-handed and right-handed states for the transformation as under:

$$D_\mu\chi_L \to e^{i\alpha(x).\mathbf{T}+i\beta(x)Y}D_\mu\chi_L \tag{2.3}$$

$$D_\mu\psi_R \to e^{i\beta(x)Y}D_\mu\psi_R \tag{2.4}$$

With the result of these gauge transformation, $\bar{\psi}D_\mu\psi$ remain invariant where:

$$\mathbf{W}_\mu \to \mathbf{W}_\mu - \frac{1}{g}\partial_\mu\,\alpha - \;\alpha\times\mathbf{W}_\mu \tag{2.5}$$

$$B_\mu \to B_\mu - \frac{1}{g'}\partial_\mu\beta. \tag{2.6}$$

Subsequently the Lagrangian can be constructed as :

$$\begin{aligned}\mathscr{L}_{EWK} = \bar{\chi_L}(x)i\gamma^\mu(\partial_\mu + ig\mathbf{T}.\mathbf{W}_\mu + \frac{ig'}{2}YB_\mu)\chi_L(x)+\\ \bar{\psi_R}(x)i\gamma^\mu(\partial_\mu + i\frac{ig'}{2}YB_\mu)\psi_R(x)+\\ \frac{1}{4}\mathbf{W}_{\mu\nu}\mathbf{W}^{\mu\nu} + \frac{1}{4}B_{\mu\nu}B^{\mu\nu}\end{aligned} \tag{2.7}$$

Where $\mathbf{W}_{\mu\nu}$ and $B_{\mu\nu}$ are the kinetic energy terms of gauge fields and can be written as in equations. 2.8 and 2.9 respectively.

$$\mathbf{W}_{\mu\nu} \equiv \partial_\mu\mathbf{W}_\nu - \partial_\mu\mathbf{W}_\mu - g\mathbf{W}_\mu\times\mathbf{W}_\nu \tag{2.8}$$

$$B_{\mu\nu} \equiv \partial_\mu B_\nu - \partial_\nu B_\mu \tag{2.9}$$

T being the generators of $\mathrm{SU}(2)$which are related to the Pauli matrices as:

$$T^a = \frac{1}{2}\sigma^a; a = 1, 2, 3 \tag{2.10}$$

The Lagrangian in equation 2.7 describes of electroweak particle (massless lepton). In order to keep the theory invariant under the transformations as described in equtions 2.5 and 2.6, mass term has been chosen to be zero i.e. the theory remains unable to describe the kinematics of the massive gauge bosons.
Some years later, Salam *et al.* introduced a mechanism to include the mass term in the Lagrangian ($\mathscr{L}_{EWK}$ in equation. 2.7) keeping the theory anomaly free. More details are described in the next Section.

2.3 The Brout-Englert-Higgs Mechanism (BEH)

Glashow's proposal of unification of EM and weak forces described the kinematics of massless particle only as described in Section 2.2.2. However, gauge fields must acquire mass in order to correctly describe weak interaction and to be consistent with experimental findings. The Higgs mechanism was postulated as a part of SM, which introduces the existence of a new scalar Higgs field. This field transformation ensures the renormalizability of the theory and gauge invariance. By this formulation the concept of breaking of symmetry and origin of masses of the fermions and bosons ($W^{\pm}$, Z boson) are introduced [1, 2, 3, 4, 5, 6]. The Higgs mechanism introduces a doublet (under isospin) of complex scalar field

$$\phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} = \begin{pmatrix} \phi^{\dagger} \\ \phi^0 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} \phi^1 + i\phi^2 \\ \phi^3 + i\phi^4 \end{pmatrix} \tag{2.11}$$

which can be substituted in electroweak lagrangian given below

$$\mathscr{L}_{EWSB} = (D_\mu\phi)^{\dagger}(D_\mu\phi) + \mathrm{V}(\phi^{\dagger}\phi) \tag{2.12}$$

here V is called field's self interaction or potential term , D_μ is described in Equation. 2.2. The lagrangian described in Equation 2.12 is gauge invariant under the $\mathrm{SU(2)_L} \otimes \mathrm{U}(1)_\gamma$ transformation. The kinetic energy term in the lagrangian is written in terms of the covariant derivatives whereas the potential only depends upon the product $\phi^{\dagger}\phi$. ϕ is the scalar field and can be codified in terms of quantum numbers which are isospin (I_3), hypercharge (Y) and electric charge (Q):

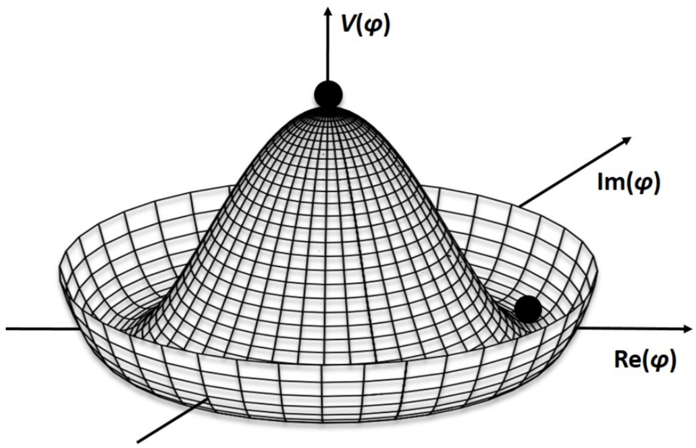


FIGURE 2.2: The Higgs potential as in Eq.2.13

	I_3	Y	Q
$\begin{pmatrix} \phi^{\dagger} \\ \phi^{0} \end{pmatrix}$	$\begin{pmatrix} \frac{1}{2} \\ \frac{1}{-2} \end{pmatrix}$	$\begin{pmatrix} 1 \\ 1 \end{pmatrix}$	$\begin{pmatrix} 1 \\ 0 \end{pmatrix}$

while the potential term in Equation 2.12 can also be expressed as

$$\mathrm{V}(\phi^{\dagger}\phi) = \mu^2\phi^{\dagger}\phi + \lambda(\phi^{\dagger}\phi)^2 \tag{2.13}$$

here μ is mass term and the parameter λ represents the strength of the interaction. As $\mu^2 < 0$ and $\lambda > 0$ and the potential shapes becomes like a "mexican hat"very interesting, which is essential for Higgs mechanism shown in Figure 2.2. There occurs a minima for

$$\phi^{\dagger}\phi = \frac{1}{2}(\phi_1^{\dagger} + \phi_2^{\dagger} + \phi_3^{\dagger} + \phi_4^{\dagger}) = \frac{-\mu^2}{2\lambda} \equiv \frac{v^2}{2} \tag{2.14}$$

Naturally the whole system will be residing in arbitrarily chosen vacuum state; however, this state is away from the symmetry center. So, the minimum does not exist at a single value of ϕ, but at circular non-zero values.The $(\phi^{\dagger}\phi^{0})$ selection that affiliated with the ground state is arbitrary and in $(\phi^{\dagger}\phi^{0})$ plane the selected point is not invariant under rotations which is called spontaneous symmetry breaking. To settle the ground state on the ϕ^{0} axis, the vacuum expectation value (VEV) of field

ϕ is

$$\langle\phi\rangle = \frac{1}{\sqrt{2}} = \begin{pmatrix} 0 \\ v \end{pmatrix}, \qquad v^2 = -\frac{\mu^2}{\lambda}. \tag{2.15}$$

The field ϕ given below, is derived by rewriting it in some generic gauge, on footings of its VEV:

$$\phi = \frac{1}{\sqrt{2}} e^{\frac{i}{v}\phi^a t_a \phi} = \begin{pmatrix} 0 \\ h(x) + v \end{pmatrix}, a = 1, 2, 3 \tag{2.16}$$

here ϕ_a and $\phi_4 = h(x) + v$ represents massless fields called Goldstone fields. New degrees of freedom is introduced along with six degrees because of polarizations vector bosons $W^{\pm}$ and Z, as they are massless and scalar. A unitary gauge transformation in the field is fixed by

$$\phi' = e^{-\frac{i}{v}\phi^a \mathbf{T_a}}\phi = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ h(x) + v \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ \phi^4 \end{pmatrix}$$

so here we have zero expectation value for the Higgs field.

The BEH Lagrangian can be written from Equation 2.12 as:

$$\begin{aligned} \mathscr{L}_{BEH} = \frac{1}{4}g^2\nu^2 W^+_\mu W^{-\mu} - \frac{1}{2}(\partial_\mu W^+_\nu - \partial_\nu W^+_\mu)(\partial^\mu W^{-\nu} - \partial^\nu W^{-\mu}) + \\ \frac{1}{8}(g^2 + g''^2)v^2 Z_\mu Z^\mu - \frac{1}{4}(\partial_\mu Z_\nu - \partial_\nu Z_\mu)(\partial^\mu Z^\nu - \partial^\nu Z^\mu) - \\ \frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}\partial_\mu h \partial^\mu h - \mu^2 h^2 + ... \end{aligned} \tag{2.17}$$

where W terms are written as:

$$W^{\pm} = (\frac{W^1 \mp iW^2}{\sqrt{2}}) \tag{2.18}$$

These terms correspond to the oppositely charged W bosons whose mass is given by:

$$m_W = \frac{1}{2}vg \tag{2.19}$$

There also emerges a term for the neutral vector Z boson:

$$Z_\mu = \frac{-g' B_\mu + g W^3_{mu}}{\sqrt{g^2 + g''^2}} \tag{2.20}$$

Z boson mass is given by:

$$m_Z = \frac{1}{2} v \sqrt{g^2 + g''^2} \tag{2.21}$$

Using W_μ^3 and B_μ, we can also write their combination orthogonal to Z_μ

$$A_\mu = \frac{g' B_\mu + g W_{mu}^3}{\sqrt{g^2 + g''^2}} \tag{2.22}$$

The Equation 2.22 corresponds to photon with the condition:

$$\frac{g' g}{\sqrt{g^2 + g''^2}} = e \tag{2.23}$$

From equations 2.19 and 2.21, we can have,

$$\frac{m_W}{m_Z} = \frac{g}{\sqrt{g^2 + g''^2}} \tag{2.24}$$

Suppose, the θ_W be weak mixing angle with definition,

$$g' = g \tan \theta_W \tag{2.25}$$

Using Equation 2.25, we can write the Equations 2.20 and 2.22 as:

$$Z_\mu = -B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W \tag{2.26}$$

$$A_\mu = B_\mu \sin \theta_W + W_\mu^3 \cos \theta_W \tag{2.27}$$

With spontaneous symmetry breaking (SSB), gauge bosons (W,Z) acquire masses as a result of interaction with BEH (or Higgs) field. Also, after SSB, BEH (or Higgs) fields emerges with a physical boson particle. Lagrangian (potential term) can be written as:

$$\mathscr{L}_V = \frac{1}{4} \lambda (v + h(x))^4 - \frac{1}{2} \lambda v^2 (v + h(x))^2 \tag{2.28}$$

This gives the Higgs mass at first order; $M_H^2 = 2\lambda v^2$

The value of v (also g) can be measured using Fermi weak constant i.e. G_F. However, λ is a free parameter in SM.

Fermions can also acquire masses if additional gauge invariant interaction term is added to term EW Lagrangian. In this way, fermions can acquire masses when they interact with BEH (or Higgs) field through Yukawa couplings. Interaction term in the Lagrangian can be written as:

$$\mathscr{L}_{Yukawa} = -G_l[\bar{\psi_R}(\phi^{\dagger}\chi_L) + (\bar{\chi_L}\phi)\psi_R] \tag{2.29}$$

Using ϕ and other inputs can further gives us the masses (m_f) of fermions and couplings (g_{Hf}):

$$m_f = \frac{G_l}{\sqrt{2}}v \tag{2.30}$$

$$g_{Hf} = \sqrt{G_l}\sqrt{2} = \frac{m_f}{v} \tag{2.31}$$

2.4 Phenomenology of proton-proton collisions at LHC

LHC is the largest colliding experiment of its kind ever built also motivated to perform test of the SM of particle physics. More details on LHC experiment will be discussed in next chapter.

In this section, however, some phenomenological part of proton-proton collision in high energy experiments, like LHC, are briefly discussed. Reuslts of scattering experiments e.g. production cross section of a certain process, are compared with prediction of perturbative QCD. It is possible through hard scattering experiments to probe at very small scale even shorter than a proton radius. Hence it helps to know the interactions of quarks and gluons (collectively called as partons) processes probe distance scale. A typical diagram of the hard scattering process in proton proton collisions is shown in the Figure. 2.3 and also in Ref. [**scattering**]. Protons are composite particles made up of three valence quarks (uud) which are bound together with strong force exchanging the gluons. There are several ways through which they can interact within a proton. Quarks can mutually iteract with other quarks or throught annihilation process they can produce gluons which can further go for pair-production of quark anti-quark pair. As a results, apart from the valence quarks, there is a sea of virtual quarks and gluons in side a proton (collectively called partons). In a collision experiment, it may happen that there is large momentum transfer from the colliding partons. This is called hard scattering and as a results many resonances can be produces e.g. Higgs. Probability of a certain

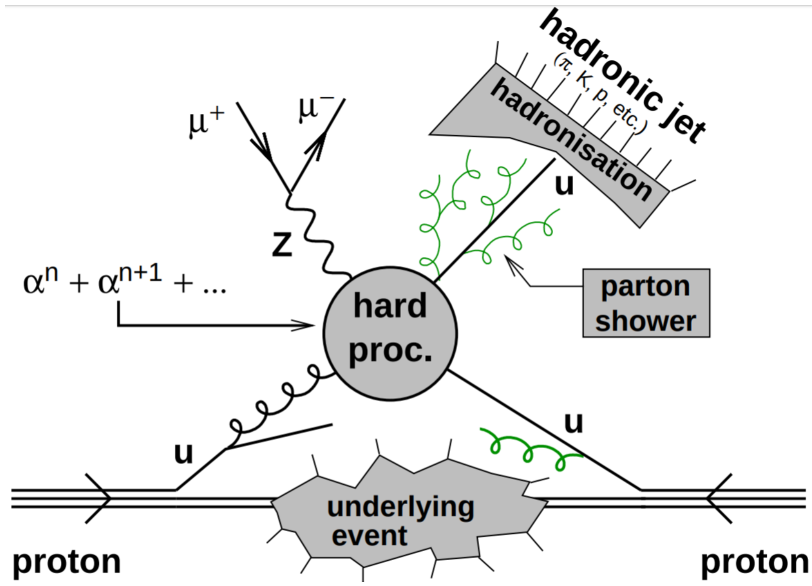
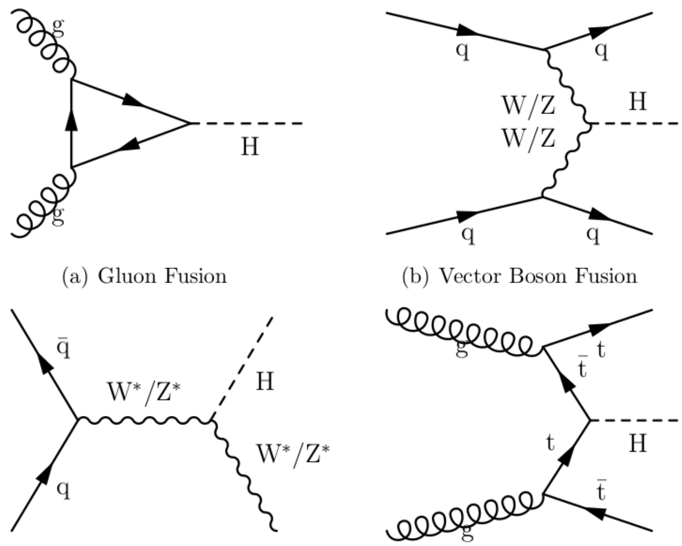


FIGURE 2.3: A schematic diagram of proton-proton collision at high energy experiments

process to occur is known as cross section of that process. The production cross sections of different process at LHC in the proton-proton collsions are shown in the Figure. 3.3 in next chapter.
Partons which do not take part in hard interaction, usually go straight and may involve in certain low momentum transfer processes. These activities of low momentum transfer are call uncerlying events. In case if more than one interactions happen in a certain proton proton collision, it is referred as multi-parton interactions (MPI).
In an accelerated environment, charged partons may radiate photons or gluons through a process known as "Bremsstrahlung". If this radiation is associate with the incoming partons, it is called initial state radion (ISR) and if is coming from outgoing parton from the interaction point, it is called final state radiation (FSR).
The quarks and/or gluons when they go through the process of hadronization (bind together to form color siglet) make a bunch of hadrons usually in the form of a cone which are called "jets" and are described in Section. 5.5.

In hard scattering experiments, probablity of a certain process to occur is know as cross section and can be calculated using the Factorization theorem. This theorem helps to disintegrate the cross section into hard and soft processes using the scale which is know as factorization scale (μ_F).
Suppose in collision of two protons, A and B, the partons a and b undergo hard



(a) Gluon Fusion (b) Vector Boson Fusion

(c) Associated Production W/Z (d) Associated Heavy Quark Production

FIGURE 2.4: Feynman diagrams of Higgs production at tree level.

scattering which produce a certain final state S. The total cross section can be given by the relation:

$$\sigma_{AB} = \int dx_a dx_b f_{a/A}(x_a, \mu_F^2) f_{b/B}(x_b, \mu_F^2) \hat{\sigma}_{ab \to S} \tag{2.32}$$

In above equation, x_a is the fraction of momentum of the parton a which it takes from proton A. Similary, x_b is the fraction of momentum of the parton b which it takes from proton B. μ_F is factorization scale and is usually of the order of Q which is total momentum transfer of hard process. $f_{a/A}(x_a, \mu_F)$ and $f_{b/B}(x_b, \mu_F)$ are the parton distribution functions (PDFs).

2.5 Higgs boson Production at LHC

Higgs can be produced at LHC in various production modes. The production modes of Higgs which are dominant in proton proton collisions at LHC are illustrated in Figure 2.4, corresponding production cross sections as a function of its mass at center-of-mass energies as shown in Figure 2.5 . A considerable increase in production cross section of Higgs is found as we move towards higher center-of-mass energies [18]. The most dominant production mode of Higgs at LHC is

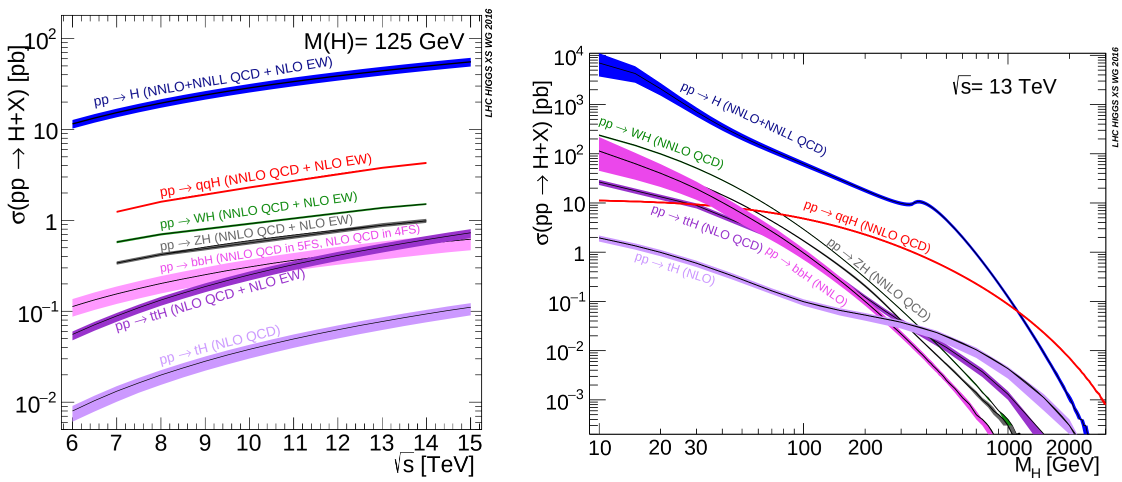


FIGURE 2.5: Production cross sections of Higgs at 13 TeV as a function of center of mass energies (left) and as a function of its mass (right).

gluon-gluon fusion as illustrated in Figure 2.4 (top left) [18]. The gluons couples to Higgs through the loop of virtual quarks, as coupling is directly proportional to fermion mass, top quark loop is most dominant. VBF is second most important production mode of Higgs at LHC and LEP as shown in Figure 2.4 (upper right). Its cross section is roughly 10 times lower than ggH at $m_H = 125$ GeV, and becomes comparable at $m_H = 1$ TeV. In VBF production mode, two $W^{\pm}$ or Z boson fusion create Higgs that have been emitted by a (anti)fermion pair collision. The VBF experimental signal is very clean and can be well distinguished from ggH with two high energy jets in detected in forward region. These jets signature helps to increase signal to background ratio.

In W/Z association production mode, Higgs is radiated by high energetic $W^{\pm}$ or Z bosons that is produced by fusion of two fermions. This production mode was leading mechanism at LEP and sub-leading mechanism at Tevatron. At LEP it becomes very probable to produce $W^{\pm}$ or Z boson with electron positron collisions. In case of hadron collider, tevatron makes quark antiquark collisions more likely as compared to LHC. The least dominant process of Higgs production is illustrated in Figure 2.4 (bottom right). In this process two gluons collide producing pair of top antitop which produces Higgs by fusion in association $t\bar{t}$.

2.6 Higgs Decay

Higgs, being an electroweak particle, can decay in variety of channels depending upon mass of final state particles, interaction strength and etc. The life time of SM 126 GeV Higgs is 1.6×10^{-22} s and it can decay to all SM particles having masses (most probably to heaviest onshell particle). Higgs boson decay branching ratio to a variety of particles are illustrated in Figure 2.6 [18]. $H \to b\bar{b}$ (at $m_H \approx 125$ GeV) is the most dominant decay mode of Higgs. But the QCD background in this channel is of many order higher than signal which makes it less sensitive channel. However, CMS and ATLAS collaborations and other collaborations [19], [20] and [21](CDF and D0) have exploited this channel, in association with a vector boson which decays to leptons, in boosted regime. B quarks are also difficult to reconstruct in a detector. When they decay in the detector, they fragment into jets of many hadrons. These jets are difficult to reconstruct perfectly. Because of this, it is necessary to exploit other channels in this region. $H \to \gamma\gamma$ is one most favorable decay channel due to signature with two high energy photons with excellent mass resolution.

As the Higgs mass at 125 GeV, Higgs boson decay to diboson(WW, ZZ) boson are also available, where one or both boson can be off-shell. WW decay further has neutrinos in final state due to which it is not possible to reconstruct the final state fully. $H \to ZZ \to \ell\ell\ell\ell$ decay is labeled as golden channel in spite of low branching ratio. The clean signature of at least four leptons makes this channel almost background free. These leptons are high energetic leptons which can reconstructed with excellence in the detector [18]. Measurements on Higgs reported in this dissertation are based on $H \to ZZ \to \ell\ell\ell\ell$ channel.

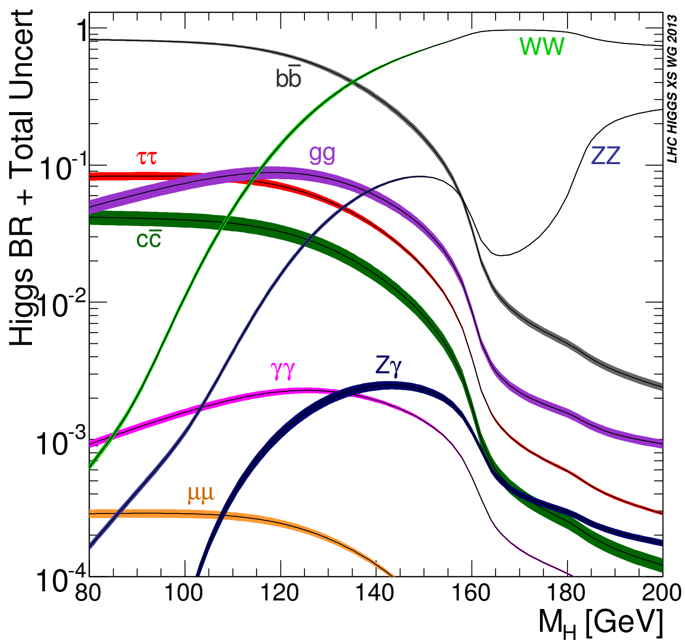


FIGURE 2.6: Higgs branching ratios for different channels (in terms of mass)

Chapter 3

The Large Hadron Collider (LHC)

LHC is located at CERN (European Organization for Nuclear Research) located at Franco-Swiss border near Geneva, Switzerland and is most powerful setup of its kind to probe resarches in particle physics. From its design, it is capable enough to accelerate and then collide two proton beams [1] at center of mass energy upto 14 TeV at peak luminosity of $\mathscr{L} = 10^{34}\text{cm}^2\text{s}^{-1}$.It was constructed from 1998 to 2008, in a 3.8 meters wide tunnel of 27 Kilometers circumference at a depth which varies from 75-150 meters underground. This tunnel was previously used for LEP collider which was constructed from 1983 to 1988. Two counter rotating proton beams (anti-proton beam is disfavored to use) in two rings are used where the beam pipes are kept at ultra-high vacuum just to preserve the beams from possible extra collisions with molecules of gas. To orient the beams in alongside the tunnel superconducting (cryo) dipole magnets are used as sketched in 3.1.

3.1 LHC beam operations

Beams of protons are inserted to the LHC experiment at injector. Protons are obtained by removing electrons using high electric field. Injector is connected to Linac2, proton synchrotron booster, proton synchrotron and the super proton synchrotron booster (PSB) in which the proton beams pass and with a subsequent gain in energy at each step (50 MeV to 450 GeV from Linac2 to PSB). From the PSB, beam is then passed to LHC tunnel where beams accelerates until each beam attains the energy of 6.5 TeVfor the subsequent collisions to be recorded at dedicated experiments or detectors of LHC. The schematic diagram of LHC accelerating mechanism is given in the Fig. 3.2

[1] LHC also does collision experiments of Heavy(Pb) ions at 2.8 TeVper nucleon. Details are beyond the scope of the dissertaton

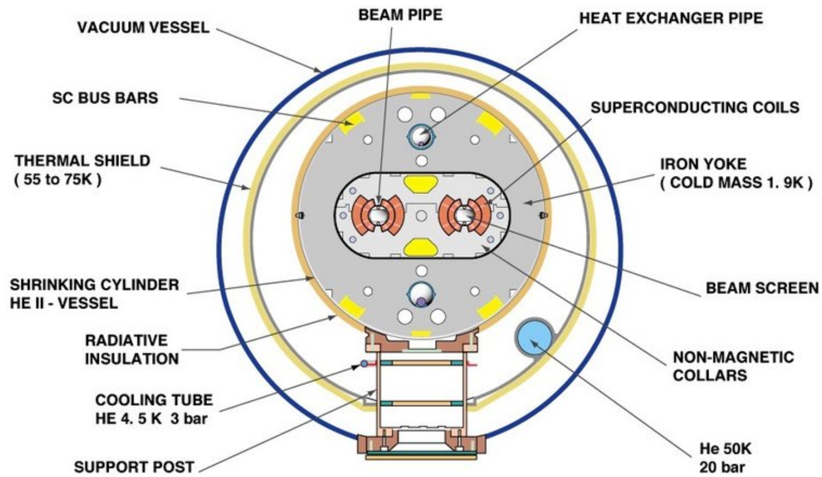


FIGURE 3.1: Cross section of the LHC cryodipole

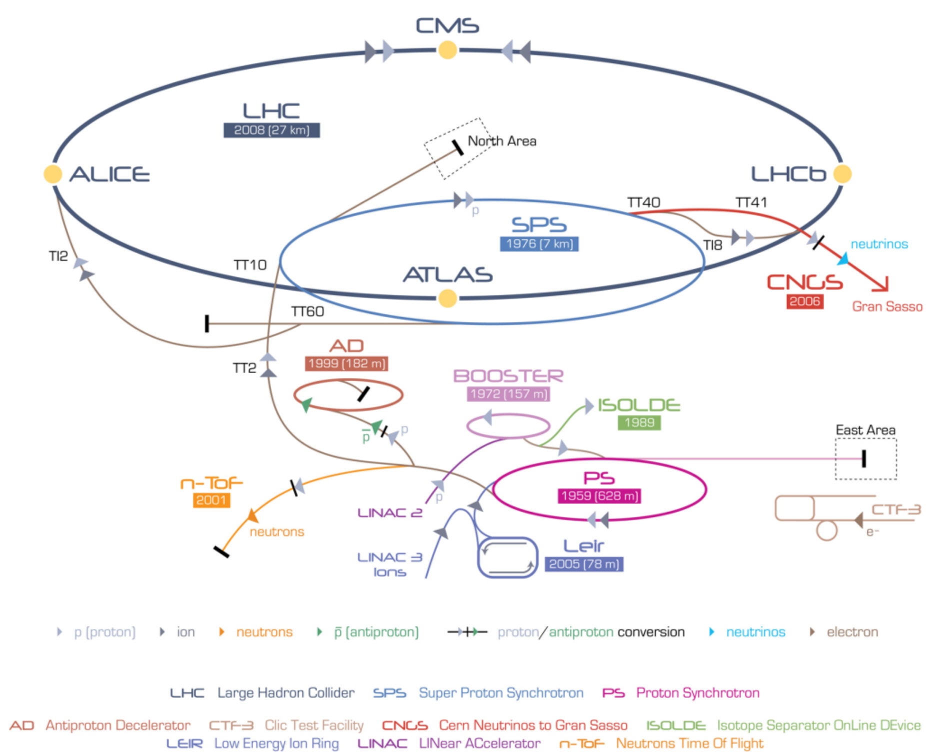


FIGURE 3.2: Accelerating mechanism of LHC experiment at CERN

For the first time, beam of protons was injected into LHC on September 10, 2008. The initial test to circulate the beams in clockwise and counterclockwise throughout the collider was also successfully completed by the same day. On 19 September 2008, due to an electric defect magnet quench was triggered about a hundred bending magnets resulting in several damages. After the repair, in November 2009, LHC became world's strongest particle accelerator after having achieved 1.18 TeV energy each beam thus defeating the Tevatron's highest achieved energy of 0.98 TeV. In 2010, LHC kept boasting the beam energies and reached center of mass energy up to 7 TeV (3.5 TeV each beam). In 2011 and then in 2012, LHC consistently delivered data at 7 and 8 TeV energies respectively. During 2013-14 , LHC machine was shutdown to do major parts upgrade on the LHC machine and detectors installed on it. In 2015, upgraded LHC started again at 13 TeV close to full stipulated operating energies (14 TeV). After successfully delivering proton-proton collisions data during Run 2 till the end of 2018, LHC machine is shutdown for Long Shutdown 2 (LS2) upgrades and is expected be operational again for Run3 in 2021 after several upgrades.

3.1.1 Machine Layout and Performance

Basic layout of LHC is shown in Figure 3.2 [22]. The LHC is designed to have two separate rings containing clockwise and counterclockwise proton beams, equipped with separate magnetic fields and vacuum chambers. These two rings only intersect each other at the points where detectors are installed. The ATLAS and CMS, two general purpose detectors of LHC, are installed opposite to each other (at P1 and P5 respectively). The other two important detectors, ALICE and LHCb, are installed (at P2 and P8 respectively) as shown in the same Figure 3.2 which are designed to study heavy ion Physics and B Physics (b quark) respectively. Two protons beams are injected in the vertical plane in both rings close to point 2 and point 8. In order to keep the beam alongside the tunnel during the process of accelerating, magnetic force is applied through diploe magnet made of $NbTi$ superconducting material. Apart from that, quadrupole magnet is used to focus the beam. Once each beam attains targeted energy i.e. 7 TeV, they are made collided at differents spots of LHC tunnel where different experiments are installed.

LHC design goal is to test SM Physics to ultimate precision and hunt for any kind of new physics up to 14 TeV. In collision process, number of events (N) per unit time (called event rate) for a certain process produced at LHC are given as

$$\frac{dN}{dt} = L \times \sigma_{process} \tag{3.1}$$

where $\sigma_{process}$ being the cross section of the process and L is the instantaneous machine luminosity. Integrating over time gives the total number of events in the particular process,

$$N = \int L dt \times \sigma_{process} = \mathscr{L} \sigma_{process} \tag{3.2}$$

Here $\mathscr{L}$ represents *integrated* machine luminosity. All possible processes' cross sections produced during proton-(anti) proton collisions at LHC machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$ are shown in Figure 3.3. It is obvious from the diagram that an increase in machine luminosity and center-of-mass energies would enable us to get more statistics of Higgs boson. The Nominal luminosity for CMS and ATLAS experiment is $\mathscr{L}$ = 10^{34}cm^{2}s^{-1}, and nominal center of mass energy of LHC is $\sqrt{s} = 14$ TeV.

Nominal Luminosity in terms of Beam Parameters

The machine luminosity can be expressed in terms of beam parameters as:

$$\mathscr{L} = \frac{\gamma_r N_b^2 f_{rev} k_b}{4\pi\epsilon_n \beta^*} F \tag{3.3}$$

here γ_r represents relativistic factor; however, N_b is bunch intensity, k_b being number of bunches in a beam; f_{rev} is the revolution frequency, β^* represents the beta function at the collisions point and F is the luminosity depletion factor due to cross angle at interaction point (IP). This factor can be written as below

$$F = 1 + (\frac{\theta_c \theta_X}{2\sigma^*})^2 \tag{3.4}$$

where θ_c, θ_z and σ^* are crossing angle at point of interaction, root mean square (RMS) length of bunch and σ^* the transverse RMS size of beam at IP. In above expression of luminosity, some parameters can be used to increased the luminosity at LHC; however, some of them cannot be used e.g. f_{rev} which is fixed by perimeter

TABLE 3.1: Parameters' values for LHC operations.

Parameter	Value
Circumference	26659 m
Center of mass energy ($\sqrt{s}$)	14 TeV
Momentum at collision	7 TeV/s
Luminosity (nominal)	$10^{34} cm^2 s^{-1}$
Bunch spacing	25 ns
Minimum distance between bunches	7 m
No. of bunches in ring	2808
No. of protons in a bunch	1.15×10^{11}
Beta function at the collision point(β^*)	0.55 m
Transverse RMS size of beam at impact point(σ^*)	16.3 μm
Peak magnetic field of dipole	8.33 T
Beam lifetime	10 h
Current in main dipole	11800 A
Stored energy in magnets	11 GJ
No.of dipoles	1232
No.of quadrupoles	858
Minimum distance between bunches	7 m
Operating temperature of dipole magnet	1.9 K

of LHC tunnel and protons' speed in the beam. The expression above is valid for two circular beams with same parameters.

The two major experiment of LHC, the CMS [23] and the ATLAS [24], are designed to operate at luminosity of $\mathscr{L} = 10^{34} cm^2 s^{-1}$. The summary of the parameters to obtain this nominal luminosity is summarized in Table 3.1.

By the end of the Run 2, LHC has delivered 150 $\mathrm{fb}^{-1}$of data. However, to fully explore the SM, particulary the Higgs sector, and even beyond SM for the rare processes, physicists are really in need of sufficient statistics (i.e. higher luminosity). High luminosity LHC is, therefore, planned to achieve a peak luminosity of 5 $\times 10^{34} cm^2 s^{-1}$ - 10 $\times 10^{34} cm^2 s^{-1}$.

3.2 Data Taking from 2015 to 2018 (Full Run 2)

After successful data taking during Run 1, LHC machine was shutdown for two years in 2013 and 2014 for major upgrades in LHC machine and detectors. LHC started again in 2015 at center-of-mass energies to 13 TeV(close to its design

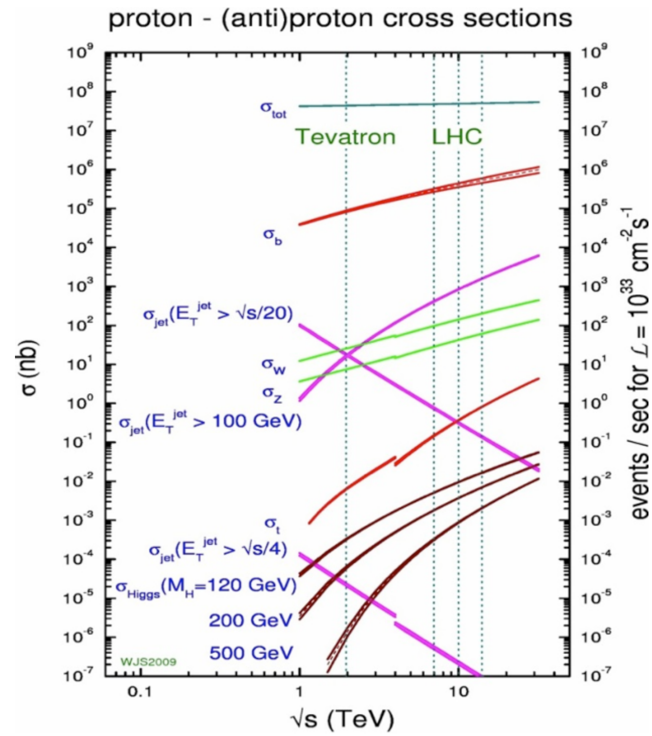


FIGURE 3.3: Cross section values for different processes corresponding to machine luminosity $\mathscr{L} = 10^{33}\ cm^2\ s^{-1}$.

value i.e 14 TeV). LHC delivered 4.2 fb^{-1}during 2015 at 13 TeV center of mass energy and subsequently, it delivered 41.0 fb^{-1}, 49.8 $\mathrm{fb}^{-1}$and 67.9 fb^{-1}during the years 2016, 2017 and 2018 as shown in the Figure 3.4. In the figures we can also see the recorded data by the CMS experiment from LHC delivered data in different run periods.

3.3 LHC upgradation during Long shutdown 2 (LS2)

After successfully delivering data during its Run2 as described in section 3.2, LHC is passing through its long shutdown 2 period. During this period, several upgardes are ongoing which can be regarded as preparations for High Luminosity LHC later on (HL-LHC). However, main upgrades in this regards will take place during LS3 after Run 3 period of LHC. One is the replacement of Linac2 with the Linac4. It is about 90 meter in length that took around 10 years to build it.It would be capable to accelerate negative hydrogen ions (H^-) for beam energy upto 160 MeV and will enable beam to double its intensity Ref. **??**. Besides, the individual

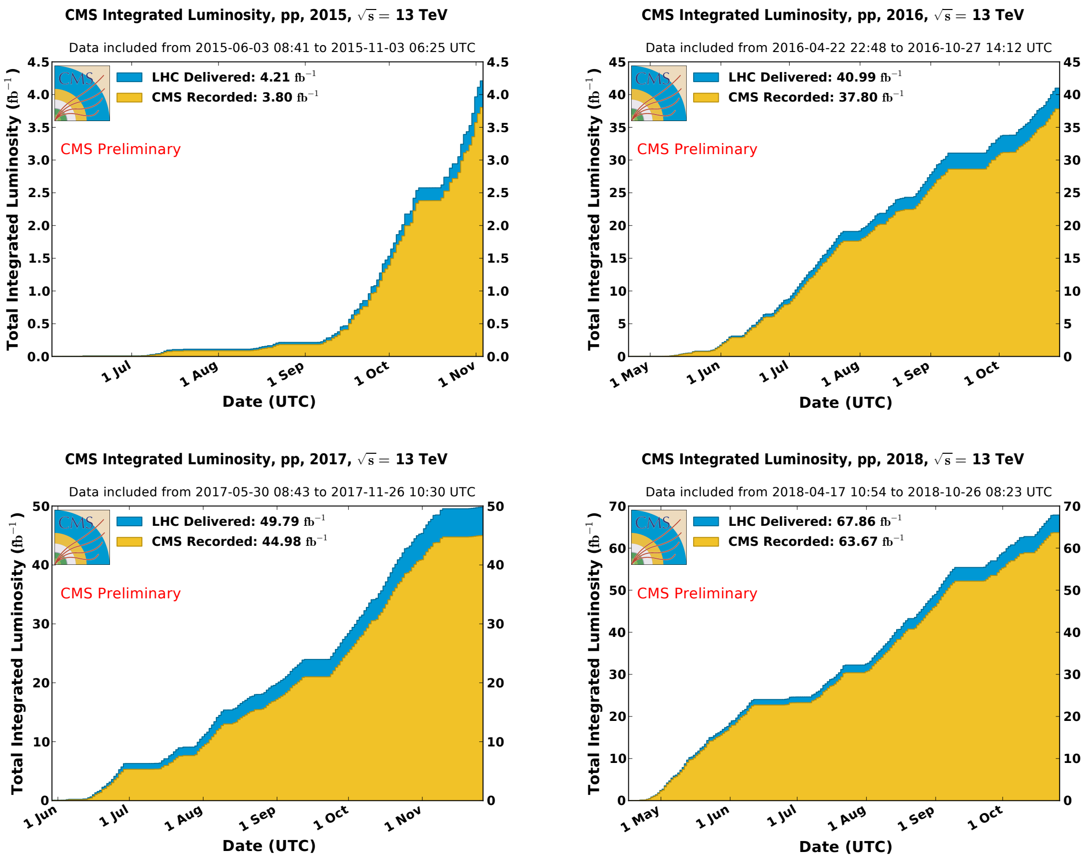


FIGURE 3.4: Integrated luminosities of LHC delivered (blue) and CMS recorded (yellow) data for year 2015 (top left), year 2016 (top right), year 2017 (bottom left) and year 2018 (bottom right).

experiments of LHC, i.e. CMS, ATLAS, ALICE and LHCb, have also planned certain upgarades during this LS2 period Ref. **??**.

ATLAS is builing muon small wheels to be integrated with the trigger system. For high pileup LAr electronics are being upgraded.

CMS is changing the photodetectors (HPD) of barrel and endcap HCAL to the silicon photo-multipliers (SiPM). Also the beam pipes made with stainless steel are being replaced with Aluminium composed material. Also, based on damage from radiation in endcaps of HCAL, scintillating tiles would be changed. Furthermore, for HL-LHC beprations, CSC muon chambers are being upgraded with new front end (FE) eleconics.

ALICE is making several upgrades for Inner Tracking System (ITS), Time Projection Chamber (TPS), Central Trigger Processor, Data Acquisition (DAQ) High Level Trigger (HLT), Muon Forward Tracker (MFT) and so on.

LHCb is improving its readout from 1 MHz to 40 MHz with trigger based fully on softwares by replaceing FE electronics and new computing infrastructure.

Chapter 4

The Compat Muon Solenoid (CMS) detector

Since the data used for the analysis in the dissertation was collected using the CMS detector, this chapter would focus on geometry of CMS, its parts (sub-detectors) and description of how events are reconstructed with the CMS detector.

4.1 The Compact Muon Solenoid (CMS) detector

LHC has set new frontiers for researches in particle physics experiments. State of the art experimental facility has annexed with equipment based on latest technology. To assure the validity of results from the experiment, LHC has its two major and general purpose experiments for cross-validation, one of which is CMS. They have same physics goals with some different design and the detector technology. Particulary the CMS is capable to provide good resolution for muon identification and momentum over wide range of dimensions and determination of the charge on muons with ultimate precision with momentum up to 1 TeV. It gives healthy inner tacker reconstruction efficiency and resolution of charged particle momentum. It has efficient trigger system and better discrimination of τ's and b-jets. Further, it is benefitted from good dielectron, diphoton and dijet mass and energy resolutions, and as well as missing-transverse-energy (MET) resolutions. This makes it capable to study large areas from SM Physics (including Higgs Physics) to search for extra dimensions and the dark matter covering physics in energy range from a few 100 GeV to TeV scale [25].

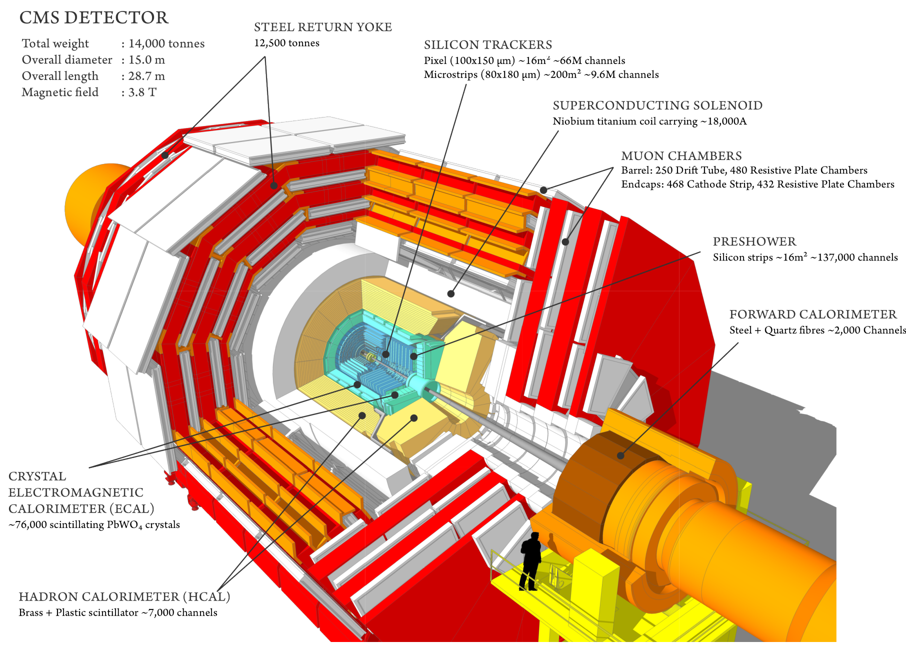


FIGURE 4.1: The CMS detector schematic diagram

4.1.1 General aspects and the geometry

Being one of the two general purpose detectors installed at LHC ring, CMS is is constructed in cylinderical shape having length about 28 m and outer ratdius of around 7 m. Its overall height is about 15 m with overall weight of about 14,000 tonnes. Overall deisgn of the CMS detector can been seen in the Figure 4.1. Learning from the earlier experiment at CERN, CMS engineered to be built in different slices at surface instead of doing everything underground. And fifteen separate slices were then lowered to underground to be fixed at the LHC tunnel.

Moving radially outwards from interaction point of beams, there are inner tracker system (ITS), electromagnetic calorimeter (ECAL) and hadronic calorimeter(HCAL) sub-detectors of CMS which are fitted inside the large solenoid magnet. Inner tracker of CMS have large track multiplicity as being closest to beam pipe where collisions happens. ECAL is build with crystals of lead tungstate which covers pseudorapidity up to $|\eta| < 3.0$. After ECAL, hadronic calorimeter is installed which is made of brass. Outermost layer of the CMS detector is the Muon System. Details of sub-detectors, trigger system and data acquisition systems will be described in next Sections.

The CMS magnet

Measuring the momentum of the charged particle with great precision is one the ultimate goals of the CMS detector. In this regards, choice of magnetic field configuration is very important feature of a particle detector like CMS. Stronger magnetic field is needed to measure high energy charged particle momentum with precision. In CMS, this is achieved by selecting the superconducting solenoid magnet.

When a charged particle moves in uniform magnetic field of a solenoid, its momentum resolution which depends upon strength of magnetic field and the radius of solenoid, can be written as:

$$\frac{dp}{p} \propto \frac{p}{Br^2} \tag{4.1}$$

Here r is the path length traversed by the charged particle inside the magnetic field (it becomes radius of the magnetic if the particle passed the full magnet). It is obvious from the relation, that for detector to obtain high momentum resolution, we need a volumetric solenoid magnet with stronger magnetic field. CMS solenoid magnet is designed to have bore of diameter 6 m and length 12.5 m producing magnetic field of 3.8 T which is 100,000 time stronger than earth magnet. It is the largest ever built magnet weighs about 12,000 tonnes. It is powerful enough in providing a strong magnetic field in its dimensions throughout the detector. It surrounds CMS ITS, ECAL and HCAL. The strong magnetic field efficiently bends the charged particles when they move through it. More the momentum of the charged particles less they bend, so high energetic charged particles require a stronger magnetic field for its momentum to be measured accurately. CMS aims to have excellent momentum resolution with such strong magnetic field specially for muons. The schematic diagram of CMS magnet is given in the Figure 4.2.

The CMS magnetic is made of coil of NbTi wire which becomes a source of uniform magnetic field when current passes through it. The NbTi becomes superconductor at $-268.5°$ C (4 K) which allows a huge amount of current (10,000 A) to pass through it. To maintain this temperature liquid helium is used. The magnetic field direction inside (tracker) and outside the solenoid (muon system) allows to make second momentum measurement for muons [26, 27, 28, 29].

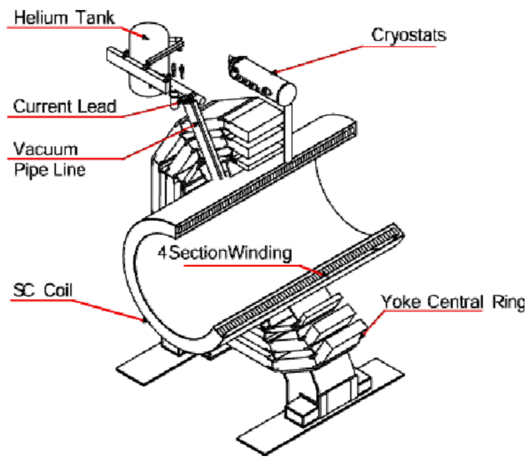
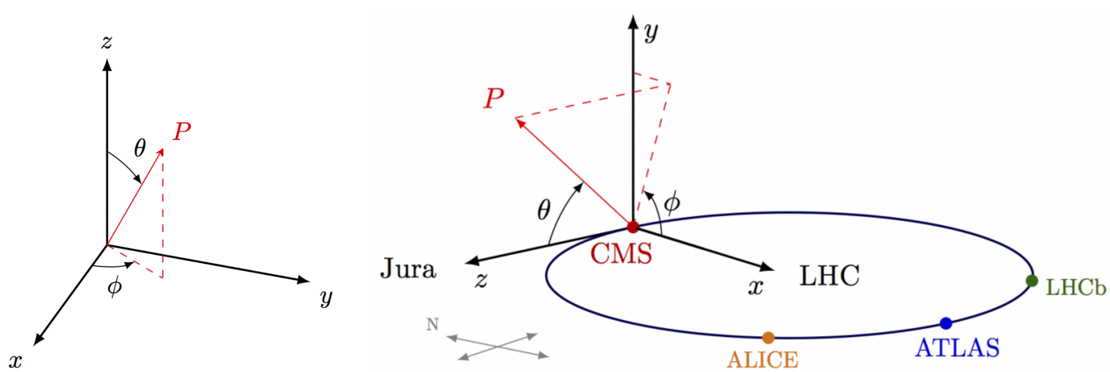


FIGURE 4.2: CMS magnet system.

4.1.2 CMS Coordinate system

Coordinate system used for CMS detector for its operations is sketched in Figures 4.3(a) and 4.3(b). The CMS detector is in cylinderical shape around beam axis as illustrated in Figure 4.1 which is represented by $Z - axis$ in Figures 4.3(b). The point where two beams collides in beam pipe is called center of experiment; the y axis is directed vertically upwards and the x axis being horizontal towards the LHC ring center.



(a) (b)

FIGURE 4.3: CMS spherical coordinte system

The r represents the radial component in transverse plane while ϕ is the the azimuthal angle measured from the x axis. θ which is measured from z axis is called polar angle. Position of particle when it passes through the CMS is described in terms of (η, ϕ), where η being the pseudo-rapidity defined as $\eta = -\ln\tan(\theta/2)$ and remains invariant under the longitudinal boosts in the environment of the partons which carry the fractions of the momentum from hard scattering collisions of e.g. proton-proton. Moreover, pseudorapidity also helps to differentiate the barrel regions from the endcap regions of sub detectors. In terms of pseudorapidity, the barrel region are defined by $|\eta| < 1.5$ whereas the endcap region corresponds to $1.5 < |\eta| < 3.0$.

4.1.3 Inner Tracking System (ITS)

The CMS ITS is a sub-detecting system of the CMS detector which is important form its task to give best tracks and hence the momenta (above 1 GeV) of the particles (electron, muon, hadrons-quark-structured particles and other short-lived particles) produced in the collisions that take place at CMS, LHC. It is the most inner sub-detector placed under influence of a strong and uniform magnetic field of CMS solenoid and it surrounds the region of beam pipe as shown in Figure 4.1. ITS also provides the position information of the primary and secondary vertices which is important to reject extra interactions and objects.

Being in cyliderical shape, it as diameter of about 2.5 m and length about 5.5 m providing $|\eta|$ coverage up to 2.5. At LHC nominal luminosity of $10^{34} cm^2 s^{-1}$, there is an average of more than 20 proton-proton interactions (called pileup, for 2018 it was above 30) leading to average about 1000 particles emerging after every bunch crossing, 25 ns. ITS is entirely made up of (200 m^2 of) silicon which is benefitted from being super efficienct in the environment of huge flux of particles coming out of the interaction point. A schematic diagram of CMS ITS is shown in the figure 4.4 (as stood before phase 1 pixel upgrade).

To limit the damage mainly by radiation ITS is kept at about -20° C. Full ITS is an assembly of a pixel detector and silicon strip tracker.

The Pixel Detector

Part of the tracker system which is nearest to the interaction point is the Pixel detector. Silicon modules of the pixel detector, are facilitated with pixel cells of size $100 \times 150 \mu m^2$ each providing the $|\eta|$ range up to 2.5. It helps to reconstruct the secondary vertex and delivers three spatial points precisely throughout its geometry [25].

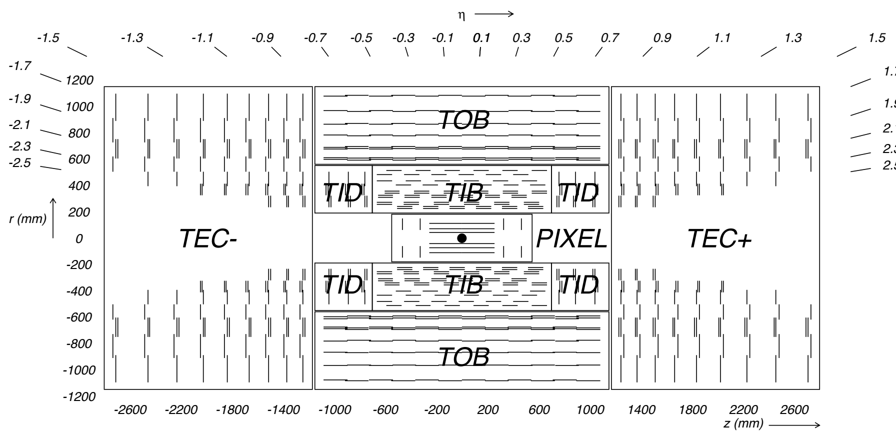


FIGURE 4.4: Schematic diagram of CMS tracker system. Lines represent detector modules (before phase 1 pixel upgrades).

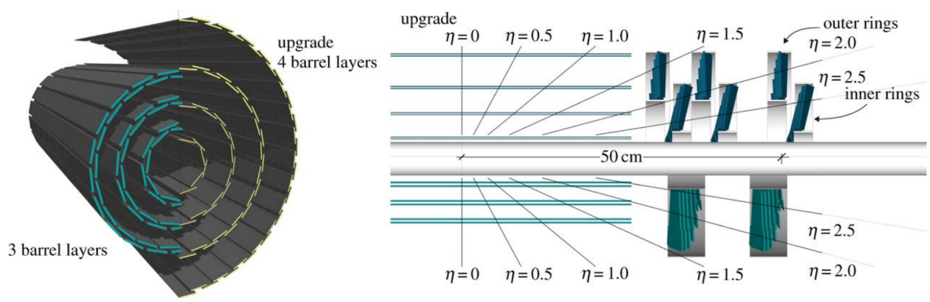


FIGURE 4.5: Schematic diagram of pixel detector's upgrade. Old pixel detector is shown by the lower half detector, and the new pixel detector is represented by the upper half.

During data taking of Run 1 operations of LHC, some dynamic inefficiencies (about 5% at an event pile-up of 40) were reported in the first layer of barrel of pixel sub-detector, due to the limited size of the readout bandwidth. Subsequently, a new pixel detector was installed in March 2017 Ref. **??**, equipped with one additional barrel layer and one additional forward disk. In the new pixel detector, the innermost layer is closer to the interaction point and the outermost one is further away from it as it is shown in Fig. 4.5

It also uses new Readout Chips (ROCs) benefitted from reduced data loss at higher collision rates with respect to Phase 1 operations (at 160 Mbit/s). Bandwidth of the readout electronics was increased approximately by a factor of eight (320 Mbit/s). In addition, DC-DC power converters were installed to allow reusing the existing fibers and cables. Hence, the upgraded pixel detector has a higher

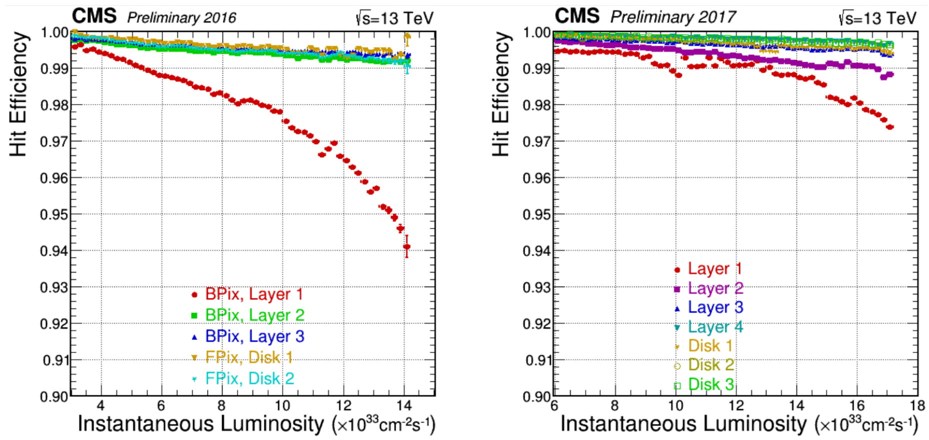


FIGURE 4.6: Pixel detector's hit efficiency as function of instantaneous luminosity corresponding to 2016 data (left) and to 2017 data (right)

rate capability and number of channels, providing increased tracking efficiency, reduced fake rate, and improved impact parameter resolution. In the figure 4.6, clear improvments in the dynamic efficiency of 2017 (with pixel upgrades) with respect to the 2016 (without pixel upgrades) can be seen in Ref. [**Sonneveld:2018ntf**].

Overall, the pixel detectors is of 1 m^2 size containing 66 million pixels.

The Silicon Strip

Next layer to the pixel detector is called Silicon Strip Tracker which is installed at radial distance from 20cm to 116cm. This detector surrounds both sides of pixel with Its tacker inner barrel (TIB, 4 layers) and tracker inner disk (TID, 3 layers) surrouds the pixel detector and extend upto 20 cm-50 cm where silicon microstrips of thickness 320μm are installed. Both are surrounded the Tracker outer barrel (TOB, 6 layers) which is 220 cm long and provides z coverage up to 118 cm. In barrel and endcap regions, the pitch of strips changes from 80μm to 120μm and 100μm to 141μm respectively. Finally, the most outer part of CMS tracker system is tracker outer barrel (TOB) installed at radial distance from 55cm to 116cm containing 500μm thick 6 barrel layers with strip pitch of 183μm and 122μm. Beyond z=118 cm, there are tracker endcaps (TEC+, TEC-, 9 discs) which cover 124 cm< $|z|$ <282 cm and 22.5 cm< $|r| < 113.5$ cm providing coverage of $|\eta|$<2.5.

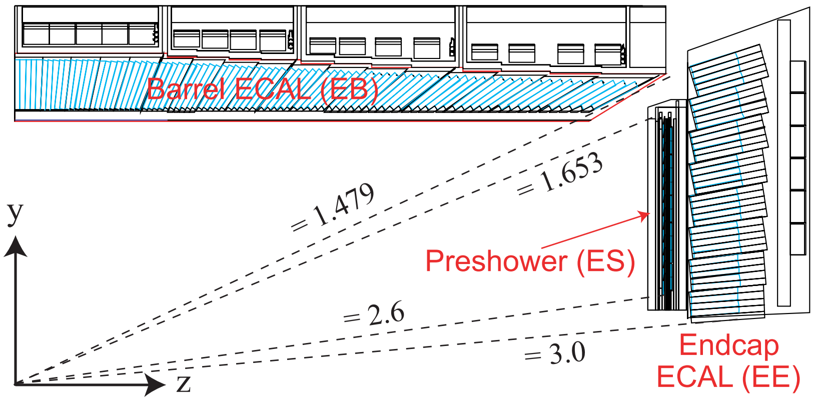


FIGURE 4.7: CMS ECAL pseudo-rapidity coverage.

4.1.4 Electromagnetic Calorimeter (ECAL)

ECAL is the cylindrical shaped homogeneous calorimeter built with 75848 crystals of lead tungstate ($PbWO_4$) placed inside the CMS solenoid magnet. Its barrel ergion (EB) contains 61200 $PbWO_4$ crystals and provides pseudorapidity coverage of $|\eta| < 1.479$. Endcap region of the ECAL (EE) is built with 7324 crystals on either side in endcap regions and provides pseudorapidity coverage up to $\eta < 3$. The region of $1.4442 < |\eta| < 1.56$ is regarded as crack region between barrel and endcaps as can been seen in Figure 4.7 taken from [25]. This region is excluded for physics analysis.

When high energy particle (e.g. electron or photon) coming out of the collision point enters ECAL, it produces bunch or cascade of particles (electron through bremsstrahlung and photon through pair production) with relatively lower energies. This is called EM showering that continues until electrons dominate in ionization and photon energy becomes less than pair-production threshold.

Photodetectors Avalanche photodiodes (APDs) in barrel region and the Vacuum phototriodes (VPTs) in endcap regions are installed which efficiently work under high magnetic field environment [30]. Response of ECAL comes with emission of light that is directly proportional incident particle's energy (through ionization).

A preshower detector is installed in endcap regions to distinguish between single photon and closeby diphoton candidate e.g neutral pion decay ($\pi^\circ \to \gamma\gamma$). Providing pseudorapidity coverage of $1.653 < |\eta| < 2.6$, it helps to efficiently determin the position of an electron and a photon.

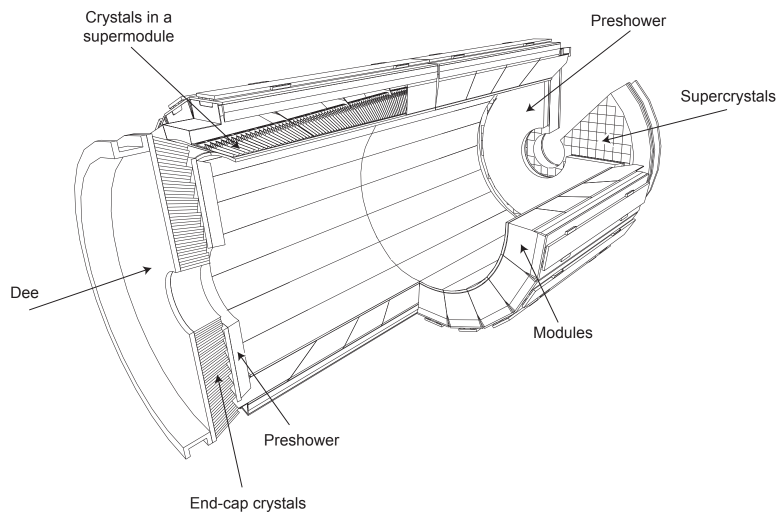


FIGURE 4.8: CMS ECAL layout

Lead tungstate is benefitted as it helps to make ECAL compact with high granularity due to the fact that it is denser (8.28 g/cm^3) with less radiation length (X○ of 0.89 cm). Also, lead tungstate has small Molière radius, which defines the area of deposited energy, helping to minimize the effect of pileup while measuring the energy. Other important factor is that crystals of lead tungstate have decay time of scintillation about 25 ns which is also the time of LHC bunch crossing.
ECAL operating temparature is around 18°C and light emitted from the scintillation process is temperature dependent. So, in order to maintain the temperature at nominal value, cooling system is installed which is based on water circulation through Aluminium pipes and is capable enough to maintain the temperature of detector upto $18 \pm 0.05^{\circ}$ C. Layout of the ECAL of CMS detector is shown in the Figure. 4.8.

Energy resolution of ECAL, which is the ability of detecting system to accurately measure energy of incident radiation, can be written as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{A}{\sqrt{E}}\right)^2 + \left(\frac{\sigma_n}{E}\right)^2 + c^2 \tag{4.2}$$

where A is stochastic term (defines the intrinsic statistical fluctuations in shower, in energy deposits of preshower, and the photo(electron)-statistics contributions), σ_n is the electronics noise term and c is constant that covers the constant factors such as non-uniformity in detector for light collection and etc. Contribution of

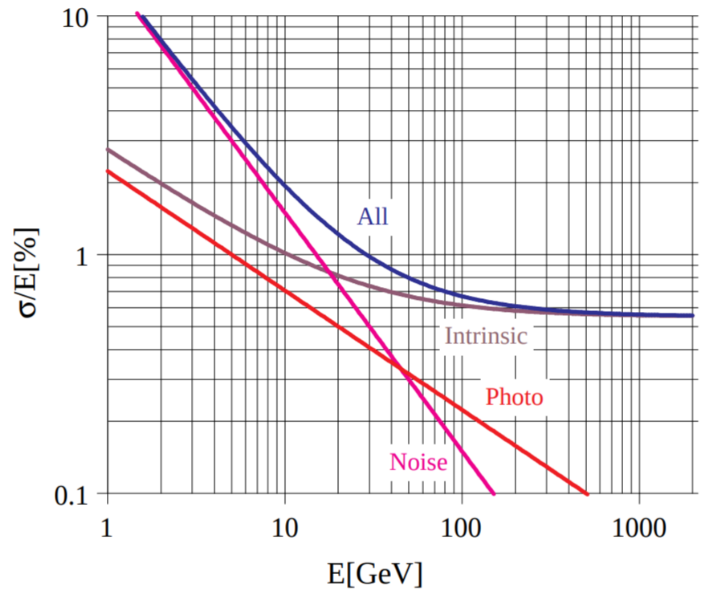


FIGURE 4.9: CMS ECAL energy resolution curve with individual term contributions.

each term to the energy resolution is shown in the Figure. 4.9.

A recent study on relative energy resolution of ECAL using 2017 data and MC has been performed using Z events decaying to a pair of electrons. The resolution plots for bremsstrahlung electrons (R_9>0.94 and R_9<0.94)[1] is separately shown for data and MC in the Figure 4.10 **[Zghiche_2019]**. For $\mathrm{H} \rightarrow 4\ell$ analyses, excellent energy resolution for leptons is important for precise measurements for which ECAL of CMS plays an important role as it did for Higgs discovery.

4.1.5 Hadron Calorimeter (HCAL)

It is the part of CMS detecting system, which helps for the detection and as well as energy measurements of strongly interacting particles such as quarks (or the ones in composition of quarks), gluons hadronic jets (like neutrons, pions and etc.). Being hermetic, HCAL also helps to measure missing transverse energy (MET) which can be associated to invisible particles like neutrinos which is important studies like top quark production. HCAL assembles with CMS system between the solenoid magnet and ECAL as can be seen in the Fig. 4.1 at radial distance from 1.77 m to 2.95 m. Benefitting from pretty short interaction length, at most brass is chosen as absorber throughout the HCAL. Brass is a non-magnetic

[1] R_9 is ratio of the energy contained in a 3×3 crystal matrix and the supercluster energy

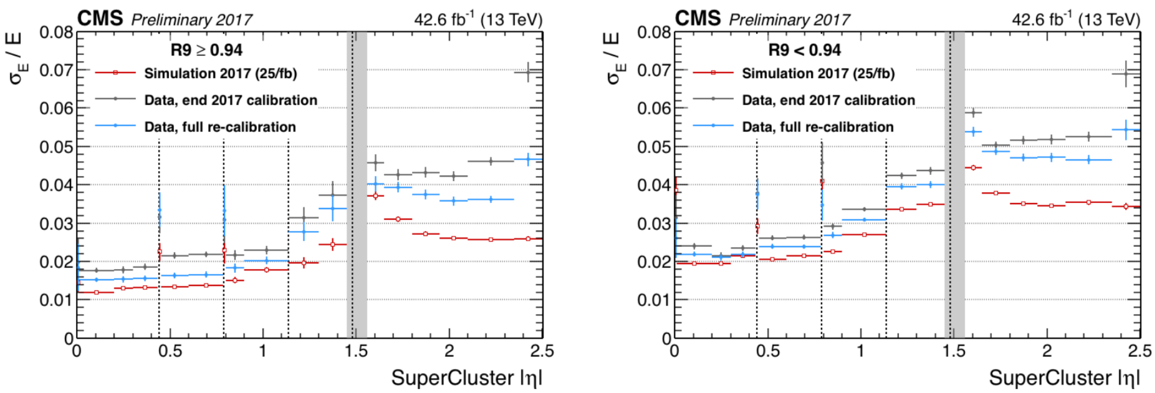


FIGURE 4.10: CMS ECAL energy resolution.

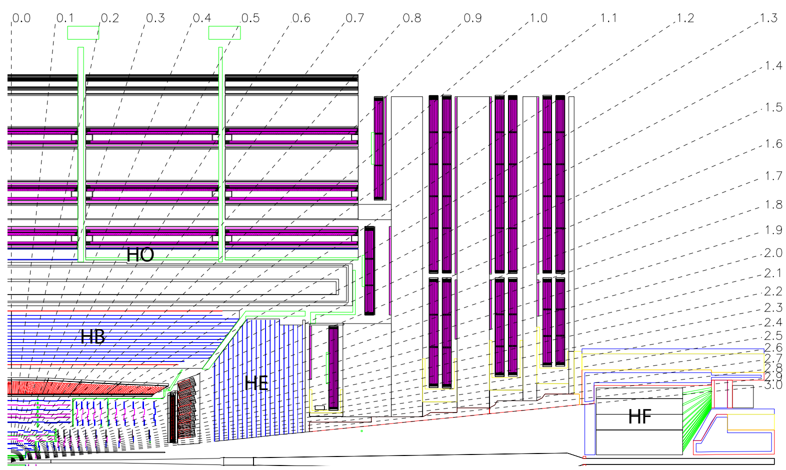


FIGURE 4.11: CMS HCAL layout.

low cost material. HCAL provides pseudorapidity coverage upto $|\eta| < 3.0$ beyond which, 11.2 m away from the interaction point, froward Hadron Calorimeter (HF) is employed on both sides of the detector providing additional coverage upt $|\eta| <$ 5.2. HF helps to detect particles from proton-proton collisions emerging at very small angles with respect the beam axis.

Schematic diagram of HCAL is shown in the Figure 4.11 [25].

HCAL is splitted into barrel and endcap regions, HB and HE which give pseudorapidity coverage of $|\eta| < 1.33$ and HE of $1.3 < |\eta| < 3$ respectively.

HCAL is also splitted into barrel and encap regions, HB and HE respectively. HB consistes of 32 towers provides coverage of pseudorapidity of $|\eta| < 1.4$ and HE of $1.3 < |\eta| < 3$. This results in 2304 towers in a segmentation of $\Delta\eta \times \Delta\phi$ = 0.87 $\times 0.87$.

HB serves as a single longitudinal sampling readout. HB is split into two halves (HB^+ and HB^-) which consist of 36 similar wedges each weighing 26 tonnes made up of brass plates. It may happen that tail of showers of hadrons which leak from rear of calorimeters may penetrate into ECAL and/or magnet. Therefore, additional calorimeter called Hadron Outer (HO) is installed outside the solenoid that used the coil of the magnet as absorber. With this effective thickness of HCAL becomes above 10 interaction lengths. HO has scintillator of 10 mm thickness and helps to improve resolution of HCAL for MET.
Pseudorapidity coverage of $3 < |\eta| < 3$ is provided by sub part of HCAL whihc is called Hadron Forward (HF). It suffers huge flux of the particles due to which sampling material is required to be hard against the radiations. For this purpose, steel/quartz fibre was used. HF is located 11.2 away from interaction point with 1.65 m absorber depth. With the incident particle flux, Cerenkov radiations are emitted through quartz fibers which is then channeled to photomultipliers (PMs).

HCAL is constructed with alternate layers of plastic and brass (the absorber and scientillator respectively) which helps to measure the arrival time, position and energy of a particle. Plates of brass are 79 mm thick each with 99 mm gaps for the scintillators installation. When a particle passes through scintillator, optical fibers collect the emitted lights and pass it to readout box. i Total energy is calculated by summation of the light over many layers in depths called tower. The scintillator in HCAL are arranged in the from of 70,000 tiny tiles.

The energy resolution of the HCAL (i.e. HB, HO and HE) is given as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{90\%}{\sqrt{E}}\right)^2 + (a)^2 \tag{4.3}$$

and for that of HF it is given as

$$\left(\frac{\sigma}{E}\right)^2 = \left(\frac{170\%}{\sqrt{E}}\right)^2 + (b)^2. \tag{4.4}$$

Where a and b are constants having values 0.045 and 0.09 repectively.

There were subsequent upgrades in HCAL as in 2017, PMTs were replaced with new multi-anode PMTs and additionally electronics replaced for better differentiation of real hit from the muons which hit the PMT window. HE of HCAL was upgraded and photo detectors were replaced with latest silicon photo multipliers (SiPMs) for noise elimination and better photo detection efficiency [**pixel_hcal_upgrade**].

4.1.6 CMS Muon System

Identification of muons originating from the collision point is a great challenge. Having long lifetime (i.e. 2.19 $\times 10^{-6}$, which is even higher in relativitic regime), they tend to cross the whole detector without decaying. Also, having higher mass (about 200 times the mass of electrons) it tends to lose very little energy via bremsstrahlung. So, they leave track however they deposit very little amount of energy in the calorimeters. Hence a dedicated muon system was proposed and installed as outermost sub-detector of CMS [31]. The schematic diagram of CMS muon sytem (as stood by Run 2 operation of LHC) is shown in the Figure 4.12. It is benefitted from three different strategies for detection and measurements on muons.

- Drift Tubes (DT), only in barrel region
- Resistive plate chamber (RPC), in barrel and endcap regions both
- Cathode Strip Chamber (CSC), only in endcap region

The detection of the muons is very important for many physics processes particularly for the SM Higgs when it decays to ZZ/ZZ^*. Best mass resolution is obtained when each of Z decays to a pair of opposite sign muons which intereact weakly with the detector leaving their traces both in muoon system and as well other parts of the CMS detector. Due to this reason Higgs decaying to ZZ/ZZ^* is regarded as golden channel for Higgs properties measurements. Some theories beyond SM also require muons to be reconstructed and identified with ultimate precision which is obtained with the CMS muon system.

Muon system of CMS, benefitted from mainly gas detection technology, is designed in cylinderical shape that gives pseudorapidity coverage upto $|\eta| < 2.4$. RPC covers both barrel and endcaps, DT covers the barrel and CSC covers the endcap only. Overall goals of CMS muon system include muon identification, standalone and global muon resolution, charge assignment, triggering and capable of withstanding in high radiation and background enviroment.

Drift Tubes (DT)

Drift chambers play role of tracking detector in barrel region motivated to work under low event rate and relatively less strength of magnetic field.

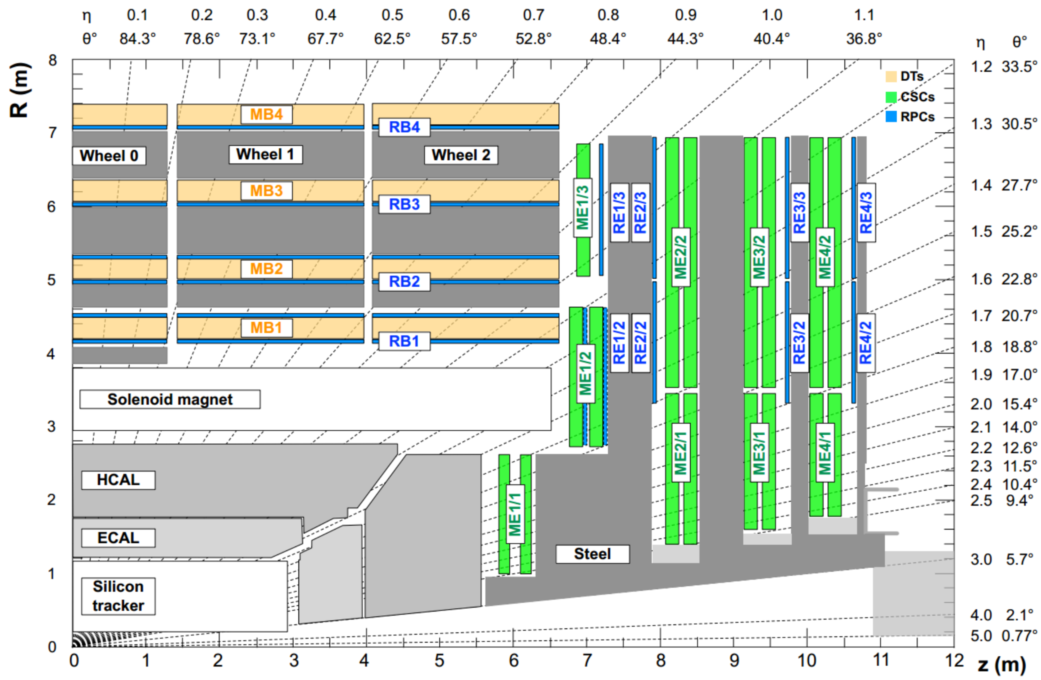
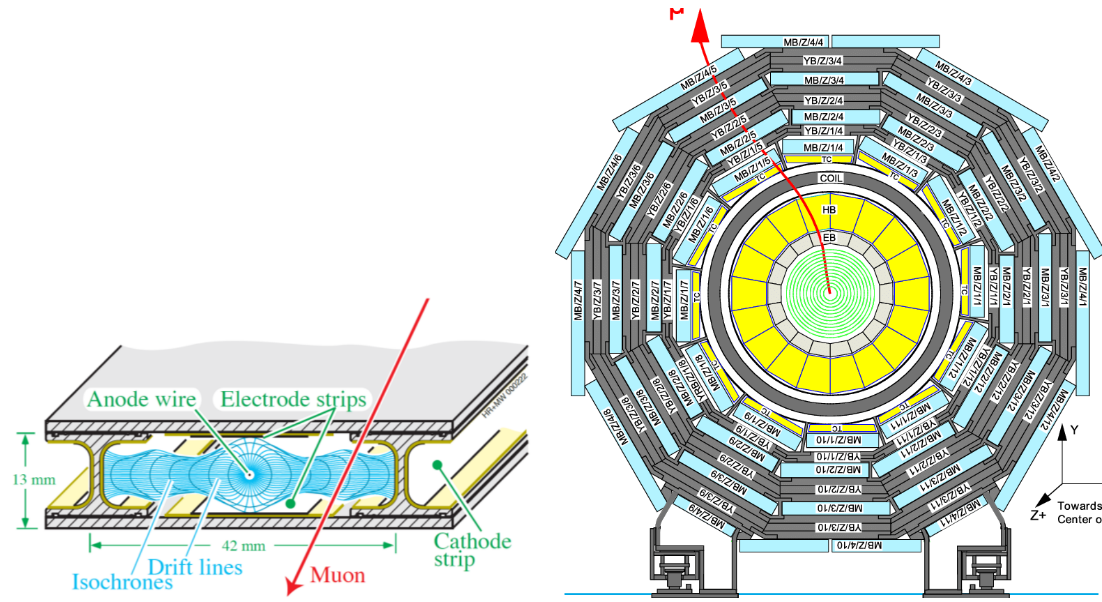


FIGURE 4.12: CMS muon system layout (as stood by Run2 operations of LHC).

The DTs make cover the barrel part of muon system alongwith the RPCs, as shown in the Figure. 4.12, providing pseudorapidity coverage of $|\eta| < 1.2$. DT system is annexed to the muon system in 4 stations, MB1, MB2, MB3 and MB4, in radially outward direction at 4.0, 4.9, 5.9 and 7.0 m respectively from IP. 250 DT chambers (composed of 12 Aluminium layers each) are fixed on 5 wheels along z-axis. 12 planes in a DT chamber (of average size 2m$\times$2.5m), except the ones from MB4, are build with 3 independent units which are called super-layer (SL). A set of two SLs measures coordinates in the bending plane (ϕ), the third SL helps to provide track coordinate along the beam (z). SL is further composed of DTs in 4 layers **[Paolucci:927394]**. Each DT cell has an anode wire made of steel (of radius 25 m) togther with walls of the cell as cathodes which produce the (drift) electric field. A DT cell is filled with mixture of CO_2 (15%) Ar (85%) gases which are benifitted from their low cost and provision of better drift velocity of saturation and the quenching under the influence of high operating voltage of 1.8kV. A schematic diagram of a drift cell and one of wheel bearing DT chambers is shown in the Figure . 4.13 and in Ref. **[article_dt]**. As a charged muon moves throught a DT cell, it causes ionization in gas. The librated electrons then move towards anode under the influence of the electric field. Drift time of these electrons helps to determine distance between anode wire and the muon.



(a) (b)

FIGURE 4.13: (a) Schematic diagram of drift cell and produced electric field. (b) Cross sectional view of DT chambers in barrel shown one of the five wheels

To keep drift velocity steady, the filled gas is kept at a constant pressure inside the chambers. The pressure in each wheel is continuously monitored by sensors.

Resistive Plate Chambers (RPC)

RPCs, the parallel plate gaseous detectors, are important part of CMS muon system installed in both barrel and endcap region. RPCs are very fast responsive used for triggering and they offer good time and spatial resolutions. RPCs are capable to provide a time resolution of about a nano-second. A schematic diagram of a RPC chamber taken from **[article_rpc]** is shown in the Figure. 4.14.

RPCs are made of two high resistive plastic positive and negative charged plates, kept parallel to each other, with gas filled inn between them (95% of $C_2H_2F_4$ and 5% of C_4H_{10}). RPCs have readout system between two gaps and operate in the avalanche mode which is produced when muon passes through by librating electrons during primary and secondary ionization of molecules of gas. With the excellent time resolution, RPCs are capable to assign bunch crossing with ultimte precision.

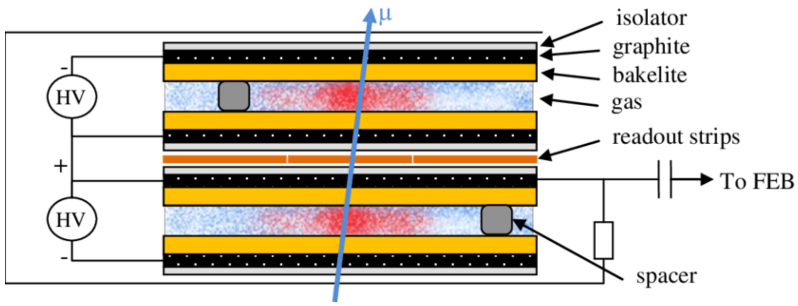


FIGURE 4.14: Cross sectional view of a RPC chamber.

Cathode Strip Chambers (CSC)

The cathode strip chambers (CSC) are the sub-detecting part of the CMS muon system, as shown in Figure 4.12, covering the high pseudorapidty range upto $|0.9| < |\eta| < 2.4$ and capable to provide the track and triggering of muon in this region alongwith the RPCs. It can efficiently operate at high event rate and helps to assign the bunch crossing quite efficiently. A total 540 CSCs are installed in 4 CSC stations (ME1, ME2, ME3 and ME4) and each station is further distributed in 2-3 rings with 18-36 CSCs fixed on each ring.

Each CSC is assembly of 6 independent layers, making 6 gaps of gas (Ar 50%, 40% CO_2, 10% CF_4) with each layer have radial cathode strips and transverse anode wires. Muon passing through a CSC chambers ionizes the gas molecules thus producing the electron avalanche towards the anode wires made of copper. Produced gaseous ions which move towards the strips for the detection of image charge. These signals are collected by readout sytem, amplified and then passed to the electronic boards on chambers. In each chamber, muon hit information can be measured along the track of muon in the 6 layers. To generat a triggere primitive, hits in 4 layers out of 6 is required. Fitting the position of successive hits in the chambers gives the precision in muon momentum measurement. A layout of CSC and signal generation process is shown in the Figure. 4.15.

In a chamber layer, Cathode Front-End Boards (CFEBs) [2] receive and then process the signal from strips. However, from anode wires, signal is read by Anode Front-End Boards (AFEBs) and processby Anode Local Charged Track (ALCT)

[2] Electronics or readout system of ME/1/1 chambers were upgraded during long shutdown 1 (LS1), hence differs from other chambers.

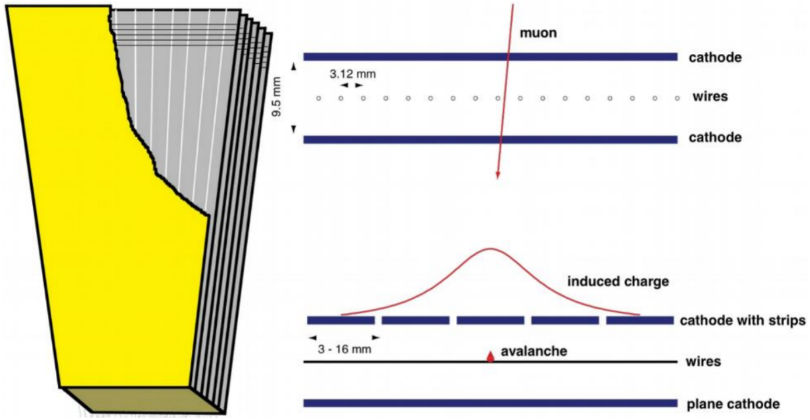


FIGURE 4.15: CSC layout (left) and induced current (right) when a charged particle pass through it.

board. These boards then communicate with the Trigger Mother Board (TMB) located in the vicinity of the endcap disks. Data is then transmitted to Data Mother Board (DMB) which sends it out Front End Driver (FET) crates whose front end electronics has interface with CMS data acquisition system. Overview of the chamber electronics and communication mechanism is elaborated in Figure. 4.16.

Improvements in CMS muon system during Run1 - Run2 and during ongoing long shutdowon 2

From Run1 to Run2 operations of LHC, increasing luminosity was challenging for muon system of CMS. In order to cope with high event rate environment and utilize the gain in the luminosity, muon system of CMS underwent some upgrades or modifications [**Sirunyan:2018fpa**]. To minimize the misidentification rates, and improve efficiency and redundancy, RPC and CSC were extended to install RE4 and ME4 as fourth stations in the mainstream. Due to this, CSC and RPC Trigger Track-Finder were rebuilt to accommodate the additional information from ME4 and RE4 planes. Readout electronics and trigger system were also enhanced. To increase the bandwidth, optical fibre links in DT and CSC were added. During Run1, strips in ME1/1a ring were ganged together into a group of 3. Subsequently, new electronics, Digital Cathode Front End Boards (DCFEBS), were installed in ME1/1 chambers of CSC to provide readout to every strip separately allowing ME1/1 to effectively perform in high luminosity environment.
During ongoing long shutdown 2 (LS2), there several upgrades are in process for

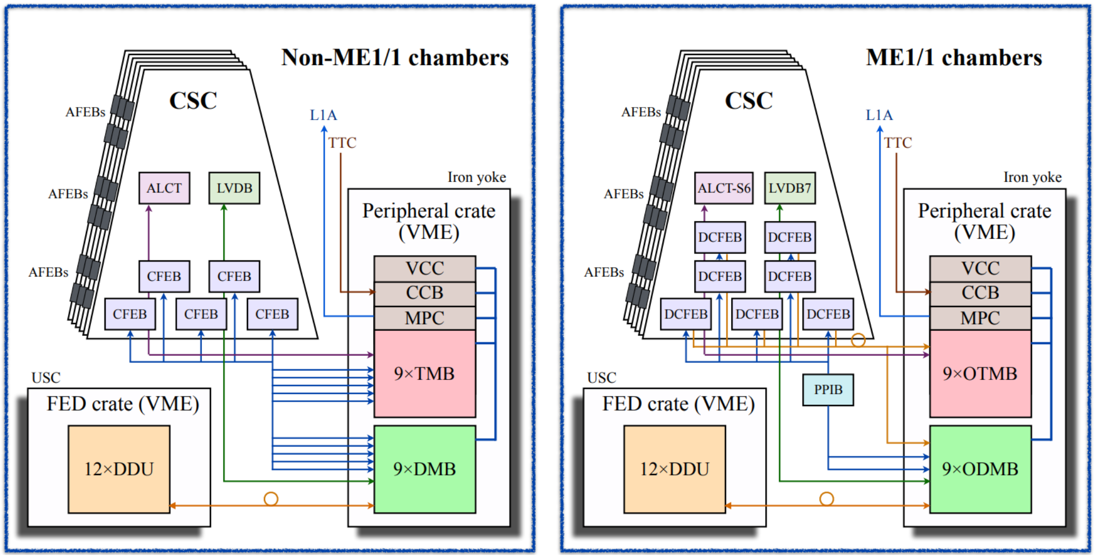


FIGURE 4.16: Overview of electronic boards of a CSC chamber and involved in communication of data. Readout system of ME/1/1 chambers were upgraded after Run 1, hence differs from other chambers.

upcoming physics programs under high luminosity LHC. CSC will undergo upgrade of readout system of ME1/1, ME2/1, ME3/1, and ME4/1. Total 180 chambers will be refurbished to replace CFEBs with upgraded DCFEBs and ALCT mezzanine boards. Also for ME2/1, ME3/1 and ME4/1, LVDBs will also be upgraded.

Other main improvement for the muon sytem for future programs is installation of Gas Electron Multiplier (GEM) chambers in forward regions. Due to upcoming higher collision rate during Run 3 and so on, muon system needs to improve its capibilities for tracking and the triggering to effidiently work extreme environment which can be achieved with GEM chambers. In this program, first two muon stations will be supported with GE1/1 and GE2/1 (covering $1.5 < |\eta| < 2.4$) which have layers for high precision measurement. Additionally, ME0 station will be installed to extend coverage of muon system upto $|\eta| < 3$. GE1/1 among these, will be installed during LS2 and would take part in Run 3 operations; however, GE1/2 and ME0 would be installed after Run 3 operations during long shutdown 3 (LS3) after 2022 when RPC will also upgraded as “Improved RPC (iRPC)" with new geometry, materials and frontend electronics. Updated schematic diagram of CMS muon system with addition of GEM chambers can be seen in the Figure. 4.17 (also in Ref. [**Radogna:2016esa**, **Costantini:2711362**]) where the actual position of

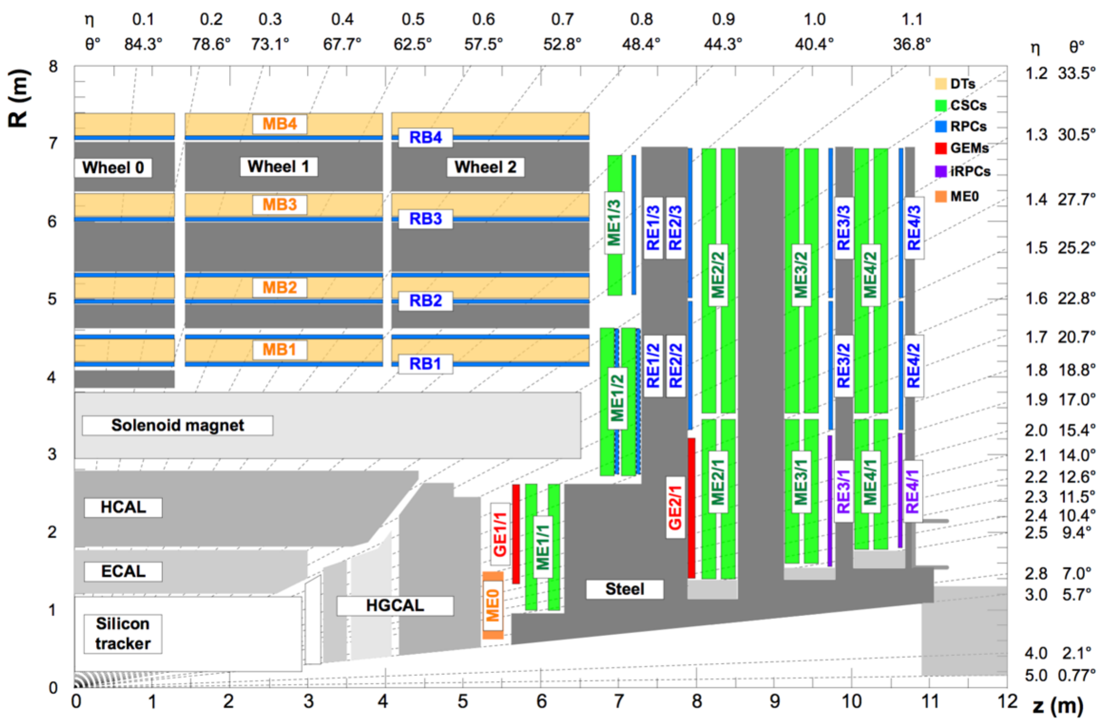


FIGURE 4.17: Updated schematic diagram of CMS muon system with addition of GEM chambers.

these chambers in muon system can be understood.

4.2 CMS Trigger System

CMS is big and general purpose experiment purposed to explore data that LHC has produced during the proton-proton collisions. LHC produces around 40 million collisions in a second (40 MHz); however, not all the collisions are interesting for physics purposes. Further, for the data acquisition (DAQ) system, it is not effective to read every event produce at high rate. Also, storing the uninteresting eventsprouced in LHC collisions will require a lot of storage. Hence, there is a dire need to reduce the rate of produced events during LHC collisions which is done by the trigger system which select only those events which are crucial for physics purposes. Trigger system of CMS is a sophisticated two-level system; Level 1 (L1) trigger and high level trigger (HLT).

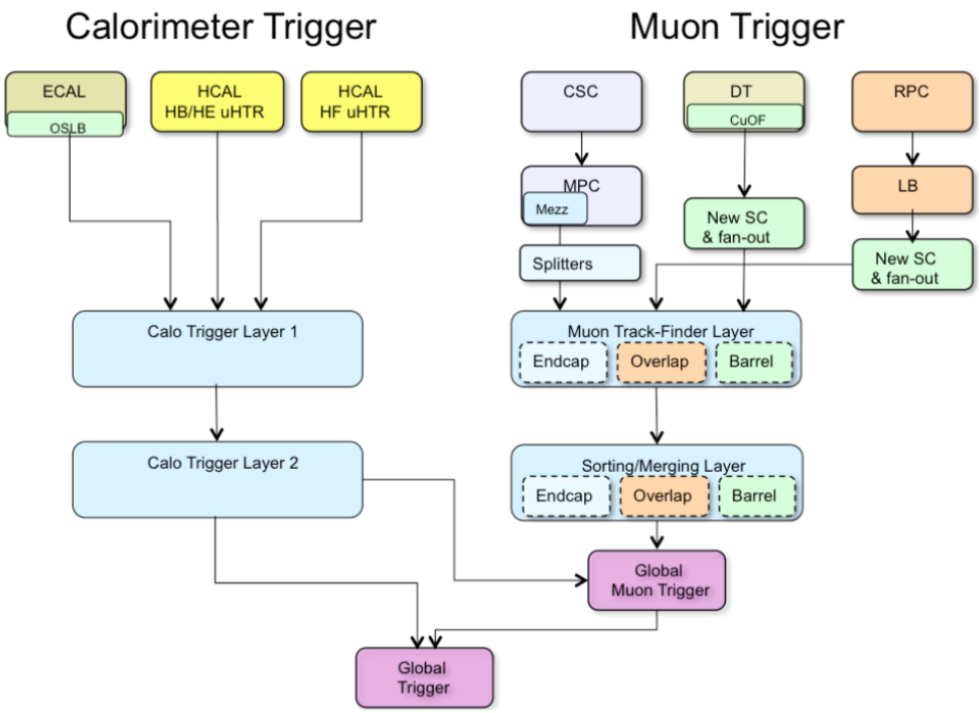


FIGURE 4.18: The configuration of L1 trigger.

4.2.1 The L1 Trigger of CMS

The L1 trigger uses CMS hardware and taking informations from the calorimeters and the CMS muon system to make decisions (in 4 s) in several steps to accept or reject an event. L1 trigger reduces event rate from 40 MHz to 100 kHz. A sketch of L1 trigger is shown in the Figure. 4.18.

L1 trigger is build with programmable electronics, that exploits coarse information of energy (ECAL and HCAL) and momentum (DT, RPC and CSC) of electrons, muons, jets, photon and MET. L1 trigger can be classified into calorimeter and muon triggers. Calorimeter has ECAL and HCAL as independent regional triggers that provide information to the global Calorimeter Trigger (GCT). Likewise, muon trigger has local trigger corresponding to CSC, DT and RPC each which then combine to make global muon trigger (GMT). GCT and GMT then feed to Global Trigger (GT) to make final decision about the event whether to accept or reject.

In order to cope with higher event rate for Run 2 operations, a new trigger architecture, the Time Multiplexed Trigger (TMT) was planted. TMT uses sophisticated algorithms (i.e. FPGAs) in μTCA to exploit full granularity of the calorimeters and is benefitted from technological improvements in hardware. rebuilding the CSC Trigger Track-Finder to accommodate the additional information from ME4/2 and ME1/1

High-Level Trigger System (HLT)

HLT performs event selections in a similar way that we perform for the offline processing of events. Primary goal of the HLT system is to take events (coming at 100kHz) from L1 trigger and reduce it to get an output rate of 1 kHz. This is done is done in several steps. For a process of interest, to reconstruct or identify an object (like electron, muon or so) a certain criteria is applied over each event. HLT is built with a farm of single processor computers which is known as event filter farm which assembles fragments of an individual event with the help of the builder units to construct the complete events. An assembled event is then fed to the filter unit which transforms the raw information into the data structure which is specific to the detector and does event reconstruction and filterig of trigger. These builder and filter units in a machine and linked through a shared memory. Filtering process exploits the precise data coming out of the trigger wherein event selection is based on the quality of the offline reconstruction algorithms. Available CPU power allows to take events at rate of 100 kHz which sustains for average processing time per event viz 90 ms which can be increased with an increased CPU power.
Performance of HLT is based on *HLT paths* which consist of a set of algorithms to perform in a predefined order to perform physics object selections followed by reconstruction step. Each HLT path applies in growing order of complexity as simpler algorithms run first to impose basic selections. Different HLT paths are based on diffrent algorithms which are designed quite similar to ones used for offline reconstruction in analysis. This helps to use CMS software which is used for offline analyses.
Events passing L1 trigger are sent to HLT menu having certain paths which are assigned to make decision. HLT operates at a specific instantaneous luminosity which is done by dividing the passed events with a paramer, n, called prescale factor. HLT path with pre-scale value n selects one event out of n passing events. Events organized in streams of HLT decisions are then passed to storage manager

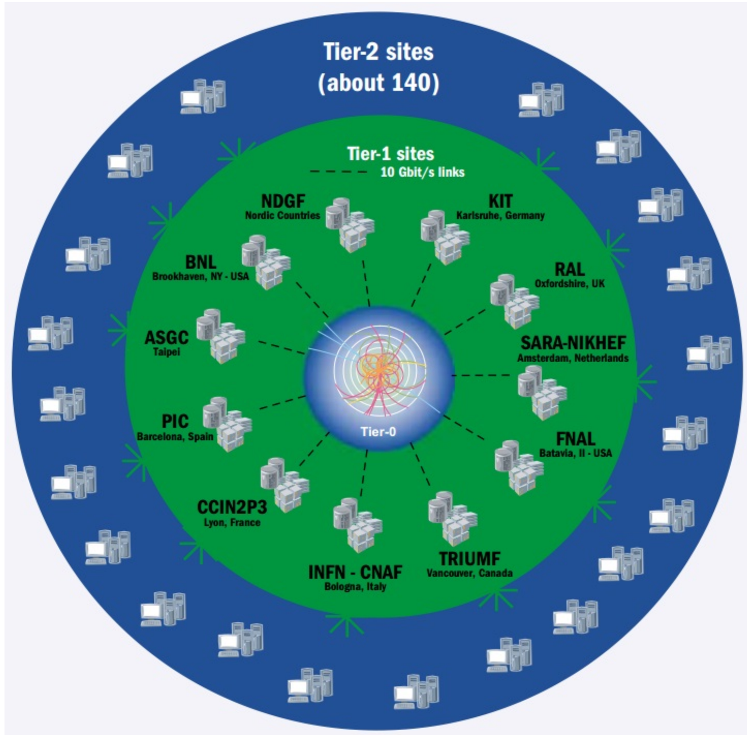


FIGURE 4.19: A sketch of WLCG Tier architecture.

softwares which store them on local disks and finally transfer to Tier-0 of CMS for its permanent storage.

4.3 Worldwide LHC Computing Grid

There is a dire demand of the four experiments of LHC (CMS, ATLAS, ALICE, LHCb) for a facility to store the prompt data obtained from LHC collisions. Further to process and analyse the huge amount of data to extract physics results is also challenging. LHC produces about 15 petabyte data each year duing its operation. To cope with this huge amount of data, setting a multi-tiered computing infrastructure was approved by CERN council in 2002. This is known as Worldwide LHC Computing Grid (WLCG) and which links around 170 computing centers in more than 40 countries. WLCG is mainly tasked for providing infrastructure of organized computing facilities to the four main experiments of CERN. To fulfill the goals, it is connected with around 150 thousand cores of processors with storage capicity of 50 petabytes on the disks **[Fisk:2010jd]**.

Schematic diagram of WLCG Tier system is shown in the Figure 4.19.

4.3.1 3-Tier Architecture

Computing models at WLCG are mainly based on 3 tier levels. Tier-0 facility is situated at CERN. It receives prompt data (RAW data) produced from LHC collisions as a primary backup and does the reconstruction of the events and store in the form of prompt-RECO dta. It then exports the RAW data to to large computing centers at Tier-1 where the events are set to be reconstructed again with refined calibrations and at each Tier-1 site, a detailed reconstruction of RAW data is performed to get AOD(Analysis Object Data) version of data which most compact form of data. It also serves to distribute the AOD data to Tier-2 sites where the data can be used for end-user analysis specific reprocessing and simulations for all experiments.

Chapter 5

Physics Objects

As described in section 4.2.1, that events organized in streams of HLT decisions are enventually transferred to the storage system (Tier-0) of CMS for its permanent storage. That data is in RAW format and needs to be fully reconstructed from this storage system. With potential reconstruction procedure, this RAW data can be associated to physical particles such as simple or general tracks, muons, photons, electrons and jets etc. These reconstructed particles then help to explore $e.g.$ Higgs decay to four lepton channel for its properties measurement like spin, parity, cross sections, mass and etc. The reconstruction of the SM Higgs boson in the decay chain $\mathrm{H} \rightarrow \mathrm{ZZ} \rightarrow 4\ell$ requires very efficient lepton reconstruction and identification in order to be sensitive to a low mass Higgs. In this kinematics region, there is need to attain an optimal reconstruction efficiency keeping misidentified lepton rate as low as possible. Further, the precise measurements on Higgs are lepton kinematics dependent and thus needs high precision momentum measurement.

5.1 Reconstruction of Track and Vertex

Reconstructing a track is procedure of exploiting hits in the detector parts used to estimate the momentum and/or position of a particle producing the hits. It is a challenging task and in the CMS experiment there a dedicated software known as Combinatorial Track Finder (CTF). It uses combinatorial Kalman filter approach for the recognition of patterns and fitting the track.
CTF algorithm repeats in multiple steps to search the track on the basis of its relatively larger p_T and closeness to the interaction point [**Chatrchyan:2014fea**].

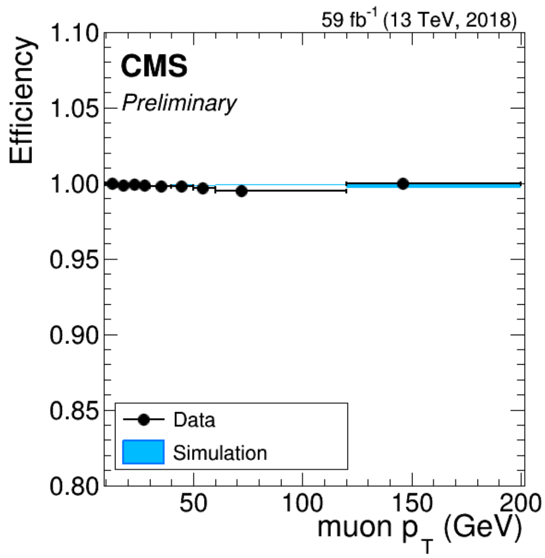
This step is called iterative tracking in which each iteration is done in following steps:

- **Seeding:** It is to provide with the help of 2 or 3 hits initial track candidates (seeds).
- **Tracking Finding:** It is to search for the additional hits based on the seed trajectories based on the Kalman filter approach.
- **Estimate of trajectory:** It is to use the Kalman-filter and smoother to estimate the best parameters of each trajectory.
- **Track Selection:** Based on the quality of the trajectories, those tracks which do not originate from some charged particle are discarded.

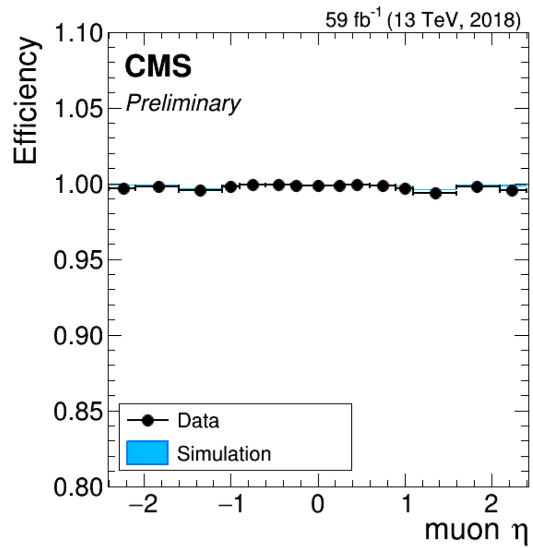
It is repeated for 6 times which are done to figure out the tracks originating from outside of the luminous region of the proton proton collisions.

The completely reconstructed tracks give the information of the proton-proton interaction vertices. Reconstruction of the primary-vertex is done in three steps:

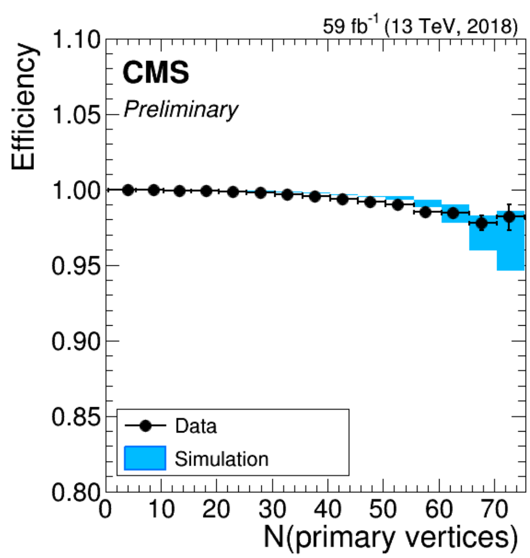
- **Selection of Tracks:** It is done by setting requirements on significance of impact parameter (SIP) from beam spot, number of hits in ITS and normalized χ^2.
- **Track Clustering:** It is done using information of transverse component of the impact parameter with respect to the beam spot. The clustering algorithm is capable to reconstruct any number of proton proton interaction in the bunch crossing. This performed using deterministic annealing algorithm (DA). This algorithm clusters the tracks into groups which can be associated to a single vertex.
- **Fitting of the track:** It is performed over the tracks in each cluster using "adaptive vertex fitter" to get the position of the primary vertex (the best vertex parameters). It needs a procedure to link a primary vertex with the hard interaction. Since in hard interaction particles are produced at high p_T, for example in $Z \to \mu^+\mu^-$ process, the vertices can be associated with sum of p_T^2 of the reconstructed tracks. The vertex with highest Σp_T^2 can be associated to the hard intraction from the multiplicity of primary vertices. Rest of the interactions, other than hard interaction, are called pileup.



(a)



(b)



(c)

FIGURE 5.1: The tracking efficiency as a function of p_T pseudo-rapidity and number of primary vertices for the standalone muon for the "all tracks" collection are shown in the sub-figure top left, top right and bottom respectively. Data (black dots) and Madgraph Drell Yan i.e. $Z \to \mu^+\mu^-$ (light blue rectangles). The uncertainties shown are statistical.

The efficiency to reconstruct a lepton track is called the tracking efficiency. For instance, to reconstruct a muon track in the inner detector is measured using as probes tracks reconstructed in the muon system alone. Measured muon tracking efficiency is shown in the Figure. 5.1. It is evident from Figures. 5.1(a) and 5.1(b) that tracking efficiency is nearly 100% in terms of p_T and η; however, with increase in number of primary vertices (pileup) it decreases a bit as shown in the Figure. 5.1(c).

These efficiencies are measured using "Tag and Probe " method for which details are described in next Section 5.1.1. This procedure is also applied when measuring the selection efficiencies of electrons and muons.

5.1.1 Tag and Probe Analysis

Simulation studies are important to validate the measurements on data. However, there are possibilities that the studies based on the simulations can be biased due to some insufficient event description and the lack of understanding of the detector. Further, for an analysis we always optimize the the event selection and in an ideal selection, we efficiently eliminate the background events with optimal signal events. Every real selection affects signal efficiencies which should be estimated. These efficiencies can be extracted directly from the simulation, but there is a danger such as bias which can appear as mentioned above. Such bias from MC, can be estimated by measuring the selection efficiencies directly from the experimental data. In high purity environment, one can measure the efficiencies by cut and cout method; however, if significant background is involved one needs to measure the efficiencies by "fit method" i.e. by using the Tag-and-Probe (TnP) method.

The TnP Tool is a generic tool developed to measure any user defined object efficiency (e.g. in this dissertation the reconstruction and/or identification, impact parameter cuts isolation and trigger) from data or MC by expoiting the di-object resonances like Z, J/ψ or Υ [**pog_tnp**]. Z boson decaying to a lepton pair is considered to be standard candle for the measurement of these efficiencies (except for reconstruction and identification of muons at low p_T (i.e. $p_T < 20$ GeV) where J/ψ resonances are used). Brief description of the procedure is given as under:

- Resonances are reconstructed as pairs with one leg passing a tight selection (called "tag") and one passing a loose selection (called "probe").
- According to whatever efficiency to be measured, passing probes (from all probes) are defined if they also pass the selection. Those which do not pass the selection are call "failing probes".

- Di-leptons mass distributions are built using "tag + passing probes" and "tag + failing probes", and are separately fit using "signal + background" model.
- The ratio of the signal yields in these two lineshapes give the subject "efficiency".
- The procedure is repeated in bins of probe variables (i.e. p_T, η and so on) to visualize the efficiencies in response of varying these variables.

An example of a TnP fit is shown in the Figure. 5.2. And schematic workflow of the TnP method while performing the procedure in measurements using the CMS central framework is shown in the Figure. 5.3.

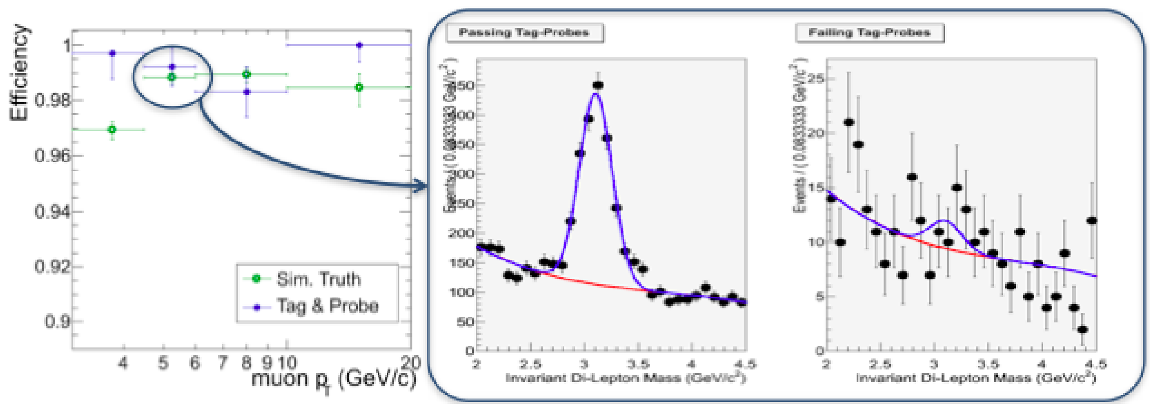


FIGURE 5.2: Efficiency plots in bins of p_T alongwith an example of TnP fit of di-lepton distributions. In the fit plot, black points are data points with statistical uncertainty bars, red lineshape is background fit and blue lineshape is "signal+background" fit in case of "tag + passing probes" (left) and "tag + failing probes" (right)

5.2 Electrons and Photons

5.2.1 Electron and Photon Reconstruction

Both of electron and photon are important ingredients of the analysis upon which the dissertation is based. Both interact electromagnetically and can be reconstructed and identified at CMS very efficiently which is also the requirement physics analyses. Photon being chargeless deposits its energy only in the electromagnetic calorimeter (ECAL), as described in Section. 4.1.4, leaving no track in the

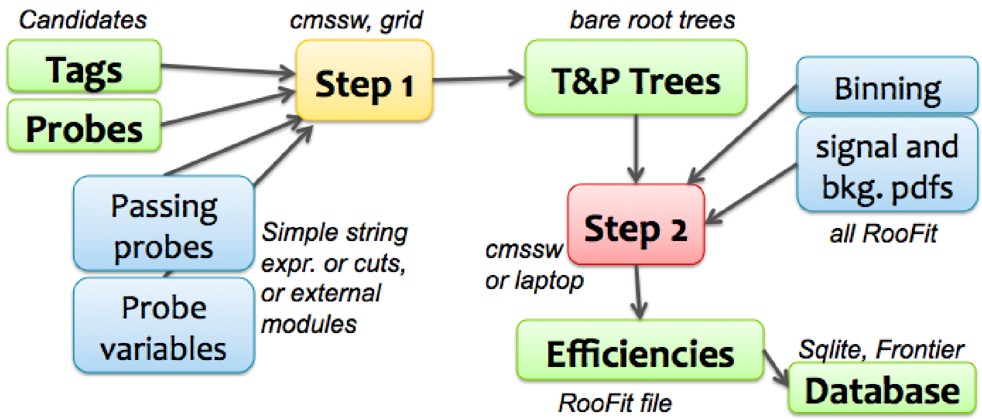


FIGURE 5.3: Tag and Probe workflow example using the central CMS framework.

1531 ITS of CMS. On the other hand, electrons leave tracking information in the ITS and
1532 well as in the ECAL. A schematic diagram of the behavior of photon and electron
1533 as they fly out of the interaction point from ITS to ECAL is shown in the Figure 5.4.

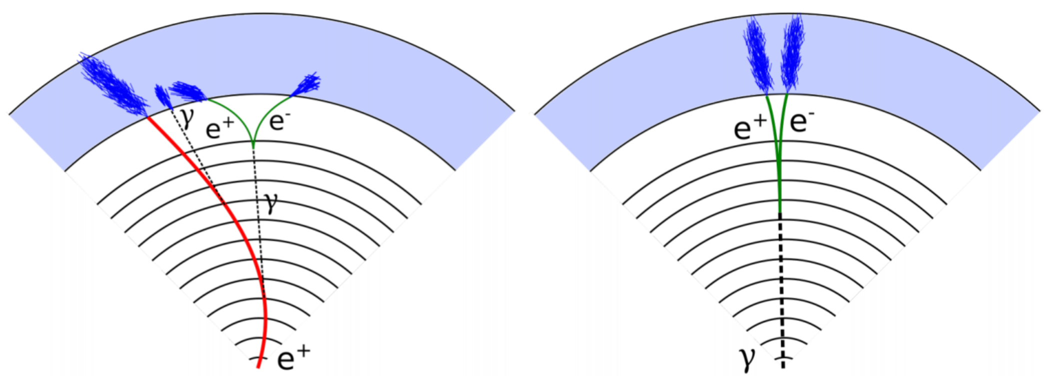


FIGURE 5.4: Pattern of positron (photon) to the detector shown left (right).

1534 There are many ways to identify the true or uncoverted photons e.g. by using:

1535 • Tracker, ECAL and HCAL isolation

1536 • Hadronic to electromagnetic ratio

1537 • R_9 ratio

Electrons can be reconstructed and identified when a supercluster is associated to a track in the silicon detector. For track reconstruction of electron, as described in Section 5.1.1, ECAL based seeding is efficient for $p_T > 10$; however, for lower p_T Tracker based seeding (that makes use of boosted decision tree (BDT) for preselection of tracker clusters) is more efficient. p_T of the electron is measured from these tracker tracks produced when it pass through it. Being charged particle, it has curved track under the influence of superconducting magnet due to which it radiates to photons. Therefore, Gaussian Sum Filter (GSF) algorithm, which takes into account the bremsstrahlung losses, is used to reconstruct the track.

Preselection of the electron candidates is made through loose cuts on track-cluster matching observables. This done to maintain highest possible selection efficiency of electrons and to reject QCD background significantly. For the analysis, electrons are subjected to have $p_T^e >$ 7 GeV within $|\eta^e| <$ 2.5, and satisfy a loose primary vertex constraints on longitudinal and transvers components of impact parameter $viz.$ $d_{xy} < 0.5$ cm and $d_z < 1$ cm respectively. Such electrons are called **loose electrons**. Details of electron reconstruction can be explored in Ref. [32].

The data-MC discrepancy is corrected using scale factors as is done for the electron selection with data efficiencies measured through technique outlined later (see Section 5.2.5). These studies for reconstructions are carried out by the EGM POG and the results are only summarised here.

The electron reconstruction scale factors are shown Fig. 5.5 and are applied as a function of electron p_T and super cluster η.

5.2.2 Electron Identification and Isolation

Electron identification and isolation is performed new Gradient Boosted Decision Tree (GBDT), a multivariate classifier algorithm. It exploits the informations from electromagnetic cluster, cluster-track matching, the variables which exclusively depend on tracking measurements and as well as particle flow isolation sums. The list of observables used can be seen in the Table 5.1.

For both signal and the background, the classifier is trained on 2017 MC samples of Drell-Yan(DY) plus jets. Fig. 5.6 shows the output of the multiclassifier discriminant for prompt electrons from DY events and fakes from jets in DY events.

The impact of the transition from the TMVA(v1) to the xgboost(v2) training library is shown in Fig. 5.7. The working points shown are adopted to get the

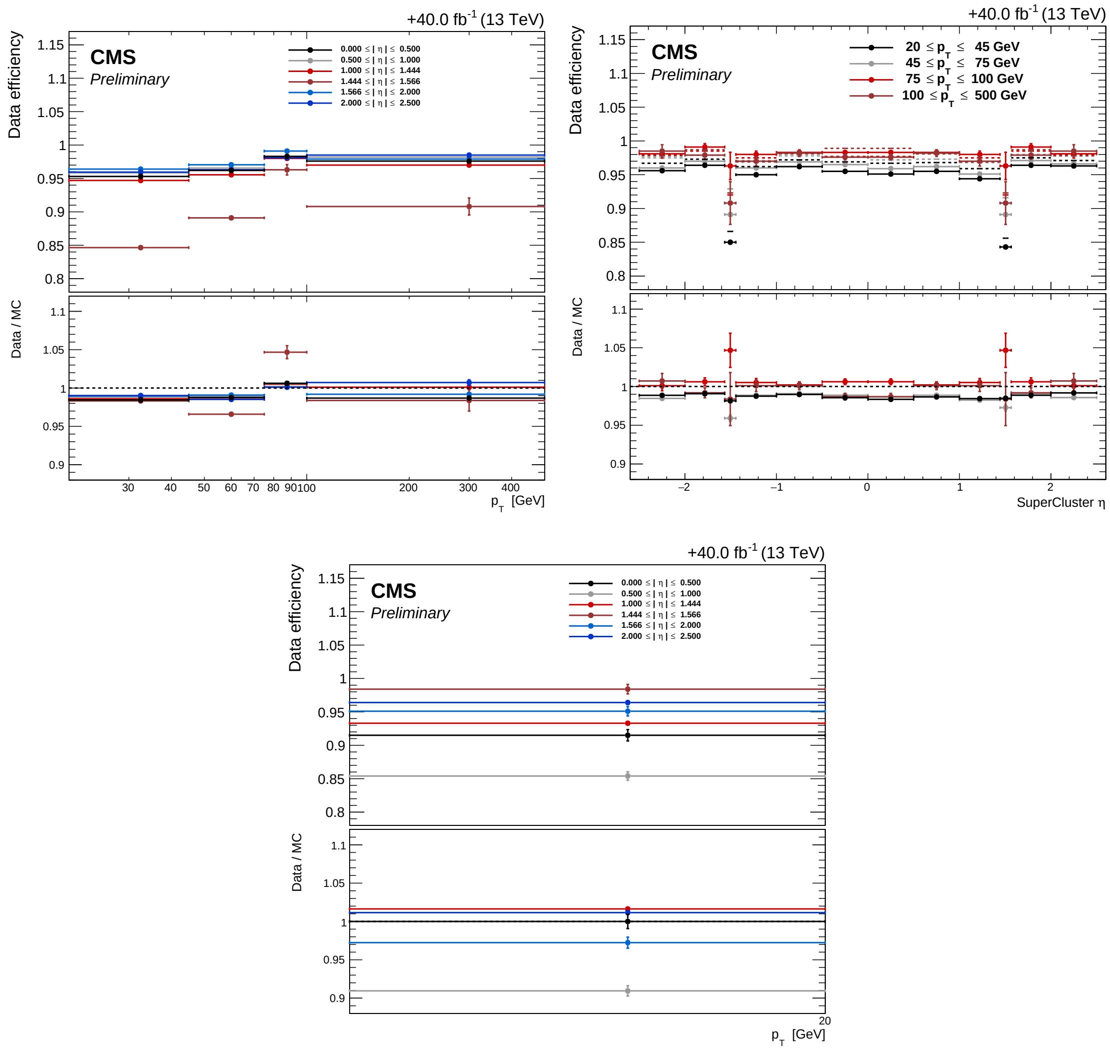


FIGURE 5.5: Reconstruction efficiencies of electron in data versus p_T (left), η (right) for electrons having $p_T > 20$ GeV (top) and $p_T < 20$ GeV (bottom) with corresponding data/MC scale factors as provided by the EGM POG.

same signal efficiency as the cut on MVA ID and a cut on the PF isolation, in each p_T and η bin.

Table 5.2 lists the cut values applied to the BDT score for the chosen working point. For the analysis, loose electrons have to pass this MVA identification and isolation working point.

observable type	observable name
6*cluster shape	RMS value of spectrum of energy-crystal number along η and φ; $\sigma_{i\eta i\eta}$, $\sigma_{i\varphi i\varphi}$
	width of super cluster along η and ϕ
	'ratio of hadronic energy behind the supercluster of electron to supercluster energy, H/E
	circularity $(E_{5\times5} - E_{5\times1})/E_{5\times5}$
	sum of the seed and adjacent crystal over the super cluster energy R_9
	for the endcap training bins: pre-shower energy fraction E_{PS}/E_{raw}
2*track-cluster matching	energy-momentum agreement E_{tot}/p_{in}, E_{ele}/p_{out}, $1/E_{tot} - 1/p_{in}$
	position matching $\Delta\eta_{in}$, $\Delta\varphi_{in}$, $\Delta\eta_{seed}$
5*tracking	fractional loss of momentum loss $f_{brem} = 1 - p_{out}/p_{in}$
	number of hits of KF and the GSF track N_{KF}, N_{GSF} $(\cdot)$
	reduced χ^2 of KF and the GSF track χ^2_{KF}, χ^2_{GSF}
	number of expected innre hits (missing) $(\cdot)$
	probability transformation in conversion vertex fit χ^2 $(\cdot)$
3*isolation	Particle Flow photon isolation sum $(\cdot)$
	Particle Flow charged hadrons isolation sum $(\cdot)$
	Particle Flow neutral hadrons isolation sum $(\cdot)$
1*For PU-resilience	Mean energy density in the event: ρ $(\cdot)$

TABLE 5.1: Overview of variables used at inputs to identification classifier. The Variables not used in the Run I MVA are specified with $(\cdot)$.

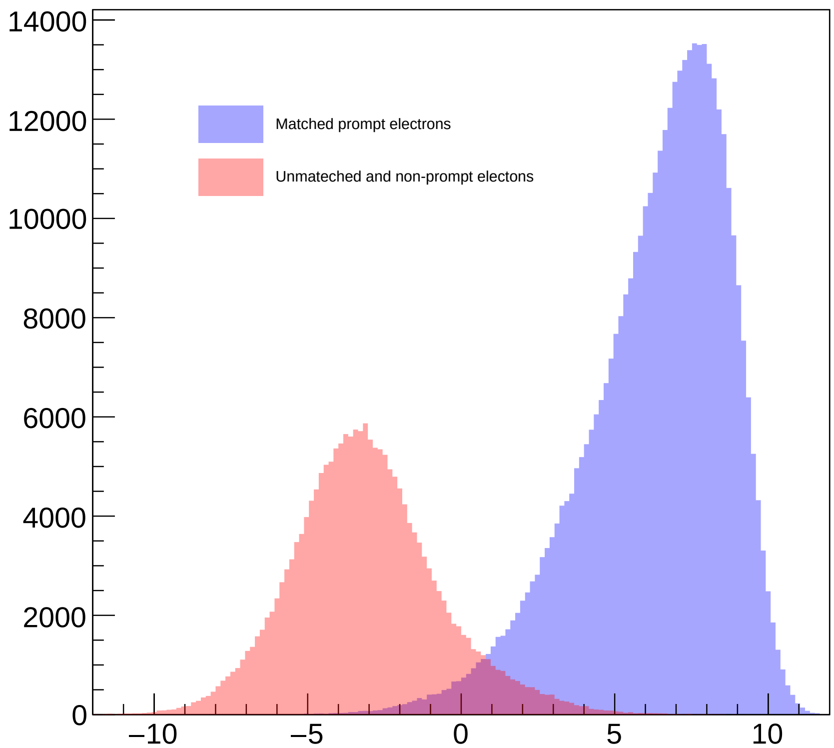


FIGURE 5.6: Output of the multiclassifier discriminant for prompt electrons matched to truth electrons from Z decay (blue) and for fakes (red). Events are all taken from DY events MC sample.

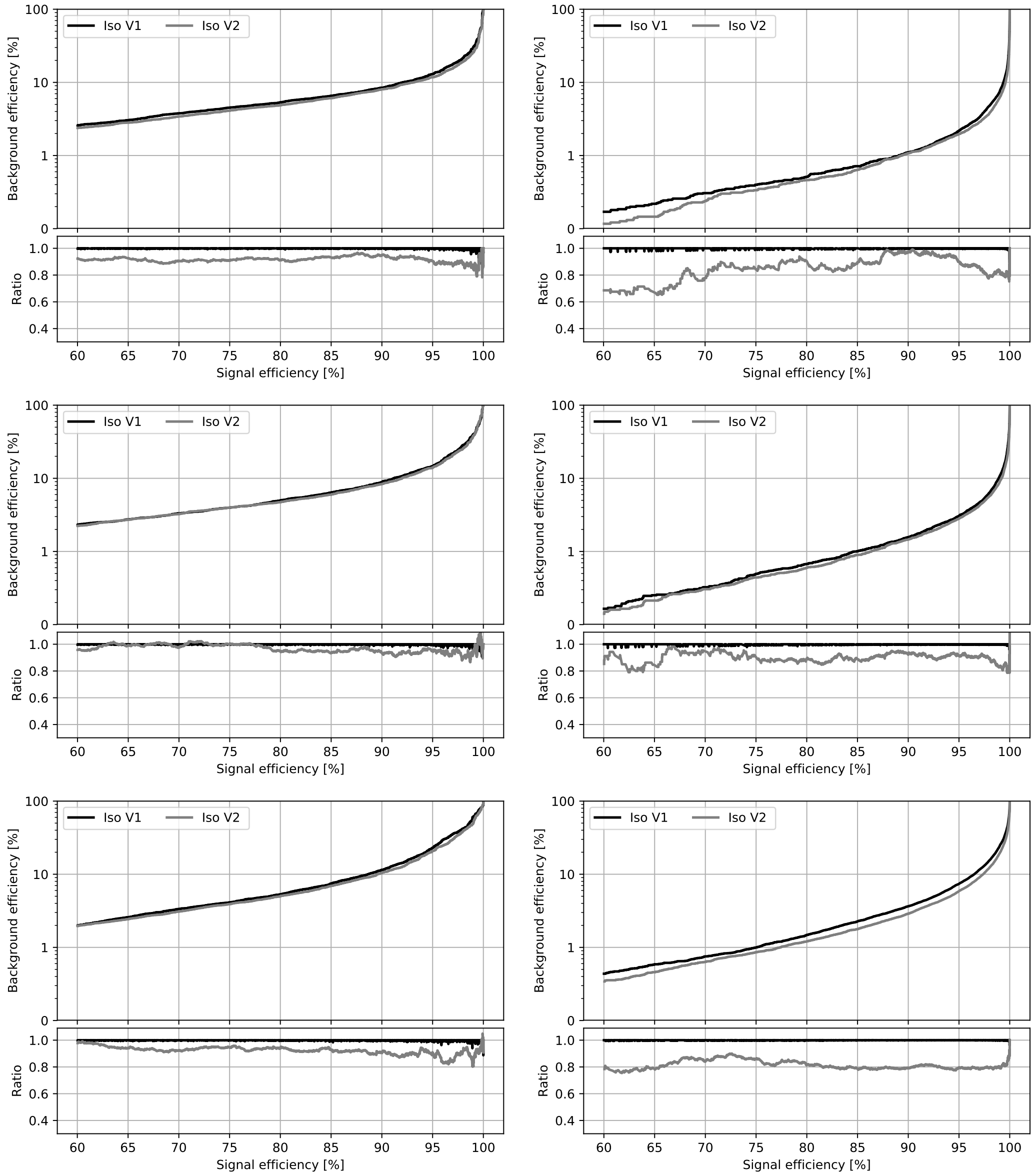


FIGURE 5.7: The receiver operating characteristic curves (background efficiency vs signal efficiency) of the MVA's trained on the 2017 simulation and applied on the 2018 simulation. The training combines identification and isolation variables. Performance are shown for electrons with $5 < p_T < 10$ GeV (left), $p_T > 10GeV$ (right) and $|\eta| < 0.8$ (top), $0.8 < |eta| < 1.479$ (middle) and $|eta| > 1.479$ (bottom). V1 and V2 versions of training are compared, exploiting TMVA and xgboost training libraries respectively.

	$\lvert\eta\rvert < 0.8$		
	Cut on BDT score	Signal eff.	Background eff.
$5 < p_T < 10$ GeV	1.264	81.04%	4.4%
$p_T > 10$ GeV	2.364	97.1%	2.9%
	$0.8 < \lvert\eta\rvert < 1.479$		
	Cut on BDT score	Signal eff.	Background eff.
$5 < p_T < 10$ GeV	1.178	79.3%	4.6%
$p_T > 10$ GeV	2.078	96.3%	3.6%
	$\lvert\eta\rvert > 1.479$		
	Cut on BDT score	Signal eff.	Background eff.
$5 < p_T < 10$ GeV	1.330	72.97%	3.6%
$p_T > 10$ GeV	1.080	95.7%	6.7%

TABLE 5.2: Minimum value of BDT score for electron to pass identification, together with the corresponding signal and the background efficiencies.

5.2.3 Electron Impact Parameter Selection

For lepton to be consistent with some common primary vertex, we require that they have an associated track that corresponds a small impact parameter from the event primary vertex. We use the significance of impact parameter to the vertex of an event, i.e. $|\mathrm{SIP}_{3\mathrm{D}} = \frac{\mathrm{IP}}{\sigma_{\mathrm{IP}}}|$, where IP being three dimensional lepton impact parameter w.r.t the primary interaction vertex, and the σ_{IP} being the uncertainty associated to this. Hence, a lepton satisfying $|\mathrm{SIP}_{3\mathrm{D}}| < 4$ is regarded as "primary lepton".

5.2.4 Electron Energy Calibrations

To measure the electron energy scale, a Crystal-ball function is used to fit di-electron mass distribution around the nominal Z boson peak.

It is measured in bins of p_T and $|\eta|$ for corrections to be applied on $Z \to \mathrm{ee}$ sample to align the dielectron mass spectrum in data to that in MC, and to minimize its width.

The resolution of $Z \to \mathrm{ee}$ mass in the Monte Carlo is made to match with data by applying a pseudorandom Gaussian smearing to electron energies, with varying parameters of Gaussian in p_T and $|\eta|$ bins. This has effect of convoluting the electron energy spectrum with a Gaussian.

The energy scale for the full 2018 dataset is shown in Fig. 5.8(a) and agrees decently with the MC with the corrections released by EGAMMA POG.

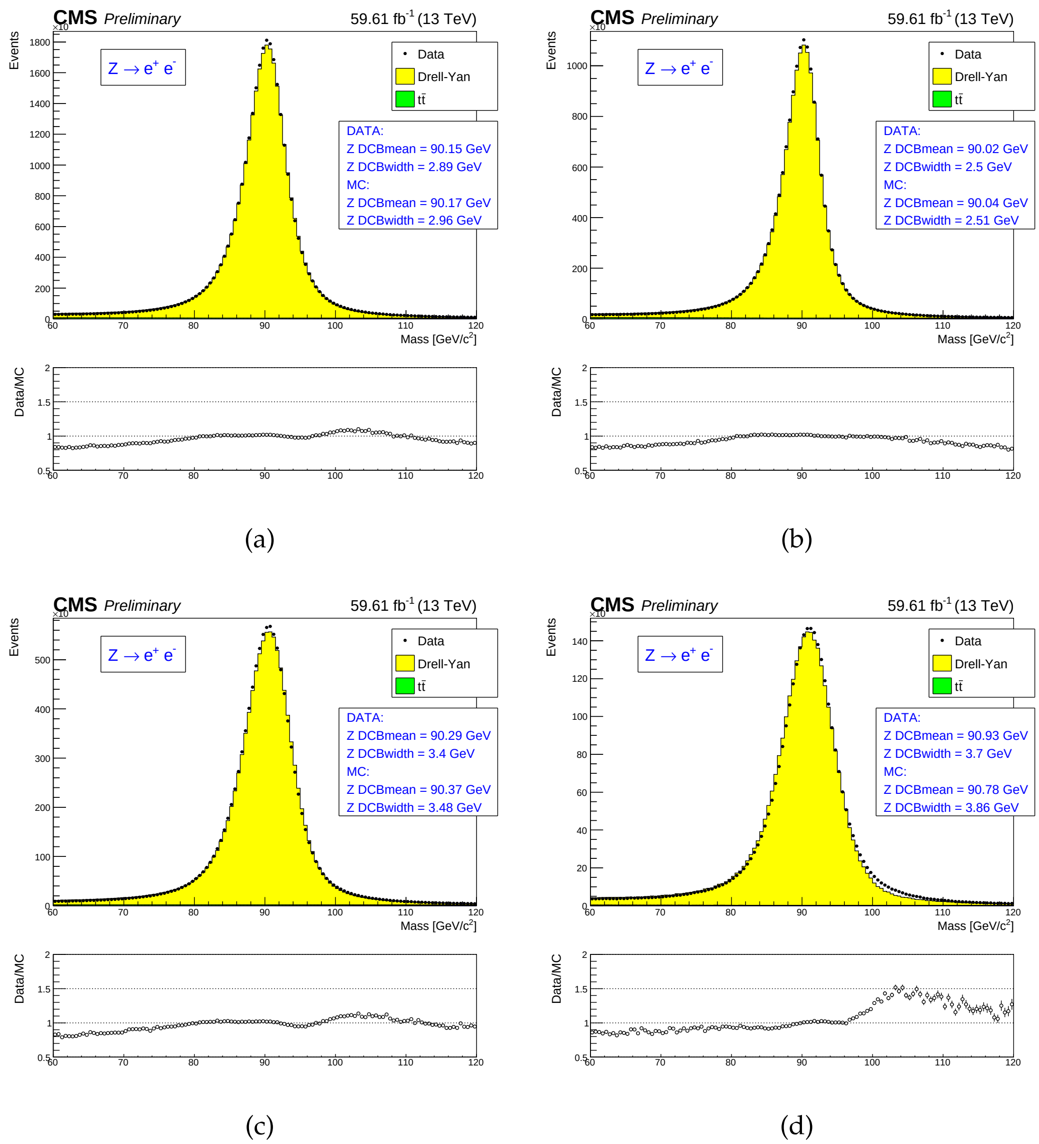


FIGURE 5.8: (a) electron energy scale measured in the $Z \to ee$ control region for all electrons,(b) for both electrons in barrel, (c) for one electron in barrel and other in endcaps, and for both of the electrons in endcaps (d). In these figures, the results based on fits using Crystall-ball are shown.

5.2.5 Electron Efficiency Measurements

The Tag-and-Probe study was performed on the EGM dataset using the golden JSON of 58.83, 41.4 and 35.9 fb^{-1} respectively for 2018, 2017 and 2016. More details on this technique can be found in Ref. [33].

Tag electrons need to satisfy the following quality requirements:

- trigger matched to HLT_Ele32_WPTight_Gsf_L1DoubleEG_v*
- $p_T > 30$ GeV, super cluster (SC) $\eta < 2.17$

For the bin between 7 and 20 GeV, additional criteria are required: the tag has to pass a cut on the MVA ID> 0.92, $\sqrt(2 * PFMET * p_T(tag) * (1 - cos(\phi_{MET} - \phi_{tag}))) < 45GeV$.

Probe electrons only need to be reconstructed as GsfElectron while the FSR recovery algorithm is not applied in efficiency measurement.

The nominal MC efficiencies are evaluated from the LO MadGraph Drell-Yan, while the NLO systematics use the MadGraph_AMCatNLO sample.

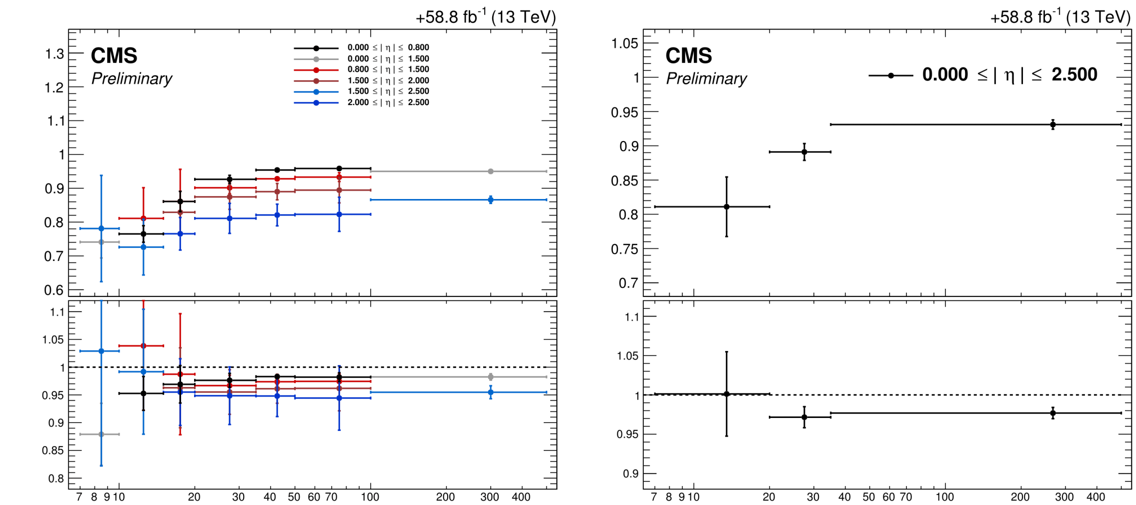
For the efficiency measurements a template fit is used. The m_{ee} signal shape of the passing and failing probes is taken from MC and convoluted with a Gaussian. The data is then fitted with the convoluted MC template and a CMSShape (an Error-function with a one-sided exponential tail). This change follows from the usage of the T&P tool developed by the EGM POG.

The measurement of electron selection efficiency is made as a function of the probe electron p_T and its SC η. For electrons falling in the ECAL gaps, it is measured separately. Figure 5.9 and 5.10 shows the p_T and η turn-on curves measured in data.

The EGM recommendations about evaluation of Tag-and-Probe uncertainties on efficiency measurements are followed. Specifically, we consider

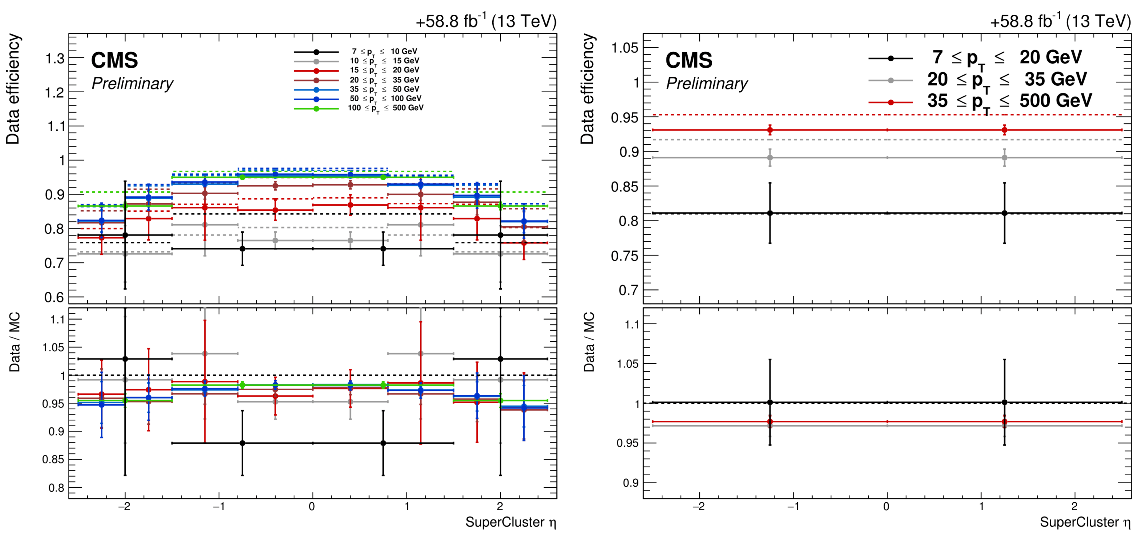
- Variation of the signal shape from a MC shape to an analytic shape (Crystal Ball) fitted to the MC
- Use of analytic function (Crystal Ball) to fit the signal as alternative of MC template
- Using an NLO MC sample for signal templates

Assessment of total uncertainty for measurement of the scale factors is taken as quadratic sum of statistical uncertainties (returned from the fit) and the aforementioned systematic uncertainties. Overall electron SFs and the associated total uncertainties can be seen in the Figure. 5.11.



(a) (b)

FIGURE 5.9: Electron selection efficiencies vs p_T measured using the Tag-and-Probe technique, non-gap electrons (left) and gap electrons (right), together with the corresponding data/MC ratio bottom part of each plot.



(a) (b)

FIGURE 5.10: Electron selection efficiencies vs η, non-gap electrons (left) and gap electrons (right), together wit the corresponding data/MC ratio at the bottom part of each plot.

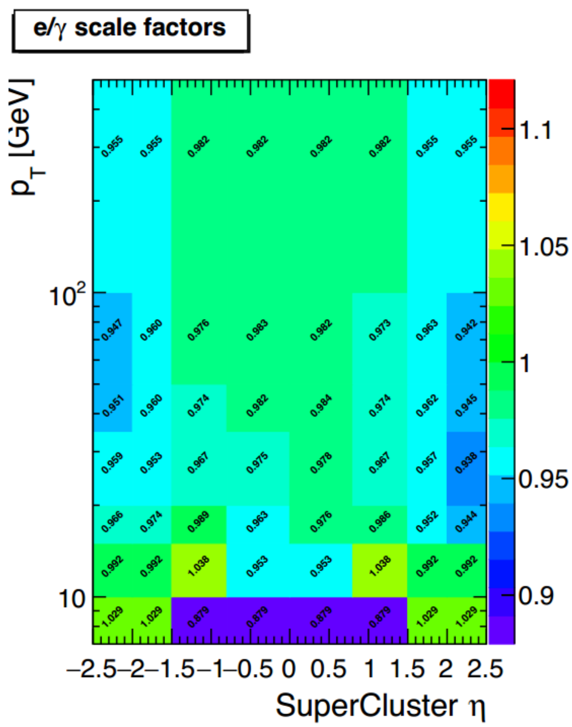


FIGURE 5.11: Overall electron SFs (left) and associated uncertainties (right) in two dimensional display as function of p_T and supercluster η.

5.3 Muons

5.3.1 Muon Reconstruction and Identification

Muons can be reconstructed as global muon which exploits the information from muon track in muon system and a tracker track. It can also be reconstructed by matching the tracker track with its track stubs (i.e. with at least one the muon track segment created by CSC or DT hits).

As per requirement of analysis, we define **loose muons** as the muons that satisfy $p_T > 5$, $|\eta| < 2.4$, $dxy < 0.5$ cm, $dz < 1$ cm, where dxy and dz are defined w.r.t. the PV and using the 'muonBestTrack'. Muons have to be reconstructed by either the Global Muon or Tracker Muon algorithm. Standalone Tracks of muon reconstructed only in the muon system are rejected. Muons with `muonBestTrackType==2` (standalone) are discarded even if they are marked as global or tracker muons. More details on the reconstruction of muons can be seen in Ref. [33].

Loose muons with p_T below 200 GeVare considered identified muons if they also pass the PF muon ID (note that the naming convention used for these IDs differs from the muon POG naming scheme, in which the "tight ID" used here is called the "loose ID"). Loose muons with p_T above 200 GeVare considered identified muons if they pass the PF ID or the Tracker High-p_T ID, the definition of which is shown in Table 5.3. This relaxed definition is used to increase signal efficiency for the high-mass search. When a very heavy resonance decays to two bosons, both bosons will be very boosted. In the lab frame, the leptons originating from decay of a highly boosted Z will be nearly collinear, and the PF ID loses efficiency for muons separated by approximately $\Delta R < 0.4$, which roughly corresponds to muons originating from bosons with $p_T > 500$ GeV.

It may happen that a single muon is reconstruted incorrectly as two muons or even more. To cope with this situation, additional "ghost-cleaning" is performed. Those tracker Muons which are not Global Muons, are needed to be arbitrated. If two muons are sharing their segments upto 50% or more, lower quality muon is removed.

5.3.2 Muon Isolation Requirement

Muon isolation used is based on Particle-Flow. So-called $\Delta\beta$ correction is used in order to subtract the pileup contribution for muons, whereby $\Delta\beta = \frac{1}{2}\sum_{\text{PU}}^{\text{charged had.}} p_T$

TABLE 5.3: The requirements for a muon to pass the Tracker High-p_T ID. Note that these are equivalent to the Muon POG High-p_T ID with the global track requirements removed.

Plain-text description	Technical description
Muon station matching	Matching of muon to the segments in two muon stations at least **NB: It means that muon is an arbitrated tracker muon.**
Good p_T measurement	$\frac{p_T}{\sigma_{p_T}} < 0.3$
Vertex compatibility $(x-y)$	$d_{xy} < 2$ mm
Vertex compatibility (z)	$d_z < 5$ mm
Pixel hits	One pixel hit at least
Tracker hits	Hits in at least six tracker layers

gives an estimate of the energy deposit of neutral particles (i.e. photons and hadrons) from the pile-up vertices. Relative isolation for muons is then defined as:

$$\text{RelPFiso} = \frac{\sum^{\text{charged had.}} p_T + \max(\sum^{\text{neutral had.}} + \sum^{\text{photon}} - \Delta\beta, 0)}{p_T^{\text{lepton}}} \tag{5.1}$$

The isolation working point for muons was optimized in Ref. [33] and the working point was chosen to be the same as electrons, namely RelPFiso$(\Delta R = 0.3) < 0.35$.

5.3.3 Muon Impact Parameter Selection

Selection on muon significance of impact parameter(SIP) is used same as for the electrons which is described in Section 5.2.3:

- $|\text{SIP}_{\text{3D}} = \frac{\text{IP}}{\sigma_{\text{IP}}}| < 4$

5.3.4 Muon Energy Calibrations

Similar to electrons the momentum scale of muon is measured in data by fitting a Crystall-ball function to the di-muon invariant mass spectrum around the Z peak in the $Z \to \mu\mu$ control region. From Fig. 5.12 shows a very good comparison of data with simulation.

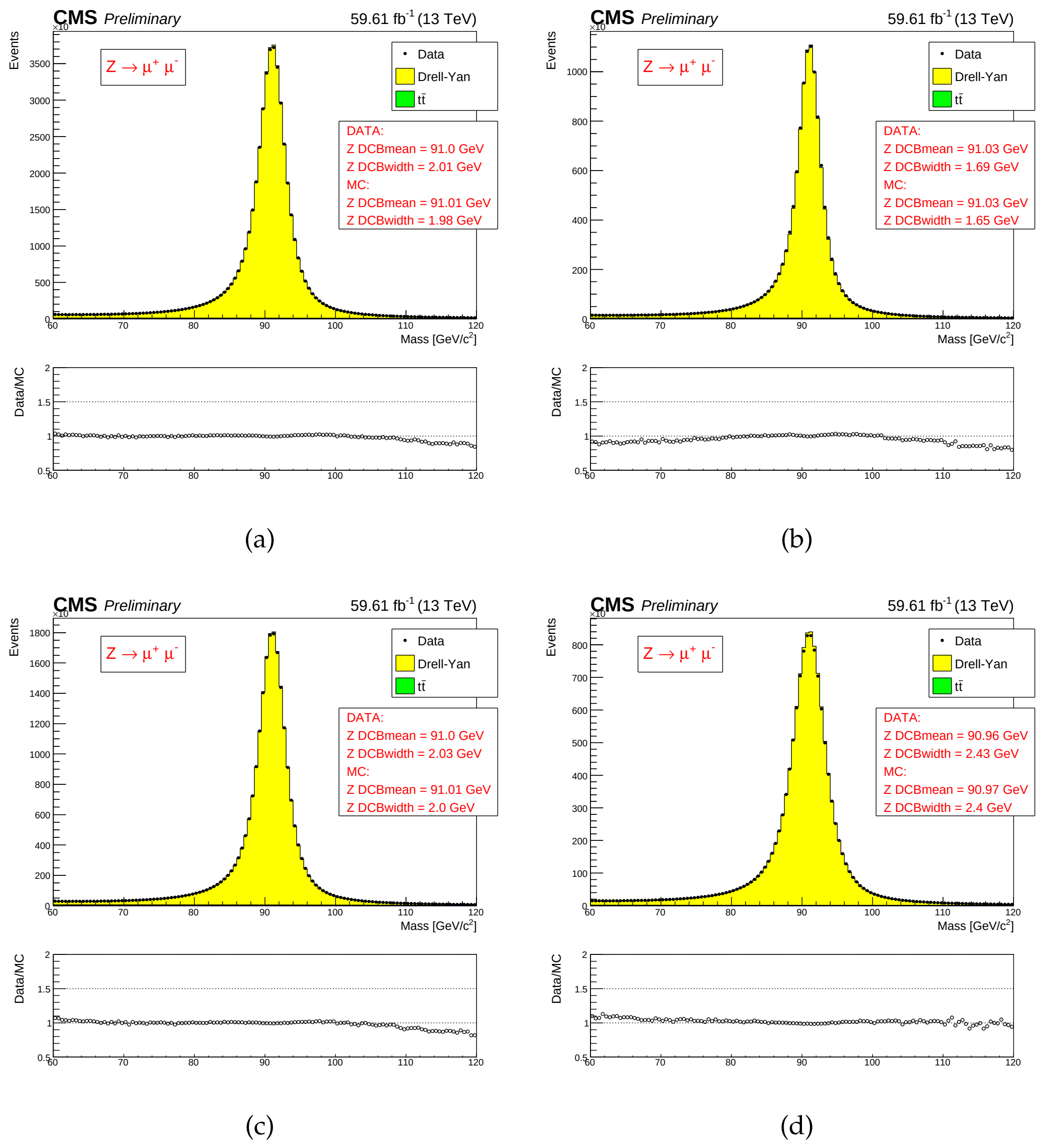


FIGURE 5.12: (a): muon energy scale measured in the $Z \to \mu\mu$ control region for all muons, for both of muons frome barrel region (b), for one muon from barrel, one from endcaps (c) and for both muons in the endcaps (d). The results in the figures are extrated from fits using the Crystall-ball function.

5.3.5 Efficiency Measurements of muons

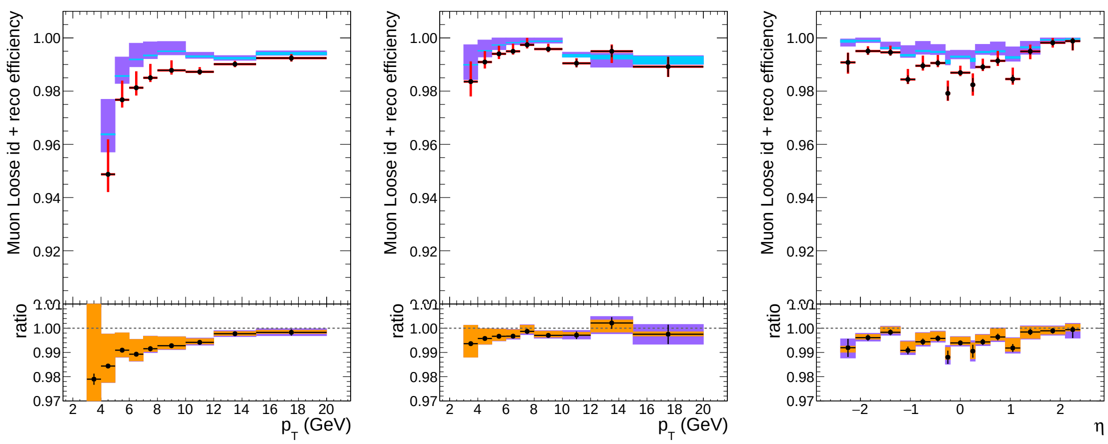
Efficiencies of muons are measured using same method used for electron measurements i.e. Tag-and-Probe (T&P) method performed on $Z \to \mu\mu$ and $J/\psi \to \mu\mu$ events in bins of p_T and η. More details on the methodology are described in Section. 5.1.1 and also in Ref. [34]. Measurements are extracted using 2018

RunA,B,C,D data while the measurements corresponding to 2016 and 2017 datasets have already been reported in Ref. [35] and Ref. [36] respectively. For the measurement of muon reconstruction and identification efficiency at high p_T (p_T>20GeV), and impact parameter requirements and isolation efficiencies for all p_T, Z sample is used. to measure the muon reconstruction and identification efficiency at high p_T, and the efficiency of the isolation and impact parameter requirements at all p_T. To measure the reconstruction and identification efficiency at low p_T, J/ψ sample is used as it benefits from a better purity in that kinematic regime. In this case, events are collected using `HLT_Mu7p5_Track2_Jpsi_v*` when probing the reconstruction and identification efficiency in the muon system, and using the `HLT_Mu7p5_L2Mu2_Jpsi_v*` when probing the tracking efficiency.

Muon reconstruction and identification efficiency measurement Results for muon reconstruction and identification efficiencies for high p_T muons ($p_T > 20$ GeV) have been derived by the Muon POG. The probe in this measurement are tracks of inner tracker, and the passing probes are those which are also regarded as a global or tracker muon and passing the Muon POG Loose muon identification. Results for low $p_T$muons were derived using events, with the same definitions of probe and passing probes. The systematic uncertainties on the measurements are estimated by including several variations e.g. varying the analytical signal and background shape models used to fit distribution of dimuon invariant mass. Furthermore, number of bins in dimuon distribution and mass window are also increased and decreased to include systematic effects. Details on the procedure can be found in Ref. [33]. The efficiency and scale factors used for low p_T muons are the ones derived using single muon prompt-reco dataset.

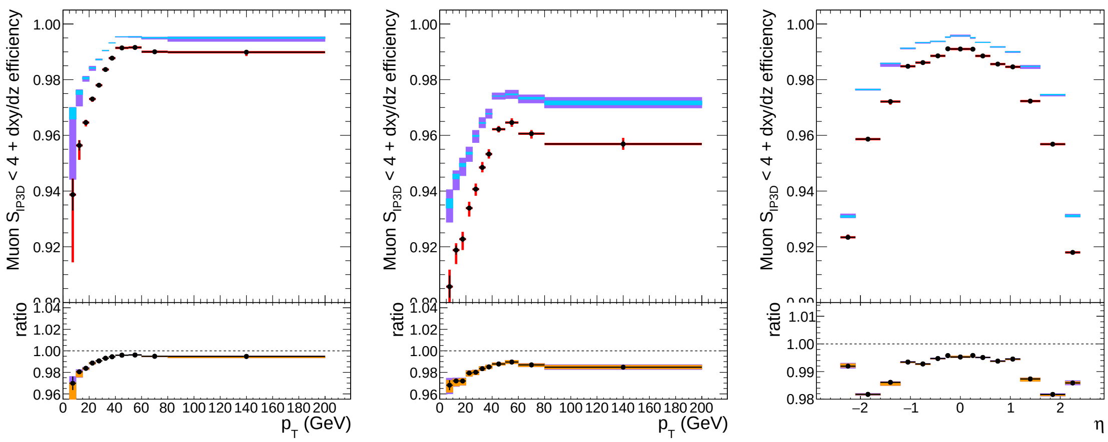
The data and simulation efficiencies and their ratios in bins are shown in Fig. 5.13.

Efficiency measurement of muon impact parameter requirement The measurement is performed using Z events. Events are selected with `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers. For this measurement, the probe is a muon passing the POG Loose identification criteria, and it is considered a passing probe if satisfies the SIP3D, dxy, dz cuts of this analysis. The results are shown in the Fig. 5.14.



(a) (b) (c)

FIGURE 5.13: Reconstruction and identification efficiencies of muons at low p_T, measured with the $T\&P$ method on events, in bins of $p_T$in barrel (left) and endcaps (center), and in terms of η for $p_T > 7$ GeV (right). In upper panel of each diagram, the larger uncertainty bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of two efficiencies, black uncertainty bars are for statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.



(a) (b) (c)

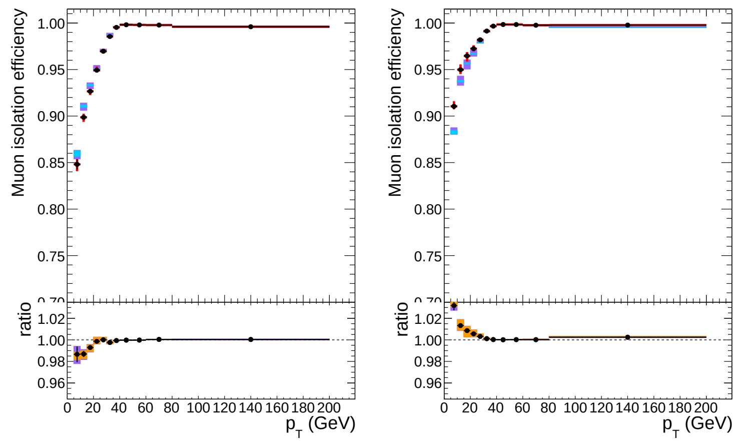
FIGURE 5.14: Efficiency of the muon impact parameter requirements, measured with the $T\&P$ method on Z events, in bins of $p_T$in barrel (left) and endcaps (center), and in terms of η for all p_T (right). In upper panel, the larger uncertainty bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of two efficiencies, black error bars are for statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

Efficiency measurement of muon isolation requirements Isolation efficiency is measured using events from the Z decay for any p_T. The events are selected with either of `HLT_IsoMu27_v*` or `HLT_Mu50_v*` triggers (for 2017 `HLT_IsoMu27_v*` is used and for 2016 measurements, OR condition is required among `HLT_IsoMu20_v*`, `HLT_IsoTkMu20_v*`, `HLT_IsoMu22_v*`, `HLT_IsoTkMu22_v*`, `HLT_IsoMu24_v*`, `HLT_IsoTkMu24_v*`). The isolation of the muons are calculated after recovery of the FSR photons and subtracting their contribution to the isolation cone of the muons. There are details described of this method which can be found in Ref. [37].

The results are shown in Fig. 5.15.

Tracking Tracking efficiency (i.e. efficiency to associate a track to some muon in inner detector) is almost 100% as shown in the Figure. 5.1 and hence muon Physics Object Group no more recommends tracking scale factors in the analyses.

Overall results of muon Scale Factors The product of all scale factors for reconstruction, identification, impact parameter and isolation requirements is shown in



(a) (b)

FIGURE 5.15: Efficiency of muon isolation requirement, measured with the tag&probe method on Zevents, as function of $p_T$in the barrel (left) and endcaps (right). In the upper panel, the larger error bars include also the systematical uncertainties, while the smaller ones are purely statistical. In the lower panel showing the ratio of the two efficiencies, the black error bars are for the statistical uncertainty, the orange rectangles for the systematical uncertainty and the violet rectangles include both uncertainties.

two dimensional display in Fig. 5.16. The figure also shows overall uncertainty on scale factors.

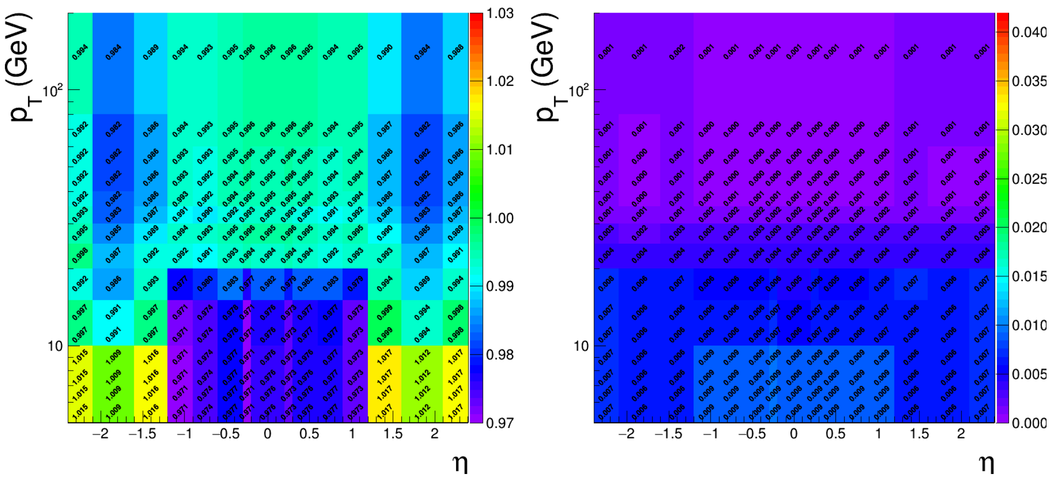


FIGURE 5.16: Left: Overall scale factors for muons, as function of p_T and η. Right: Uncertainties on overall scale factors muons shown in two dimensions of p_T and η.

5.4 Photons for FSR recovery

The procedure of FSR recovery is described as under:

1. Photons are initially collected using PF alogorithm which are called PF photons.
2. These photons are preselected by requiring $p_{T,\gamma} > 2$ GeV, $|\eta^{\gamma}| < 2.4$, and a relative Particle-flow isolation smaller than 1.8 (computed within a cone of radius $R = 0.3$).
3. Supercluster (SC) veto: All photons matching (direct association with PF candidates) with some electron that passes ID and SIP requirements are removed.
4. Association of photons is made with lepton which is closest and passes ID and SIP requirements.
5. Photons should fulfill the requirements of $\Delta R(\gamma, l)/E_{T,\gamma}^2 < 0.012$, and $\Delta R(\gamma, l) < 0.5$.
6. For a lepton, a photon with least $\Delta R(\gamma, l)/E_{T,\gamma}^2$ is considered.
7. FSR photons are excluded from sum of isolation of all leptons in an event which pass both loose ID and SIP requirements.

More details on the optimization of the FSR photon selection can be found in Ref. [33, 37].

5.5 Jets

In a high energy experiment e.g. proton-proton collision at LHC, jets are produced by quarks and/or gluons when they go through the process of hadronization and making a cone like structure in the detector like CMS. Hence, jets are physics objects which provide the experimental evidence of quarks and the gluons. As these jets of particles propagate through the CMS detector, they leave signals in components such as the tracker and the electromagnetic and hadronic calorimeters. A typical diagram of the a jet can be seen in the Figure. 5.17 and in

Ref. [**jets_at_cms**] Information from these signatures is exploited through some jet algorithms to fully reconstruct a "jet".

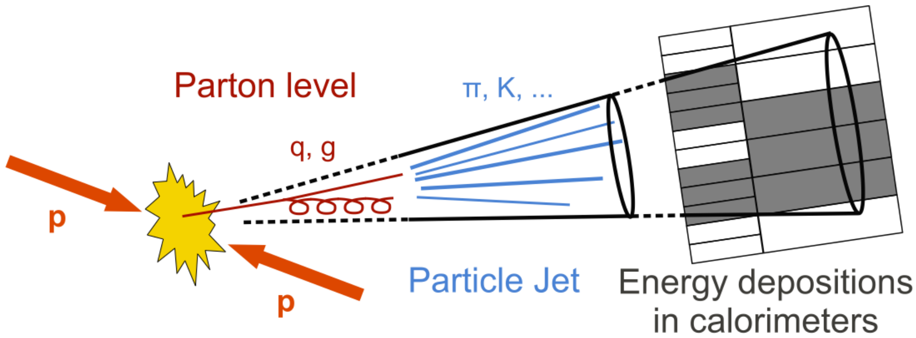


FIGURE 5.17: A schematic diagram of proton-proton collison and signature for a jet cone.

The processes of Higgs production through fusion of vector bosons (VBF) and through other mechanisms normally differ as regards the jet kinematics.

5.5.1 Jet Identification

For the analysis upon which the dissertation is built, jets are reconstructed through anti-k_T clustering algorithm out of particle-flow candidates, with distance parameter $R = 0.4$, after rejecting the charged hadrons that are associated with a pileup primary vertex.

To reduce instrumental background, the tight working point jet ID suggested by the JetMET Physics Object Group is applied [38]. In addition, jets from Pile-Up are rejected using the PileUp jet ID criteria suggested by the JetMET POG [39] [1]. In this analysis, the jets from within $|\eta| < 4.7$ area are required to have a transverse momentum greater that 30 GeV. In addition, jets are cleaned from any of the tight leptons (passing the SIP and isolation cut computed after FSR correction) and FSR photons by a separation criterion: $\Delta R(\text{jet,lepton/photon}) > 0.4$.

[1] In 2016, jet pileup ID is not applied (to be included in upcoming manuscript on the same subject).

5.5.2 Jet Energy Corrections

Energy of the jets as measured from calorimeters is not exactly what is true energy at hadron level. There are many experimental reasons, e.g non-linear response of calorimeters, addition of energies from the pileup events, possible mixing from non-instrumented regions or noise in the detector, due to which the true energy may not be associated to the jets and hence, one needs to estimate the corrections to energy of a jet. These corrections are to be applied both in data and the simulations to match with the true hadron level energy.

In the analysis, standard jet energy corrections are implemented to the reconstructed jets, which consist of L1 Pileup, L2 Relative Jet Correction, L3 Absolute Jet Correction for both Monte Carlo samples and data, and also residual calibration for data [40]. Jet Energy resolution corrections are also applied to MC.

5.5.3 B-tagging

B-hadrons or b-jets have long life time $viz.$ about $1.5ps$ (those of c-hadrons have $1ps$ or even less), and travel longer in the detector before they decay and subsequently leave the tracks of large impact parameter depending upon the momentum. These tracks are called displaced tracks and can help to reconstruct the secondary vertex (SV). A typical diagram of a heavy-flavour jet with a SV from the decay of b or c hadron resulting in charged-particle tracks is shown in the Figure. 5.18 and in Ref. [**Sirunyan:2017ezt**].

In order to distinguish whether a jet is b-jet or not *DeepCSV* algorithm is used as our b-tagging algorithm. It combines the same information as the previous tagger *CSVv2* i.e. SIP, jet kinematics and secondary vertex but uses information of more tracks. Also, the b-tag output discriminator is computed with a Deep Neural Network. In the analysis, a jet is considered as a b-tagged if it meets requirement of *medium* working point, i.e. if its `|pfDeepCSVJetTags:probb+` pfDeepCSV-JetTags:probbb`|` discriminator is greater than 0.4184 [41].

Scale factors of data to simulation efficiencies for b-tagging are provided for this working point for the full dataset varied in jet p_T, η and flavour. They are applied to simulated jets by downgrading (upgrading) the b-tagging status of a fraction of the b-tagged (untagged) jets that have a scale factor smaller (larger) than one.

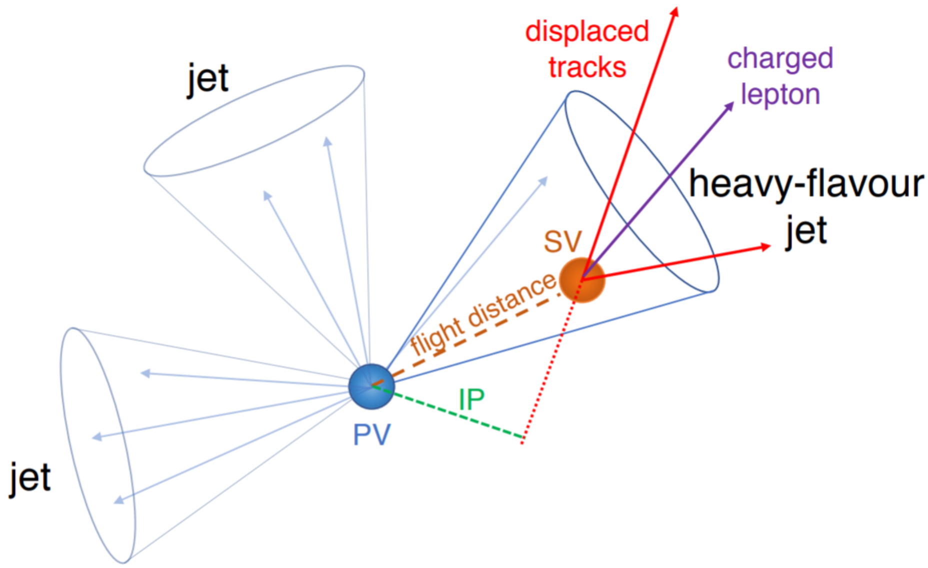


FIGURE 5.18: A schematic diagram of a heavy flavor jet with a secondary vertex.

5.5.4 Validation on data

Some validation studies are performed to probe agreement of data with simulated samples for jets. It is similar to what was done with electrons and muons. Events containing or $Z \to \mu^+\mu^-$ were selecting, following the same trigger and lepton selection criteria of the main analysis, but increasing the p_T threshold on leptons to 30 GeV and asking the dilepton invariant mass window of 70-110 GeV. Basic properties of additional jets in an event, requiring $p_T > 30$ GeV and $\eta < 4.7$, were then compared. Figure 5.19 shows comparison of data with the MC in bins of pseudo-rapidity of the leading jet. Simulated samples of Drell-Yan and $^-$are used. Uncertainties from the jet energy corrections are also displayed on ratio plots of data to MC.

The disagreement of data with MC for the MET distribution prevents us at this stage to use this variable in the analysis, in particular to better target VH events where, for instance, a Z would decay into neutrinos.

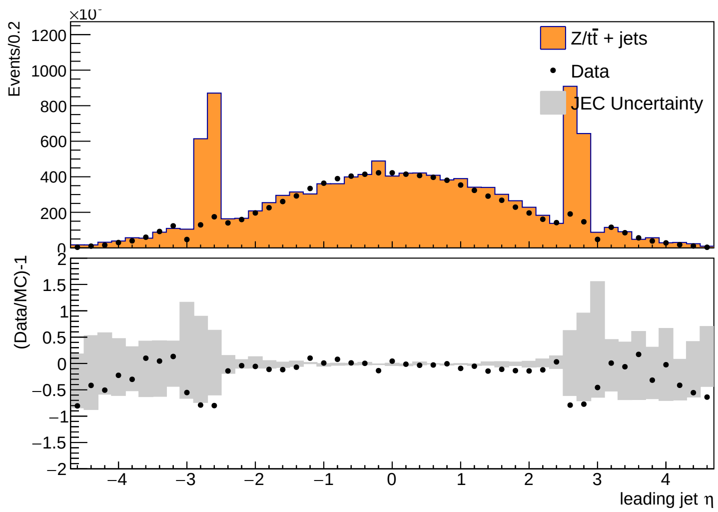


FIGURE 5.19: Comparison of data with MC for leading jet η in Z$\rightarrow \ell\ell$ +jets. MC is normalized to data. Jet ID and Jet PUID are applied. MC samples include DY and ttbar. Data/MC ratio plots are shown in the bottom of each plots, together with the uncertainties (shaded histograms) from Jet Energy Corrections.

5.6 Object Summary

The requirements on all objects used in the analysis (with 2018 data) are summed up in the Table 5.4. In addition, a "ghost-cleaning" procedure is applied to the muons, as described in Sec. 5.3.1.

A lepton is declared **loose** if it passes the reconstruction, kinematics and dxy/dz cuts and declared **tight** if it passes in addition the identification, isolation and SIP3D cut.

For 2016 and 2017 overall selections are same except a few as given below:

- For 2018 and also for 2017, BDT includes ID and as well as Isolation requirements; however, for 2016 BDT includes only ID and cut based Isolation is explicitly applied on the top of BDT.
- In 2017 and 2018, *DeepCSV* is used as B-tagging algorithm. For 2016, *CSVv2* was used as B-tagging algorithm (with chosen working point as `pfCombinedInclusiveSecondaryVertexV2BJetTags` discriminator should be greater than 0.8484)

TABLE 5.4: Chart of selection of physics objects for the $\mathrm{H} \rightarrow \mathrm{ZZ} \rightarrow 4\ell$ analysis used on the 2018 data (few differences w.r.t 2016 and 2017, as described in Section. 5.6 and rest of selections remain same in all years.)

Electrons
$\mathrm{p_T^e} > 7$ *and* GeV $\quad \eta^e < 2.5$
$\mathrm{d_{xy}} < 0.5\,\mathrm{cm}$ and $\mathrm{d_z} < 1\,\mathrm{cm}$
$\lvert \mathrm{SIP_{3D}} \rvert < 4$
BDT ID with isolation (Fall17v2 training) with cuts from Table 5.2
Muons
Global or Tracker Muon
Discard Standalone Muon tracks if reconstructed in muon system only
Discard muons with muonBestTrackType==2 even if they are global or tracker muons
$\mathrm{p_T^\mu} > 5$ GeV and $\eta^\mu < 2.4$
$\mathrm{d_{xy}} < 0.5\,\mathrm{cm}$ and $\mathrm{d_z} < 1\,\mathrm{cm}$
$\lvert \mathrm{SIP_{3D}} \rvert < 4$
The PF muon ID if $p_T < 200$ GeV, PF muon ID or High-p_T muon ID (Table 5.3) if $p_T > 200$ GeV
$\mathcal{I}^\mu_{\mathrm{PF}} < 0.35$
The FSR photons
$\mathrm{p_T^\gamma} > 2$ GeV and $\eta^\gamma < 2.4$
$\mathcal{I}^\gamma_{\mathrm{PF}} < 1.8$
$\Delta R(\ell, \gamma) < 0.5$ and $\frac{\Delta R(\ell,\gamma)}{(p_T^\gamma)^2} < 0.012\ \mathrm{GeV}^{-2}$
Jets
$\mathrm{p_T^{jet}} > 30$ GeV and $\eta^{\mathrm{jet}} < 4.7$
$\Delta R(\ell/\gamma, \mathrm{jet}) > 0.4$
Cut-based jet ID (tight WP)
Jet pileup ID (tight WP)
Deep CSV algorithm for b-tagging (medium WP)

Chapter 6

Probing four lepton final-state for Higgs Cross section measurement

6.1 Introduction

The ATLAS and CMS collaborations of CERN announced the discovery of much awaited Higgs boson in early July 2012 [7, 8] verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation while studying its properties. Later on, these experiments have also studied its comprehensive properties [9, 42, 43, 44] which are found to be consistent with the SM Higgs. There is a legacy of measurements on this topic from CMS and ATLAS collaborations in different LHC data taking eras for different decay cahannels of Higgs which can been studied in Ref. [**Khachatryan:2015yvw**, **CMS:2015hja**, **CMS:2016rqf**, **Sirunyan:2017exp**, **CMS:2018hhg**, **atlas-conf-2019-029**, **atlas-conf-2019-025** , **atlas-conf-2019-032**, **CMS:2019drq**, 45, 46, 47, 48, 49, 50, 11]. However, these measurements can be improved with newer data at higher luminosities and improved statistical techniques.

The cross section measurements (integrated and differential) are performed using collisions data of CMS (LHC) corresponding to 137.1 fb^{-1} integrated luminosity at $\sqrt{s} = 13$ TeV. The studied differential observables are sensitive to production mechanism of Higgs. Further details follow in next sections.To minimize the model dependence, results are extracted in the fiducial volume that closely matches with the detector geometry as shown in Figure 6.1.

Conribution of other model dependence is extracted by repeating the measurements for SM and other exotic models. The differences are reported as systematic uncertainty. All measurements includes the corrections for the finite detection efficiency and resolution effects, and are extracted assuming $m_H = 125.09$ GeV, as measured from $H \to 2\gamma$ and $H \to 4\ell$ decay modes at CMS experiment of LHC [51].

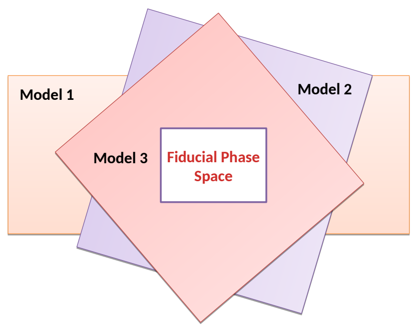


FIGURE 6.1: Fiducial phase definition among different models.

6.2 Datasets and Triggers

The analysis uses full Run 2 (2016, 2017 and 2018) CMS data corresponding to $137.1\ \text{fb}^{-1}$ intergrated luminosity.

Used datasets are listed and documented in [**CMS:2019chr**] with the corresponding integrated luminosities. The analysis relies on four different primary datasets (PDs), *DoubleMuon*, *MuEG*, *EGamma*, and *SingleMuon*, each of which combines a certain collections of HLT paths. Events are collected to avoid duplications from different PDs, using:

– EGamma if they pass the diEle or triEle or singleElectron triggers,

– DoubleMuon if they pass the diMuon or triMuon triggers and fail the diEle and triEle triggers,

– MuEG if they pass the MuEle or MuDiEle or DiMuEle triggers and fail the diEle, triEle, singleElectron, diMuon and triMuon triggers,

– SingleMuon if they pass the singleMuon trigger and fail all the above triggers.

The HLT paths used for full Run 2 collision data is listed in **[CMS:2019chr]** and also in the Appendix.

6.2.1 Trigger Efficiency

Efficiency in data of combination of triggers used for the anlysis is measured by considering 4ℓ events triggered by single lepton triggers. Among the reconstructed four leptons, is geometrically associated with an object that passes at least one single electron or muon triggers. Rest of three leptons are regarded as “probes". In a 4ℓ event, the possible tag-probe pairs are 4 and are counted in denominator while computing efficiency. For each of three leptons which are probes, all matching trigger filter objects are collected. Then the matched trigger filter objects of the three probe leptons are combined in attempt to reconstruct any of the triggers used in the anlysis. If any of the analysis triggers can be formed using the probe leptons, probes are call passing probes which are countedas numerator for the efficiency measurement. The details on the procedure can be seen in Section 5.1.1.

This technique is used to measure efficiency on MC and the comparison with data is exploited to estimate the reliability of simulation. Fig. 6.2 shows the measured efficiency both in data and MC (2018) varied with minimum p_{T} of leptons. MC efficiency seems to describe data well within the uncertainties.

6.3 Simulation Samples

Simulations of the interesting events help us build a reallistic model of experiment in high energy physics. It enables us for a detailed study of physics processes (signal and background). In high energy physics experiments, different MC event generators are used to simulate different process. For this analysis, in particular, descriptions of the production of SM Higgs through

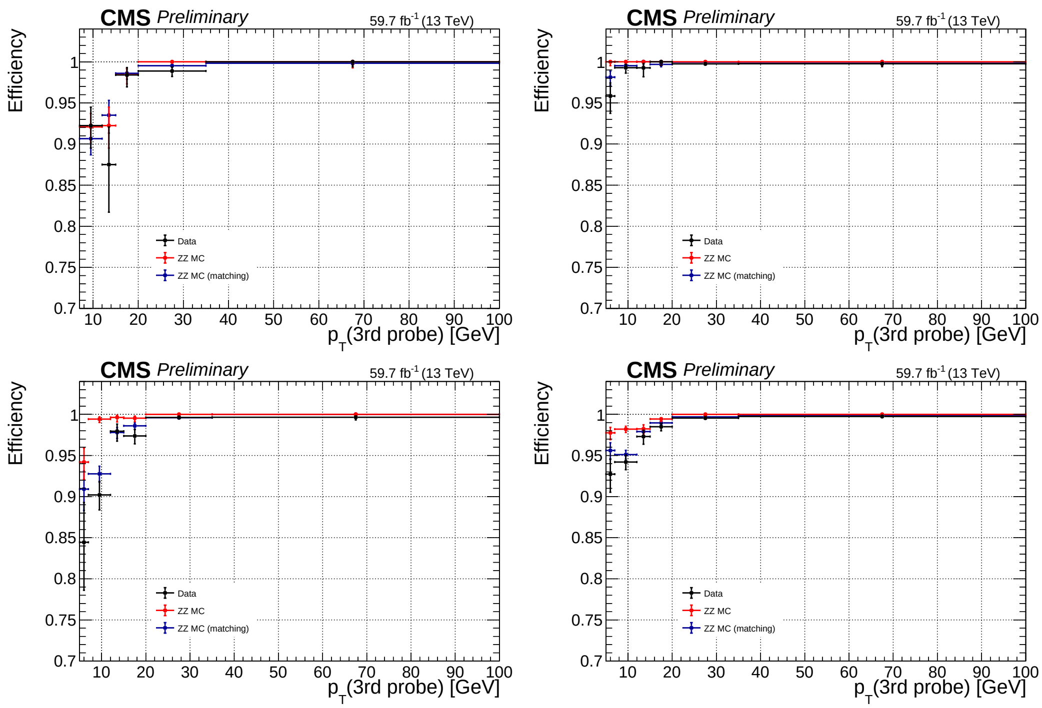


FIGURE 6.2: Measured trigger efficiency in 2018 data where the event collection is made using single electron and muon triggers: $4e$ (top left), 4μ (top right), $2e2\mu$ (bottom left) and 4ℓ (bottom right) final states.

process of gluon-gluon fusion (Fig. 6.3) are acquired using POWHEG V2 [52, 53] generator. At accuracy level of next-to-next-to-leading order (i.e. NNLO) in perturbative QCD (pQCD), $\mathrm{gg} \to \mathrm{H}$ process description is obtained by partonic level MC generator NNLOPS.

The POWHEG, provides perturbative QCD calculations at next-to-leading (NLO) accuracy with a connection to parton-shower programs. Using the POWHEG, Higgs production via $\mathrm{gg} \to \mathrm{H}$ with zero associated jets can be described at accuracy level of NLO. Finite masses of top and bottom quarks has been considered for $\mathrm{gg} \to \mathrm{H}$ production for all generators.

Signal Samples

Descriptions of the SM Higgs boson production are obtained using the POWHEG V2 [54, 55, 56] generator for the five main production modes (ggH, VBF, WH, ZH and ttH) [57], [58], [59], [60]. However, WH and ZH production modes are simulated by the MiNLO HVJ extension of the POWHEG [60]. Higgs decay to four lepton is simulated by JHUGEN generator [61]. For the WH, ZH and ttH modes, Higgs is allowed to decay to H→ $ZZ \rightarrow 2\ell 2X$ such that two leptons from 4-lepton events are produced from the decay of associated W, Z bosons or top quarks respectively. Showering of parton-level events is done using PYTHIA8.209, and in all cases matching is performed by allowing QCD emissions at all energies in the shower and vetoing them afterwards according to the POWHEG internal scale. All samples are generated with the default NNPDF 3.1 NLO parton distribution functions (PDFs) [62]. The list of signal samples with the corresponding cross sections are shown in **[CMS:2019chr]**.

These samples, for each production mode, are also used to assess model dependence of measured results as shown in the Table 6.8. The details of the method used to assess the model dependence will be described in Section 6.5. The POWHEG and NNLOPS based samples for $\mathrm{gg} \rightarrow \mathrm{H}$ production are utilized as theoretical predictions based on SM calculations for the comparison with measured results as described in in Section 6.6.

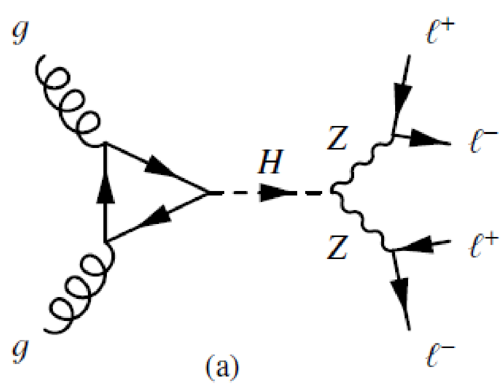


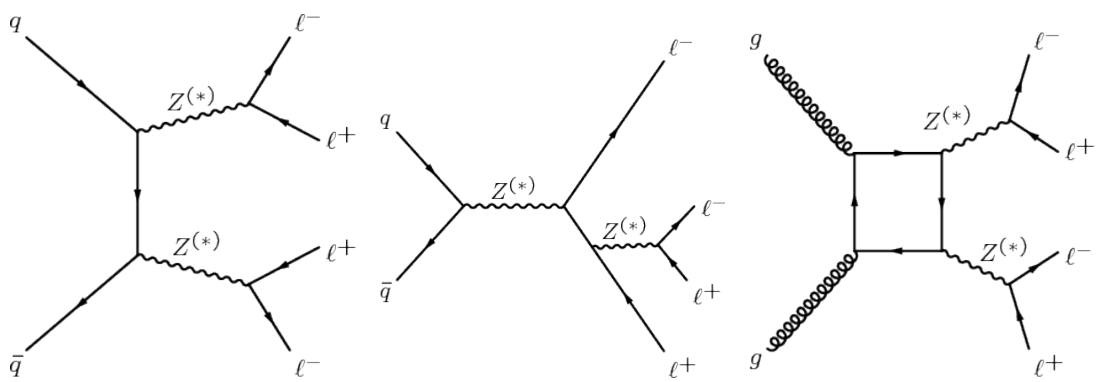
FIGURE 6.3: $\mathrm{gg} \rightarrow \mathrm{H}$ process feynman diagram.

All these MC samples are produced using POWHEG for the representation of QCD effects at NLO in Higgs production, and by using JHUGEN [61, 63, 64] to represent such exotic resonances decays to four-leptons containing all the spin correlations.

Background Samples

For the estimations of the backgrounds, POWHEG V2 interfaced with PYTHIA8 was used to generate the dominant process $\mathrm{q\bar{q}} \to \mathrm{ZZ} \to 4\ell$ at the NLO accuracy level containing s-, t- and u-channel diagrams [65]. Generatars were used with same configurations as used for the Higgs signal . Feynman diagrams of the involved background processes are shown in the Figure 6.4. The dynamical renormalization and factorization scales have been chosed equal to m_{ZZ} since the simulation covers a wide range of invariant masses of ZZ.

The $\mathrm{gg} \to ZZ$ process, as shown in the Figure 6.4(c) is simulated at leading order (LO) with MCFM [66, 67]. In order to have matching in spectra of $\mathrm{gg} \to \mathrm{H} \to ZZ$ as predicted at NLO by POWHEG, the showering of these samples has been performed with some different PYTHIA8 settings. This only allows emissions only up to parton-level scale ("wimpy" shower).



(a) s-channel (b) t-channel (c) ggZZ-box

FIGURE 6.4: $\mathrm{q\bar{q}}to\mathrm{ZZ} \to 4\ell$ process s- and t- channel fynman diagrams (a) and (b). $\mathrm{gg} \to \mathrm{ZZ} \to 4\ell$ process fynman diagram (c).

The initial-state and final-state radiations are included for all events generators, which provides hints for the existence of extra (hard) photons in the event. Interference between all amplitudes of process $H \to 4\ell$ are properly taken into consideration both in POWHEG and the JHUGEN. For the generators, the used default parton distribution functions (PDFs) are NNPDF3.1 NLO (at NLO and LO) are used.

The generated events were interfaced to PYTHIA 8.209 Tune tune CUETP8M1 to include the fact of the parton shower, underlying events and hadronization.

Processing of all generated events was made via a full simulation for CMS detector using GEANT4 [68] and their reconstruction was done by exploiting same algorithms used for the data analysis. Multiple proton-proton interactions are incorporated in the simulations by reweighting the pileup distribution to match the distribution of the observed data.

Pileup Reweighting

For each year, corresponding simulation samples are reweighted to meet pileup distribution in the data. An example of the reweighting procedure for 2018 data is shown in Fig. 6.5.

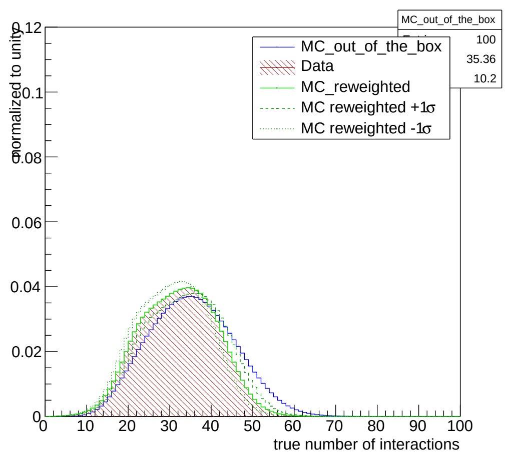


FIGURE 6.5: Pileup distribution in 2018 MC and Data shown before and after the application of PU weights. Up and down variations of 5% in the minimum bias cross section when calculating the weighs are also shown.

6.4 Analysis Strategy

As mentioned in Section 6.3, the measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. Measurement strategy and definition of the fiducial volume is optimized to:

- detect a low mass Higgs (m_H 125.09 GeV) boson decaying to 4 leptons through a pair of Z bosons
- keep the measurements independent of how the Higgs boson is produced.

For reconstruction efficiency to be independent of fiducial phase space, it is necessary to include isolation in the definition of fiducial phase space.

In case of associated production of Higgs with vector bosons where the bosons can decay leptonically, there is possibility that Higgs is reconstructed from wrong combination of leptons. This makes it necessary to consider such "wrong combination" events as a background in order to keep complete information from unbinned simultaneous fit of signal and background parameterizations to four-lepton invariant mass.

Also, signal events which pass the fiducial level selections but fail the reconstruction are accounted in the fiducial efficiency (ϵ) and the events which pass the reconstruction but fail the fiducial selections are referred to as "nonfiducial signal" events (f_{nonfid}) and are treated as a background as illustrated in Figure 6.6.

There is significant effect of imperfect resolution of the detector on the differential measurements particularly in case of jet related observables for which detector resolution is poor. A procedure, called unfolding, is applied for the correction of the reconstruction efficiencies and the effects are applied to the measurements.

The effects of imperfect detector resolution can in general have a non-negligible effect on shape of the distributions of measured observables. For this reason, for differential cross section measurements of Higgs, a procedure to correct the detection efficiencies and resolution effects is applied.
The details of the procedure would be described in next Sections.

6.4.1 Event Selection

This dissertation is based on the analysis for which event selections are similar to what was used in pervious analyses on same topic using the same final state [**CMS:2018mmw**, **CMS:2017jkd**, 69, 70]. The online event selection is

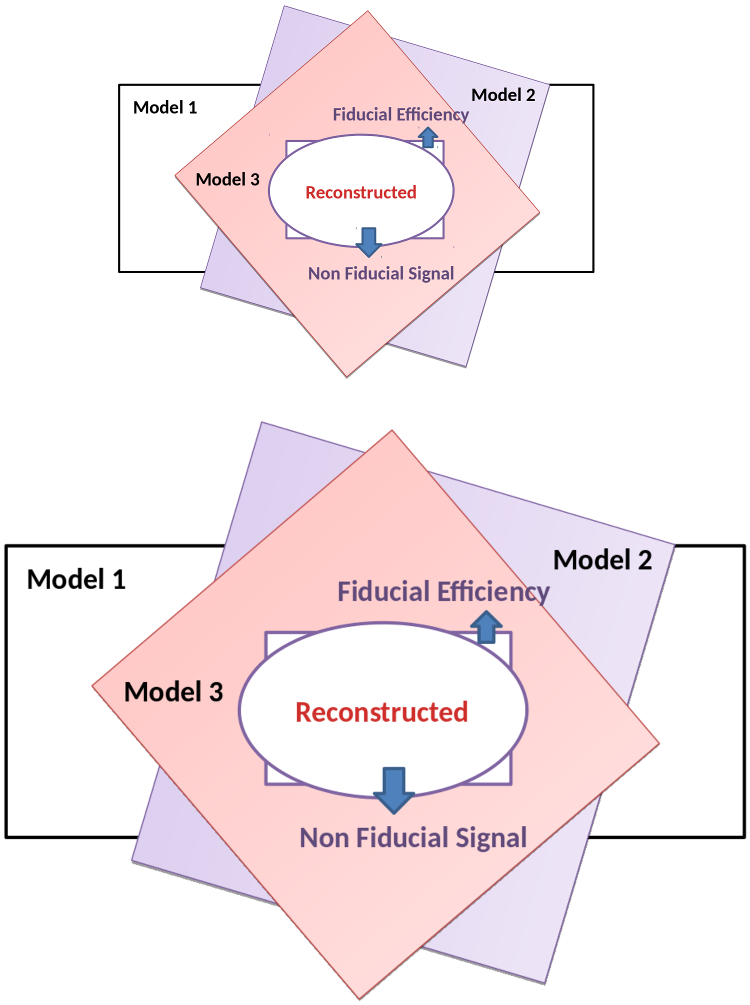


FIGURE 6.6: Fiducial efficiency(ϵ) and nonfiducial(f_{nonfid}) signal definition.

based on having fired High Level Trigger paths as described in 6.2. Unlike in the analysis based on Run I dataset, we require OR of all HLT paths so that the trigger requirement does not depend on the selected final state. The reason is Full Run II data (at $137.1\ \mathrm{fb}^{-1}$), associated production modes that can come with additional leptons are targeted which improve trigger efficiency.

Vertex Selection

The events are required to have at least one primary vertex (PV) that fullfills the criteria:

- Large degree of freedom ($N_{PV} > 4$)
- Small radius of PV being ($r_{PV} < 2$ cm)
- Restricted collisions along $z-$axis ($z_{PV} < 24$ cm).

p_T threshold requirements on the Triggers

For the double lepton triggers, the requirement on minimum p_T for leading and the subleading lepton are imposed to be 17 and 8 GeV respectively; however, for triple electron triggers, it is 15, 8 and 5 GeV in the same order respectively.

Lepton Selection

Details on the selection requirments on electron and muons are explained in sections 5.2.2 and 5.3.1, and 5.2.2 and 5.3.2 respectively. Both electrons and muons are initially required to pass POG Loose identification criteria with subsequent requirements on imparameter and isolation. These leptons are regarded as **selected leptons** and can be used for analysis.

Jet Selection

Reconstruction of the jets is mandatory to study the differential cross section measurements for the jet related observables. Jets are reconstructed with $p_T \geq 30$ GeV within $|\eta| \leq 4.7$, discussed in Section 5.5. To remove overlapping of jets overlap from leptons or selected photons, it is required to have at least $\Delta R > 0.5$ where $\Delta R \equiv \sqrt{(\Delta\eta)^2 + (\Delta\phi)^2}$ [71] with respect to any of the selected leptons and FSR photons candidates.

6.4.2 ZZ Candidate Selection

Four-lepton candidates are built from the selected leptons, that pass the $\mathrm{SIP}_{3D} < 4$ vertex constraint and the isolation cuts (defined in sections 5.2.2 and 5.3.2), where FSR photons are subtracted as briefed in Section 5.4. Cross cleaning of the leptons is made by discarding electrons which are within $\Delta R < 0.05$ of selected muons.

The construction and selection of four-lepton candidates proceeds according to the following sequence:

1. **Z candidates** are built from oppositely charged and same flavoured leptons (e^+e^-, $\mu^+\mu^-$) with additional requirement on dilepton mass as $12 < m_{\ell\ell(\gamma)} < 120GeV$. Z boson candidate is also required to include FSR photons (if any).

2. **ZZ candidates** are then built from non-overlapping Z boson candidates. The one who is reconstructed with $m_{\ell\ell}$ closer to the SM Z boson mass is regarded as Z_1 and other as Z_2. Additional requierements on ZZ candidates are as under:

   - **Ghost removal** : Selected four leptons are required to have $\Delta R(\eta, \phi) > 0.02$ among each other.
   - **lepton** p_T: Among selected four leptons, two are required to fulfill $p_{T,i} > 20GeV$ and $p_{T,j} > 10GeV$.
   - **QCD suppression**: Dilepton inivariant mass is required to pass (regardless of flavor) $m_{\ell\ell} > 4\ GeV$ for QCD induced J/Ψ veto.
   - Z_1 **mass**: $40\ GeV < m_{\mathrm{Z}1} < 120\ GeV$
   - **four-lepton invariant mass**: $m_{4\ell} > 70\ GeV$

3. **Signal region** is built from the events containing one selected ZZ candidate at leaset.

6.4.3 Choice of best ZZ Candidate

An important change with respect to the Run I analysis is to choose the best ZZ candidate after all kinematic cuts. This helps to test other selection strategies for this candidate choice and becomes important for events having more than four selected leptons to search for Higgs boson production modes where associated particles can decay leptonically (e.g. VH and ttH).

For fiducial cross section measurement, Z_1 is chosen with mass closest to the mass of SM Z boson mass while Z_2 is chosen from Z candidates whose lepton pair gives higher p_T sum.

6.4.4 Definition of the Fiducial Volume

The measurements on Higgs cross sections (Inclusive and differential) are made in a fiducial region that matches the reconstruction level selection. This is done because fraction of all $\mathrm{H} \to 4\ell$ events which pass the selection criteria as described in Section 6.4.3 can significantly differ for different Higgs production mechanisms and also for its distinct exotic models. As a treatment for the impact of model dependent acceptances, these measurements are done in the a certain fiducial volume which is defined so as to closely match with the experimental acceptance from reconstruction level selections.

Requirement of fiducial phase space is minimum four leptons having $p_T >$ 7 GeV and $|\eta| < 2.5$ in case of electron, and $p_T > 5$ GeV and $|\eta| < 2.4$ in case of muon. Besides, the leading and sub-leaing leptons are required to have $p_T > 20$ GeV and $p_T > 10$ GeV respectively. The four-momenta of each lepton is calculated at the hard scattering level and isolation is calculated by adding p_T of all stable particles within a cone size $\Delta R < 0.3$ around the lepton (by subtracting contributions from neutrinos and FSR photons). All neutrinos and FSR photons are excluded from above summation. Then the lepton's relative isolation, defined by ratio of this sum and p_T of considered lepton, is required to be less than 0.35 being consistent with reconstruction level isolation requirement [69].

Simulation based studies show that signal efficiency can significantly change among different models without requiring lepton isolation at fiducial level. The difference is more visible in $t\bar{t}\mathrm{H}$ and $\mathrm{gg} \to \mathrm{H}$ production modes as can be seen in the Table **??**. It is understood from the fact that $t\bar{t}\mathrm{H}$ events have more hadronic activity around leptons than those of the $\mathrm{gg} \to \mathrm{H}$ events, so the cut at relative isolation removes more events in $t\bar{t}\mathrm{H}$ than $\mathrm{gg} \to \mathrm{H}$. Hence, for reconstruction efficiency to be independent of fiducial phase space, it is necessary to include isolation in the definition of fiducial phase space.

For other requirements of fiducial volume, there must be two pairs of same falvor oposite sign (SFOS) leptons where Z_1 being the one with invariant mass closest to the SM Z boson and reconstructed with $40 \leq m(Z_1) \leq 120$ GeV. Z_2 is reconstructed from rest of the SFOS pairs of leptons within invariant mass range of $12 \leq m(Z_2) \leq 120$ GeV. In case of more than one Z_2 passing the criteria, leptons pair with highest $\Sigma|p_T|$ is selected. To remove the overlapping between leptons, we ask for $\Delta R(\ell_i \ell_j) > 0.02$ for any $i \neq j$. Any opposite-charge pair of leptons is required to meet the condition $m(\ell^+ \ell'^-) \geq 4$ GeV to suppress background from QCD for low mass resonance which is called J/ψ veto.

Invariant mass of selected four-leptons is required to fulfill 105 GeV $< \mathrm{m}(4\ell) <$ 140 GeV and must originate from Higgs boson decay. This helps to avoid from the contamination from off-shell gg fusion production [72, 73]. The requirements that define fiducial phase space are also gathered in Table 6.1.

TABLE 6.1: Overview of all requirements utilized to define the fiducial phase space.

Requirements on $\mathrm{H} \to 4\ell$ fiducial phase space	
Lepton kinematics and the isolation	
p_T of the leading lepton	$p_T > 20$ GeV
p_T of sub-leading lepton	$p_T > 10$ GeV
Additional electrons (muons) p_T	$p_T > 7$ (5) GeV
Pseudorapidity of electrons (muons)	$\|\eta\| < 2.5$ (2.4)
Sum of the scalar p_T of all the stable objects within $\Delta R < 0.3$ from the lepton	$< 0.35 p_T$
Event topology	
From leptons satisfying criteria above, minimum two SFOS lepton pairs	
Invarivant mass of Z_1 candidate	$40 < m(Z_1) < 120$ GeV
Invariant mass of Z_2 candidate	$12 < m(Z_2) < 120$ GeV
Separation between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$
Invariant mass of any OS lepton pair	$m(\ell_i^+ \ell_j^-) > 4$ GeV
Invariant mass of selected four leptons	$105 < m_{4\ell} < 140$ GeV

Part of the signal events passing fiducial phase space definition ($\mathcal{A}_{\text{fid}}$), and reconstruction efficiencies (ϵ) for SM production modes are registered in Table 6.2.

TABLE 6.2: Summary of different SM signal models. The MC samples are from full Run 2 yearly productions (2016, 2017 and 2018) assuming $m_{\mathrm{H}} = 125.09$ GeV. Luminosity averaged fiducial acceptance ($\mathcal{A}_{\mathrm{fid}}$) defined the fraction of events passing fiducial phase space selections, reconstruction efficiency (ϵ) represented by the ratio of reconstructed to accepted events. Also, f_{nonfid} which are reconstructed events but does not pass fiducial phase space definition at generator level. Values are obtained using 13 TeV signal models and the uncertainties shown are only statistical uncertainties.

Signal process	$\mathcal{A}_{\mathrm{fid}}$	ϵ	f_{nonfid}	$(1 + f_{\mathrm{nonfid}})\epsilon$
Individual production modes of Higgs boson				
gg→H (POWHEG)	0.402 ± 0.001	0.592 ± 0.002	0.053 ± 0.001	0.624 ± 0.002
VBF (POWHEG)	0.444 ± 0.002	0.605 ± 0.003	0.043 ± 0.001	0.631 ± 0.003
WH (POWHEG+MINLO)	0.325 ± 0.002	0.588 ± 0.003	0.075 ± 0.002	0.632 ± 0.004
ZH (POWHEG+MINLO)	0.340 ± 0.003	0.594 ± 0.005	0.081 ± 0.004	0.643 ± 0.006
ttH (POWHEG)	0.314 ± 0.003	0.585 ± 0.006	0.169 ± 0.006	0.684 ± 0.007

6.4.5 Signal Modeling

For fiducial cross section measurements, we perform a maximum likelihood fit of the signal and background parameterisations to the observed $m_{4\ell}$ mass distribution and directly extract the parameter of interest (σ_{fid}). Definition of the likelihood function and the procedrue is explained in Section 6.4.7.

However, in this section we describe the PDF used to describe the signal shapes for different masses of Higgs. The resolution function model depends only on the leptons quality, while the exact shape depends on the lepton kinematics, and so depend also on the Higgs mass. Double Crystal Ball function is used for shape description of signal resonant contribution ($\mathcal{P}_{\mathrm{res}}(m_{4\ell})$) [69] with the yield proportional to the parameter of interest (σ_{fid}). A Double Crystal-Ball function $f_{dCB}(m_{4\ell}\,|\,m_H)$ is defined as under:

$$dCB(\xi) = N \cdot \begin{cases} A \cdot (B + |\xi|)^{-n_L}, & \text{for } \xi < \alpha_L \\ A \cdot (B + |\xi|)^{-n_R}, & \text{for } \xi > \alpha_R \\ \exp\left(-\xi^2/2\right), & \text{for } \alpha_L \le \xi \le \alpha_R \end{cases} \tag{6.1}$$

where $\xi = (m_{4\ell} - m_H - \Delta m_H)/\sigma_m$.

This function has six independent parameters, and is intended to capture the Gaussian core (σ_m) of the four-lepton mass resolution function, systematic mass shift Δm_H of the peak, and the left- and right-hand tail originating from leptons emitting brem in the tracker material, present for both electrons and muons, and from the non-Gaussian mis-measurements specific to interactions of electrons with the detector material (two parameters, n and α, for each side of the mean): The prominence of the left-,right-hand tail is defined the power n_L, n_R, respectively. The parameters α_L, α_R define where the splicing of the tails and the core are made, in units of σ_m. Parameters A and B are not independent; they are defined by requiring the continuity of the function itself and its first derivatives. N is the normalizing constant.

To modell the Higgs shape, several mass points in mass points 120-130 GeV are used in the three final states. The fitting strategy used to deal with this situation is to use the linear approximation of all the Double Crystal Ball parameters varying with with m_H.

$$params = parms_{CB0} + params_{CB1} \times (m_H - 125GeV) + params_{CB2} \times (m_H - 125GeV)^2 \tag{6.2}$$

Simultaneous fits are performed using 5 mass samples to extract the $parms_{CB0}$, $params_{CB1}$ and $params_{CB2}$. The initial value for the $parms_{CB0}$ is obtained by fitting 125 GeV mass sample alone, and refitted in the simultaneous fits. Uncertainties on these parameters are included as systematic uncertainties on the measurements as described in the Section **??**.
Some examples of the fits are shown in Fig. 6.7 for the ggH production mode. The simultaneous fit at different mass points are shown in Fig. 6.8.

6.4.6 Background Modeling

Background processes for this analysis can be divided into two parts:

1. **Irreducible Background**: Mainly it is due to quark anti-quark ($q\bar{q}$) annihilation which produce a pair of Z bosons and each Z boson decays into a pair of leptons resulting four leptons in the final state. The second one is when quark anti-quark ($q\bar{q}$) annihilate to Z boson which further decays into a pair of leptons. One of the two leptons, then radiate a virtual Z boson which subsequently decays into a pair of leptons.

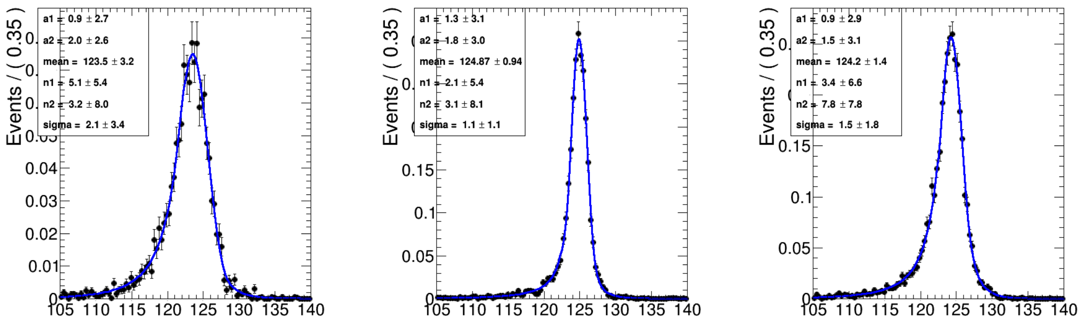


FIGURE 6.7: Probability density functions $f(m_{4l}|m_H)$ for signal with m_H**=125** GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained **from 13** TeV ggH MC samples are fitted with the model described in the text for 4μ (left), $4e$ (center) and $2e2\mu$ (right) events. The distribution show the events for p_T below 10 GeV.

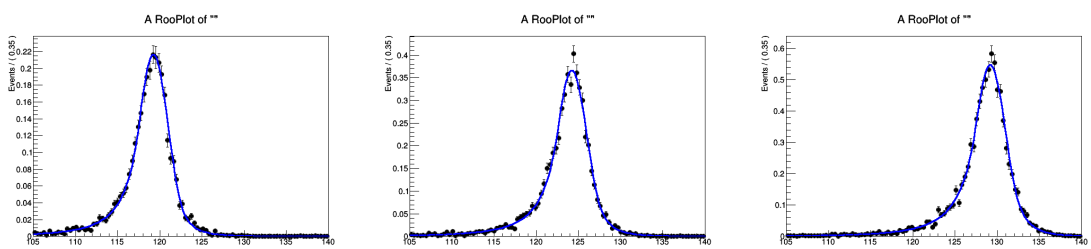


FIGURE 6.8: Simultaneous fits for $f(m_{4l}|m_H)$ for $\mathrm{gg} \to$ H signal for $p_T$below 10 GeV with m_H**=120,125,130** GeV at the reconstruction level after the full lepton and event selections are applied. The distributions obtained **from 13** TeV $\mathrm{gg} \to$ H MC samples are fitted with the model described in the text for $2e2\mu$ events.

This background has dominant contribution in the analysis. Additional source of the Irreducible background is from gluon gluon fusion process. However, in the analysis it has sub-dominant contribution. Feynman diagrams of the these background processes can be seen in the diagram 6.4.

2. **Reducible Background**: This background is instrumental nature and arises from the process where partial jets are misidentified as leptons. The main processes which contributes in this background are: production of jets in association with Z boson, production of the jets in association with WZ boson pair and the $\mathrm{t\bar{t}}$ pair production. This background is denoted as Z+X.

For fiducial cross section measurement for $\mathrm{H} \rightarrow 4\ell$, both of irreducible backgrounds ($\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ and $\mathrm{gg} \rightarrow \mathrm{ZZ}$) are estimated from Monte Carlo (MC) simulation as done earlier in Ref. [69].

$\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ Modelling

The $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ background is generated at NLO, while the fully differential cross section has been computed at NNLO [74]. Since the cross sections are not yet available in the generator we use for its estimation, the NNLO/NLO k-factors for the $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ background process are applied to the POWHEG sample used. The inclusive cross sections obtained using the same PDF and renormalization and factorization scales as the POWHEG sample at LO, NLO, and NNLO are shown in Table 6.3. The NNLO/NLO k-factors are applied in the analysis differentally as a function of $m(ZZ)$.

Additional NLO electroweak corrections which depend on the initial state quark flavor and kinematics are also applied to the $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ background process in the region $m(\mathrm{ZZ}) > 2m(\mathrm{Z})$ where the corrections have been computed. The differential QCD and electroweak k-factors can be seen in Figure 6.9.

QCD Order	$\sigma_{2\ell 2\ell'}(\mathrm{fb})$	$\sigma_{4\ell}(\mathrm{fb})$
LO	$218.5^{+16\%}_{-15\%}$	$98.4^{+13\%}_{-13\%}$
NLO	$290.7^{+5\%}_{-8\%}$	$129.5^{+4\%}_{-6\%}$
NNLO	$324.0^{+2\%}_{-3\%}$	$141.2^{+2\%}_{-2\%}$

TABLE 6.3: Cross sections for $\mathrm{q\bar{q}} \rightarrow \mathrm{ZZ}$ production at 13 TeV

$\mathrm{gg} \rightarrow \mathrm{ZZ}$ Modelling

Event simulation for the $\mathrm{gg} \rightarrow \mathrm{ZZ}$ background is done at LO with the generator 7.0 [66, 75, 67]. Although no exact calculation exists beyond the LO for the $\mathrm{gg} \rightarrow \mathrm{ZZ}$ background, it has been recently shown [76] that the soft collinear approximation is able to describe the background cross section and the interference term at NNLO. Further calculations also show that the K factors are very similar at NLO for the signal and background [77] and at NNLO for the signal and interference terms [78]. Therefore, the same K factor is used for

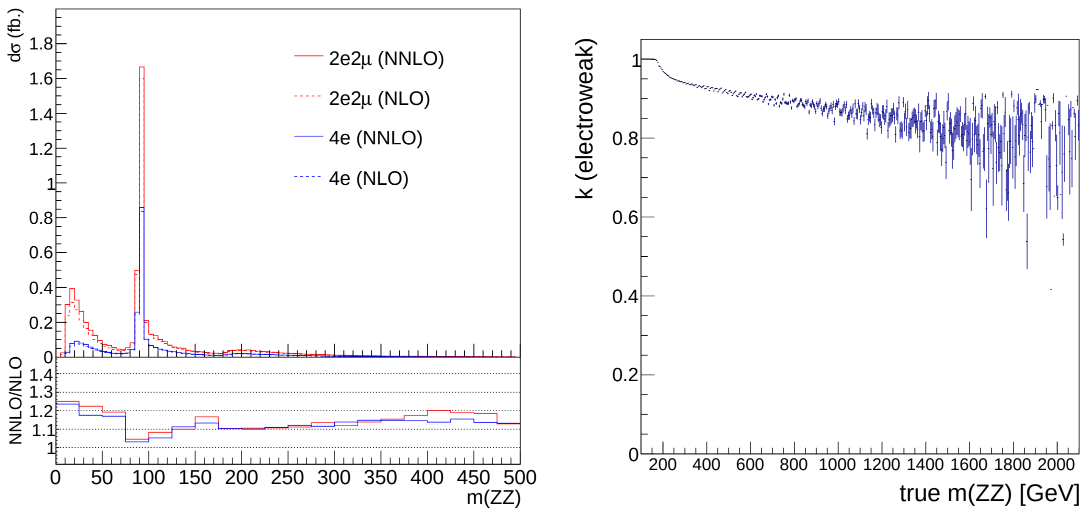


FIGURE 6.9: Left: NNLO/NLO QCD k-factor for the $\mathrm{q\bar{q}} \to$ ZZ background as a function of $m(ZZ)$ for the 4ℓ and $2\ell 2\ell'$ final states. Right: NLO/NLO electroweak k-factor for the $\mathrm{q\bar{q}} \to$ ZZ background as a function of $m(ZZ)$.

the signal and background [79]. The NNLO K factor for the signal is obtained as a function of $m_{4\ell}$ using the HNNLO v2 Monte Carlo program [80, 81, 82] by calculating the NNLO and LO $\mathrm{gg} \to H \to 2\ell 2\ell'$ cross sections at the small H boson decay width of 4.07 and taking their ratios. The NNLO as well as the NLO K factors and the cross sections from which they are derived are illustrated in Fig. 6.10, along with the NNLO, NLO and LO cross sections at the SM H boson decay width [83].

In addition, following the study in Ref. [84], in these measurements the expected yield for the gg $\to$ ZZ background has been increased by assigning the same ($m_{4\ell}$ dependent) k-factor as signal $\mathrm{H} \to 4\ell$ process to leading order (LO) cross section of background. In the region ($105 < m_{4\ell} < 140$ GeV) this factor leads to an increase of expected gg $\to$ ZZ background by factor of ~ 2.66. After inclusion of this k-factor, the gg $\to ZZ$ process is still a sub-dominant background in all measurements.

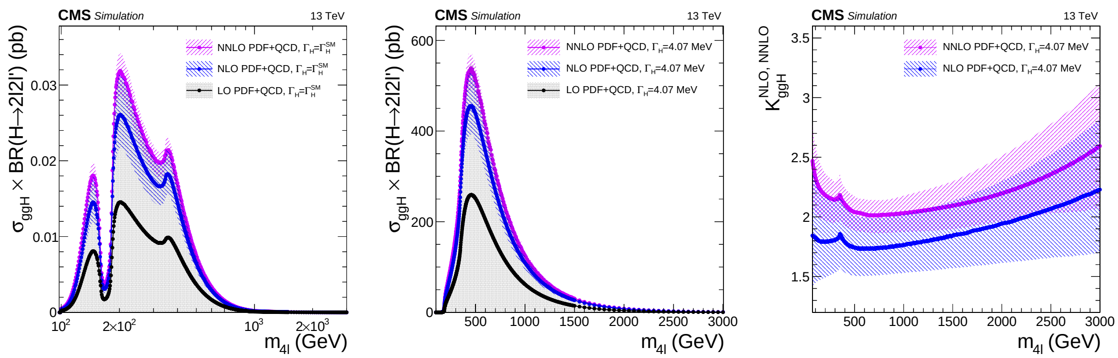


FIGURE 6.10: $gg \to H \to 2\ell 2\ell'$ cross sections at NNLO, NLO and LO at each H boson pole mass using the SM H boson decay width (top) or at the fixed and small decay width of 4.07 MeV (top). The cross sections using the fixed value are used to obtain the K factor for both the signal and the continuum background contributions as a function of $m_{4\ell}$ (bottom).

Reducible Background Estimation

In the $H \to ZZ \to 4\ell$ analysis, Reducible background (Z+X) is estimated using data with tight-to-loose "fake rate" technique is used in the control regions of data by following Ref. [69].

In the $H \to ZZ \to 4\ell$ analysis, Reducible Background (Z+X) is estimated using data with tight-to-loose "fake rate (FR)" technique. The rates of involved background processes are estimated by measuring the probabilities for fake electrons and fake muons (denoted by f_e and f_μ respectively) which do pass the **loose** selection criteria (defined in Section 5.2.1 and 5.3.1) to also pass the final selection criteria (defined in Section 6.4.2) [69]. These probabilitiess or fake rates (FR) are applied in dedicated control samples in order to extract the expected background yield in the signal region.

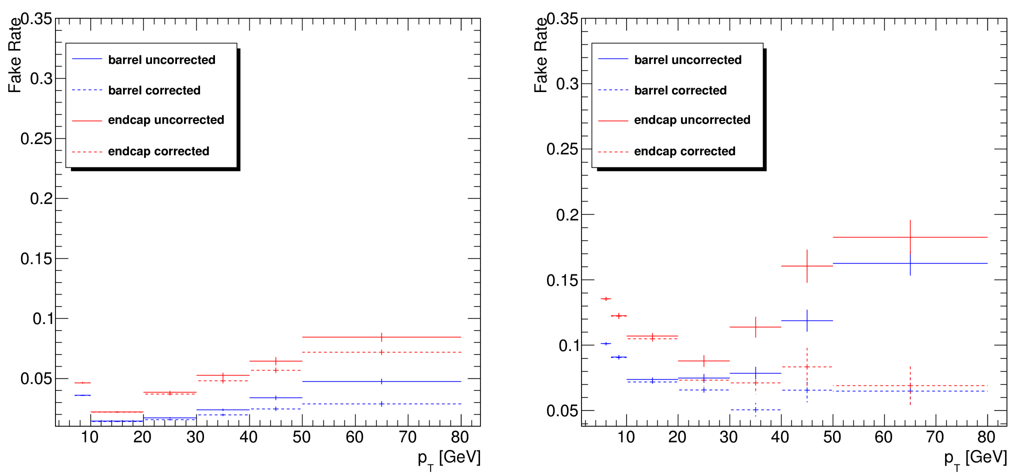
Two independent methods are presented to measure both the yields and shapes of the reducible background. The final result combines the outcome of the two approaches.

(1) Opposite Sign (OS) method Fore lepton fake ratios f_e, f_μ, samples of $Z(\ell\ell) + e$ and $Z(\ell\ell) + \mu$ events are selected which are expected to be completely dominated by final states which include a Z boson and a fake lepton. These events are required to have two SFOS leptons with $p_T > 20/10$ GeV (p_T

of leading and sub-leading leptons) passing the tight selection criteria, thus forming the Z candidate. In addition, there is exactly one lepton passing the loose selection criteria as defined above. This lepton is used as the probe lepton for the fake ratio measurement. The invariant mass of this lepton and the opposite sign lepton from the reconstructed Z candidate should satisfy $m_{2l} > 4$ GeV.

To reduce the contribution from photon (asymmetric) conversions populating low masses, FRs are evaluated using the tight requirement $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV. The fake ratios are measured in bins of the p_T loose lepton in barrel and endcap region.

The electron and muon fake rates are measured within $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV and $E_{\mathrm{T}}^{\mathrm{miss}} < 25$ GeV are shown in Figure 6.11.



(a) (b)

FIGURE 6.11: Fake rates as a function of the probe p_T for electrons (a) and muons (b) which satisfy the loose selection criteria, measured in a $Z(\ell\ell) + \ell$ sample in the 13 TeV data. The barrel selection includes electrons (muons) up to $|\eta|$ = 1.479 (1.2). The Fake rates are shown before (dotted lines) and after (plain line) removal of WZ contribution from MC.

Fake Rate Application (OS Method) Two control samples are obtained as subsets of four lepton events which pass the first step of the selection (*First Z step*, see section 6.4.2), requiring an additional pair of loose leptons of same flavour and opposite charge, that pass the $\mathrm{SIP}_{3\mathrm{D}}$ cut. The events must satisfy all kinematic cuts applied for the *Higgs phase space* selection (see 6.4.2). The first control sample is obtained by requiring that the two loose leptons which do not make the Z_1 candidate do not pass the final identification and isolation criteria. The other two leptons pass the final selection criteria by definition of the Z_1. This sample is denoted as "2 Prompt + 2 Fail" (*2P+2F*) sample. It is expected to be populated with events that intrinsically have only two prompt leptons (mostly DY, with a small fraction of $t\bar{t}$ and $Z\gamma$ events). The second control sample is obtained by requiring one of the four leptons not to pass the final identification and isolation criteria. The other three leptons should pass the final selection criteria. This control sample is denoted as "3 Prompt + 1 Fail" (*3P+1F*) sample. It is expected to be populated with the type of events that populate the 2P+2F region, although with different relative proportions, as well as with WZ events that intrinsically have three prompt leptons.

The control samples obtained in this way, orthogonal by construction to the signal region, are enriched with fake leptons and are used to estimate the reducible background in the signal region.

The invariant mass distribution of events selected in the 2P+2F control sample is shown in Fig. 6.12 for the 13 TeV dataset.

The expected number of reducible background events in the 3P+1F region, $N^{\mathrm{bkg}}_{\mathrm{3P1F}}$, can be computed from the number of events observed in the 2P+2F control region, N_{2P2F}, by weighting each event in the region with the factor $(\frac{f_i}{1-f_i}+\frac{f_j}{1-f_j})$, where f_i and f_j correspond to the fake ratios of the two loose leptons:

$$N^{\mathrm{bkg}}_{\mathrm{3P1F}} = \sum\left(\frac{f_i}{1-f_i}+\frac{f_j}{1-f_j}\right)N_{\mathrm{2P2F}} \tag{6.3}$$

Figure 6.13 shows the invariant mass distributions of the events selected in the 3P+1F control sample, together with the expected reducible background estimated from Eq. 6.3, stacked on the distribution of WZ and of irreducible background ($ZZ, Z\gamma* \to 4\ell$) taken from the simulation.

Would the fake rates be measured in a sample that has exactly the same background composition as the 2P+2F sample, the difference between the observed number of events in the 3P+1F sample and the expected background

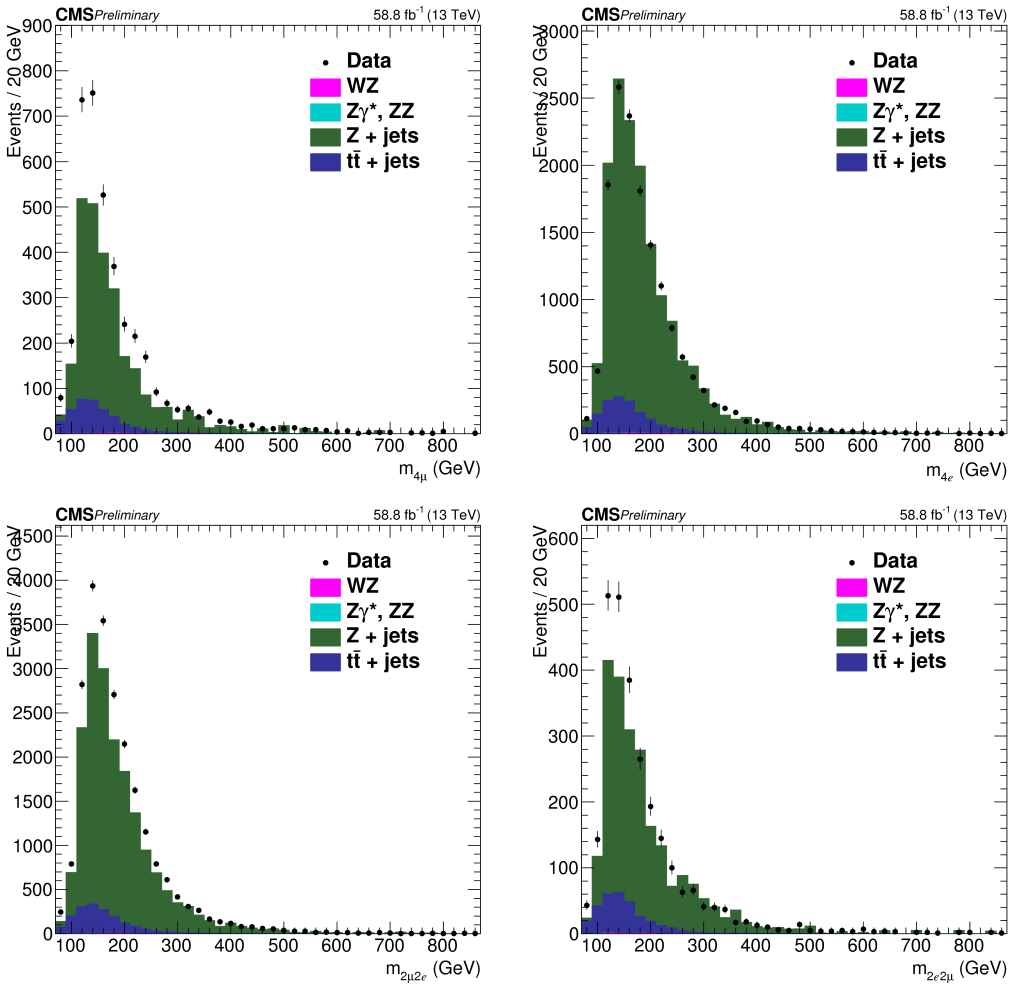


FIGURE 6.12: Invariant mass distribution of the events selected in the 2P+2F control sample in the 13 TeV dataset, (top left) 4μ , (top right) $4e$, (bottom left) $2\mu 2e$ and (bottom right) $2e2\mu$ channels.

predicted from the 2P+2F sample would solely amount to the (small) WZ and $Z\gamma_{conv}$ contribution. Large differences arise because the fake rates used in Eq. 6.3 do not properly account for the background composition of the 2P+2F control sample.

In particular, the difference seen in Fig. 6.13 between the observed 3P+1F distribution and the expectation from 2P+2F, in the channels with loose electrons ($4e$ and $2\mu 2e$), and concentrated at low masses, is due to photon conversions. This is confirmed explicitly by the simulation. The difference between the 3P+1F observation and the prediction from 2P+2F to recover the missing contribution from conversions - and more generally, in principle, to “correct"

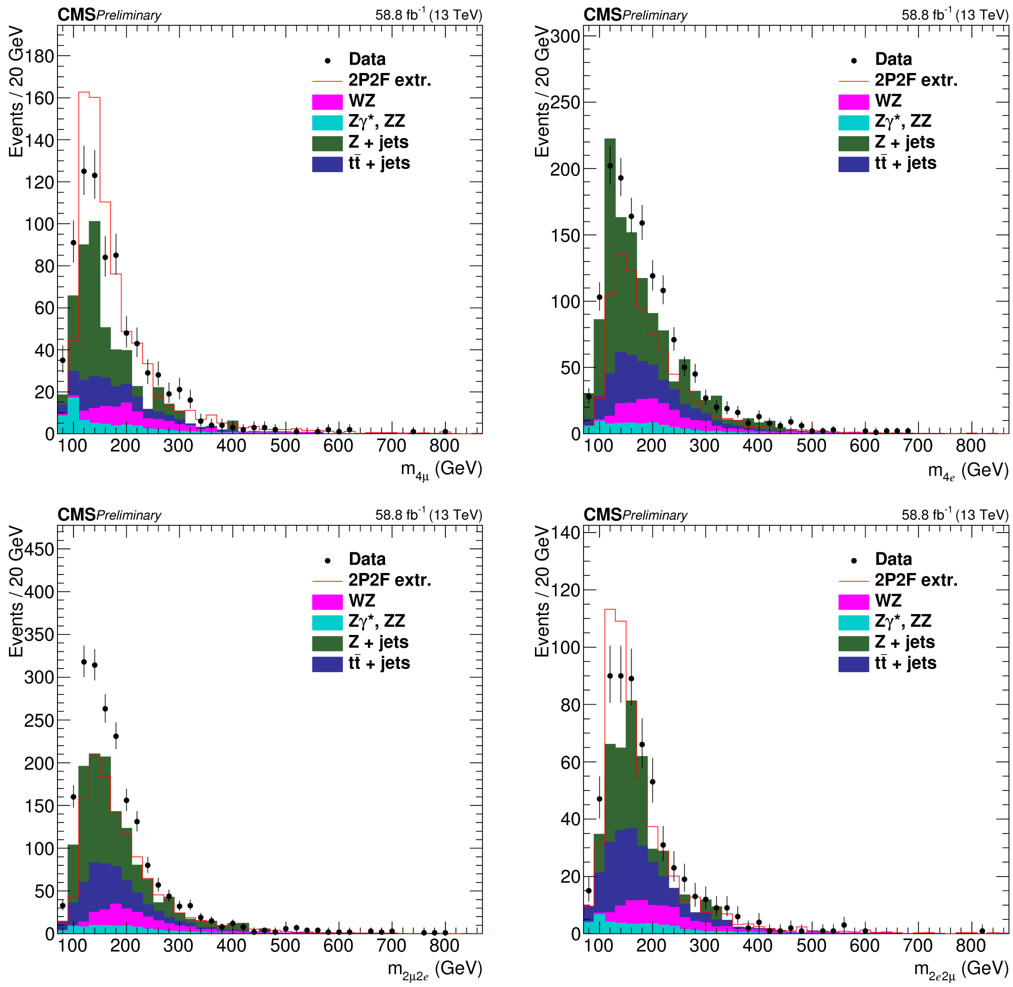


FIGURE 6.13: Invariant mass distribution of the events selected in the 3P+1F control sample in the 13 TeV dataset, (top left) 4μ , (top right) $4e$, (bottom left) $2\mu 2e$ and (bottom right) $2e2\mu$ channels.

for the fact that the fake rates do not properly account for the background composition of the 2P+2F sample. More precisely, the expected reducible background in the signal region is given by the sum of two terms :

- a "2P2F component", obtained from the number of events observed in the 2P+2F control region, $N_{\rm 2P2F}$, by weighting each event in that region with the factor $\frac{f_i}{1-f_i}\frac{f_j}{1-f_j}$, where f_i and f_j correspond to the fake ratios of the two loose leptons.

- a "3P1F component", obtained from the difference between the number of observed events in the 3P+1F control region, $N_{\rm 3P1F}$, and the expected contribution from the 2P+2F region and ZZ processes in the signal region, $N^{\rm ZZ}_{\rm 3P1F} + N^{\rm bkg}_{\rm 3P1F}$. The $N^{\rm bkg}_{\rm 3P1F}$ is given by equation 6.3 and $N^{\rm ZZ}_{\rm 3P1F}$ is the contribution from ZZ which is taken from the simulation. The difference $N_{\rm 3P1F} - N^{\rm bkg}_{\rm 3P1F} - N^{\rm ZZ}_{\rm 3P1F}$, which may be negative, is obtained for each (p_T, η) bin of the "F" lepton, and is weighted by $\frac{f_i}{1-f_i}$, where f_i denotes the fake rate of this lepton. This "3P1F component" accounts for the contribution of reducible background processes with only one fake lepton (like WZ events), and for the contribution of other processes (e.g. photon conversions) that are not properly estimated by the 2P2F component, because of the fake rates used.

Therefore, the full expression for the prediction can be symbolically written as:

$$N^{bkg}_{\rm SR} = \sum \frac{f_i}{(1-f_i)}(N_{\rm 3P1F} - N^{\rm bkg}_{\rm 3P1F} - N^{\rm ZZ}_{\rm 3P1F}) + \sum \frac{f_i}{(1-f_i)}\frac{f_j}{(1-f_j)}N_{\rm 2P2F} \quad (6.4)$$

Previous equation is equivalent to the following:

$$N^{bkg}_{\rm SR} = (1 - \frac{N^{ZZ}_{3P1F}}{N_{3P1F}}) \sum_j^{N_{3P1F}} \frac{f^j_a}{1-f^j_a} - \sum_i^{N_{2P2F}} \frac{f^i_3}{1-f^i_3}\frac{f^i_4}{1-f^i_4} \quad (6.5)$$

For channels where the Z_2 candidate is made from two electrons, the contribution of the 3P1F component is positive, and amounts to typically 30% of the total predicted background.

For channels with loose muons (4μ and $2e2\mu$), the 3P+1F sample is rather well described by the prediction from 2P+2F, as seen in Fig. 6.13b, and the 3P1F component is mainly driven by statistical fluctuations in the 3P+1F sample, which are larger than the expectation from WZ production.

Table 6.4 shows the expected number of events in the signal regions from the reducible background processes for the 13 TeV.

(2) Same Sign (SS) method This method used to predict the reducible background allows to have an inclusive measurement of the all the main reducible backgrounds at the same time.

Channel	4e	4μ	$2e2\mu$	$2\mu 2e$
2016	20.2 ± 6.2	27.0 ± 8.58	25.3 ± 8.0	22.3 ± 6.8
2017	16.1 ± 4.9	33.1 ± 10.4	24.3 ± 7.8	21.3 ± 6.5
2018	25.1 ± 7.6	50.7 ± 15.7	35.0 ± 11.0	32.7 ± 9.9

TABLE 6.4: The contribution of reducible background processes in the signal region predicted from measurements in 2016, 2017 and 2018 data using the OS method. The predictions correspond to 35.9, 41.5 and 59.7 fb^{-1} of data at 13 TeV, respectively.

The control sample is obtained as a subset of the events that satisfy the first step of the selection (*First Z* step, see section 6.4.2), requiring an additional pair of loose leptons of same sign (to avoid signal contamination) and same flavour (SS-SF: $e^{\pm}e^{\pm}, \mu^{\pm}\mu^{\pm}$) . The SS-SF leptons are requested to pass the $\mathrm{SIP_{3D}}$ cut while no identification or isolation requirements are imposed. The reconstructed invariant mass of the SS-SF leptons has to satisfy $m_{\ell\ell} > 12$ GeV, $m_{\ell\ell} < 120$ GeV, the reconstructed four-lepton invariant mass is required to satisfy $m_{4\ell} > 100$, and the QCD suppression cut (see 6.4.2) is applied. The inclusive number of reducible background events in the signal region is derived from this set of events and from the probability for the two additional leptons to pass the isolation and identification analysis cuts, obtained from the fake rate measurement presented in section 6.4.6.

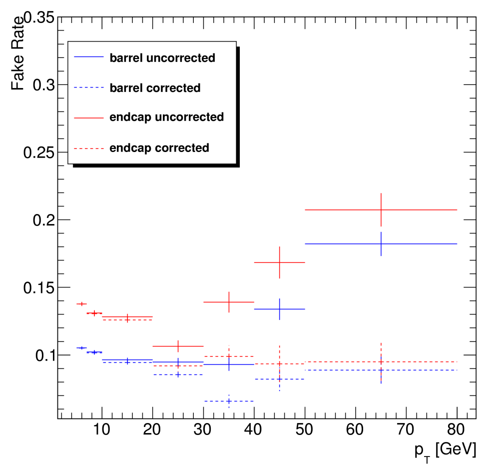
Starting from the control sample previously described, the final reducible background prediction in the signal region is given by the following expression:

$$N^{\mathrm{Z+X}}_{\mathrm{expect}} = \quad N^{\mathrm{DATA}} \times (\tfrac{\mathrm{OS}}{\mathrm{SS}})^{\mathrm{MC}} \times \quad f_1 \times f_2 \tag{6.6}$$

where:

- N^{DATA} is the number of events in the control region,
- $(\frac{\mathrm{OS}}{\mathrm{SS}})^{\mathrm{MC}}$ is a correction factor between opposite sign and same sign control samples,
- f_1 and f_2 are the fake rates of each additional loose lepton, parameterised as a function of p_T and η.

Fake Rate Determination (SS Method) The lepton fake rates are determined in the very similar way as it was described before for the Opposite Sign method. Samples of $Z(\ell\ell) + e$ and $Z(\ell\ell) + \mu$ events are selected in the same way except that the mass window around the nominal Z mass is set to 40 GeV $< M_{inv}(\ell_1, \ell_2) <$ 120 GeV, as in the signal region ("SS phase space"). Also, despite the cut on the missing transverse energy, a contamination of real leptons from WZ events is still visible at high p_T. This contribution is thus subtracted from the final fake rate, using the estimation given by the Monte Carlo. The fake rates for muon are shown in Fig. 6.14 for both muons in the barrel ($|\eta| < 1.2$) and in the endcaps ($|\eta| > 1.2$).



(a)

FIGURE 6.14: Fake rates as a function of the probe p_T for muons which satisfy the loose selection criteria, measured in a $Z(\ell\ell) + \ell$ sample in the 13 TeV data. The barrel selection includes electrons (muons) up to $|\eta|$ = 1.479 (1.2). The Fake rates are shown before (dotted lines) and after (plain line) the reoval of the WZ contribution from the simulation.

For electrons, events where a radiated photon makes an asymmetric conversion, where one low p_T leg is not identified, contribute to the $Z + e$ sample that is used to measure the electron fake rate. While the requirement $|M_{inv}(\ell_1, \ell_2) - M_Z| <$ 7 GeV ("OS phase space") largely suppresses FSR of photons radiated off the lepton legs, these radiations occur at a much larger rate in the "SS phase space". As a result of this enhanced contribution from conversions, the electron fake rates measured within the SR phase space are larger than the "OS fake rates".

However, the relative fraction of FSR conversions is not the same in the "SS phase space" and in the control sample (defined below) where the fake rates will be applied. A correction accounting for this difference must be applied to the fake rates measured within the SS phase space, in order to obtain average fake rates that are appropriate for the control sample. To determine this correction, several fake rate samples of $Z(ll) + e$ events are defined by applying different cuts on $M_{inv}(\ell_1, \ell_2)$ or on $M_{inv}(\ell_1, \ell_2, e)$. In addition to the "OS fake rate sample", where $|M_{inv}(\ell_1, \ell_2) - M_Z| < 7$ GeV ensures a minimal amount of conversions from FSR photons, one defines a sample that is maximally enriched in FSR conversions by $|M_{inv}(\ell_1, \ell_2, e) - M_Z| < 5$ GeV, as well as samples with an intermediate contamination from FSR conversions (60 GeV $< M_{inv}(\ell_1, \ell_2) <$ 120 GeV). In each sample, one determines, in several (p_T, η) bins for the loose electron:

- the fake rate ratio;
- the average value of the expected inner missing hits ($N_{miss.hits}$ in the following), variable useful to tag conversions.

In a given (p_T, η) bin, both the measured fake rate and the average $< N_{miss.hits} >$ are expected to grow linearly with the fraction of conversions. Hence, one expects a linear dependence of the fake rate with respect to $< N_{miss.hits} >$. This linear behaviour is demonstrated in Fig. 6.15, and linear fits are made in each (p_T, η) bin, which relate the fake rate to $< N_{miss.hits} >$.

Finally, one looks at the loose electrons in the control sample where the fake rate will be applied, and one measures, in each (p_T, η) bin, $< N_{miss.hits} >$. The proper average fake rate corresponding to the control sample is then determined from the linear relation derived previously. Figure 6.16 shows examples of the resulting corrected fake rates, together with the fake rates measured in the SS phase space. It can be seen that the correction for the actual fraction of conversions that is present in the control sample lowers the fake rates considerably with respect to what is measured in the SS phase space. The determination of these corrected fake rates mostly suffers from

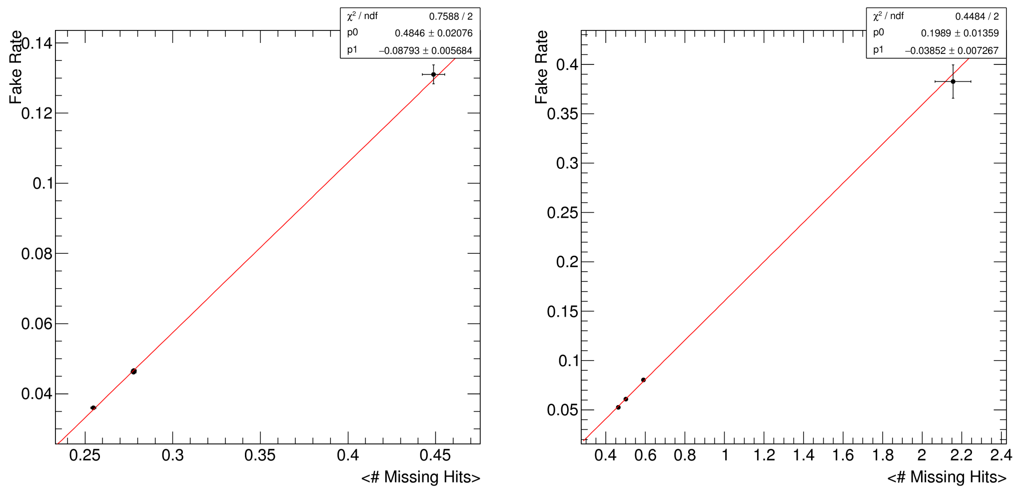


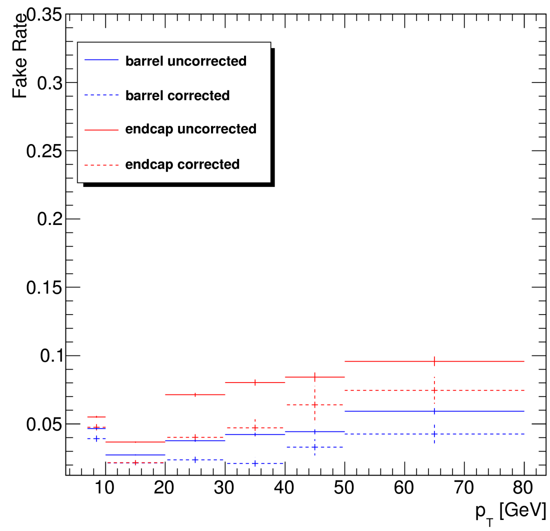
FIGURE 6.15: Examples of the correlation between the fake rate and the fraction of loose electrons for which the track has one missing hit in the pixel detector for $7 < p_T < 10$ GeV electrons in the barrel (left) and for $20 < p_T < 30$ GeV electrons in the endcaps (right). Each dot shows the measurements made in a given Z + loose e sample. Results are produced at 13 TeV with 36.8 fb^{-1} data.

the limited statistics of the control sample, which translates into a large uncertainty on $< N_{miss.hits} >$, the error on each dot being the fake rate obtained from the linear relations using the error on the $< N_{miss.hits} >$.

Fake Rate Application (SS Method) Figure 6.17 shows the invariant mass distribution of events in control samples with SS-SF loose leptons, for channels with loose electrons ($4e$ and $2\mu 2e$) and channels with loose muons (4μ and $2e2\mu$), for the 13 TeV dataset. The prediction from the Monte-Carlo simulation is also shown.

A good agreement is achieved between data and MC for the channels with loose electrons. The agreement is not as good for loose muons but this has no impact on the final result as only data are used in the end.

The differences in rates between OS and SS samples are used to compute the correction factor in equation 6.6 for the final data-driven estimation. They are given in table 6.5.



(a)

FIGURE 6.16: Average fake rates to be applied to the control sample of SS Method (plain line), compared to the fake rates measured in the SS phase space (dotted line), for electrons in the barrel (blue) and in the endcaps (red). Results are produced at 13 TeV with 2018 data.

Channel $\rightarrow$	4e	2μ2e	4μ	2e2μ
2016	1.00 ± 0.01	1.00 ± 0.03	1.03 ± 0.03	1.00 ± 0.01
2017	1.01 ± 0.01	1.04 ± 0.03	1.01 ± 0.03	1.00 ± 0.01
2018	1.01 ± 0.01	1.00 ± 0.01	1.03 ± 0.02	1.03 ± 0.03

TABLE 6.5: The OS/SS ratios used to estimate the number of ZX events with the SS method for each final state in all three years.

Combination of the Z+X background estimtaions Two methods for the reducible background estimation are presented with associated uncertainties in the previous sections. Table 6.6 shows the summary of the results of the methods. The shape systematics is absorbed in the statistical and systematic uncertainties on the estimated yields, and the total uncertainties are found to be of the order of $35-45\%$. Predictions of the two methods are combined assuming no correlation between the uncertainties (independent control regions, partially different sources systematics), and combined mean values are obtained by weighting the individual means according to the corresponding variances.

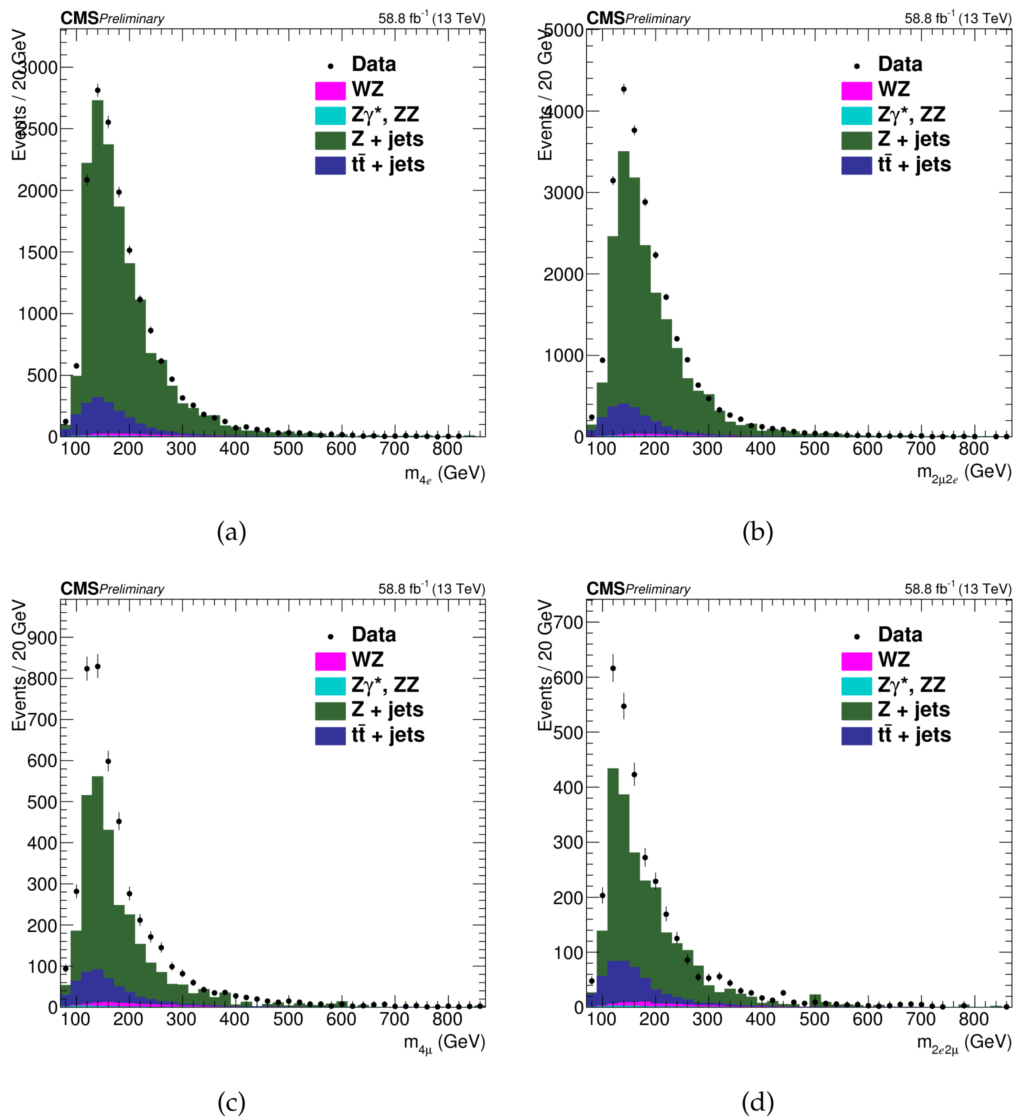


FIGURE 6.17: DATA-MC comparison of the SS-SF samples in the Z+X background control samples. (a) and (b) $4e$ and $2\mu 2e$, (c) and (d) 4μ and $2e2\mu$ in 2018 data.

Systematic Uncrertainties on Reducible Background Source of systematic uncertainty of the methods presented above potentially arises from the different composition of background processes (DY, $t\bar{t}$, WZ, $Z\gamma$) in the region where FRs are measured and are applied. OS method corrects for the resulting bias via the “3P1F component" of its prediction. SS method corrects explicitly the electron fake rates for the correct fraction of photon conversions. The closure tests presented here are used to assess a possible residual bias in the methods.

2016	4e	4μ	$2e2\mu$
Method OS	$20.2 \pm 6.2_{\text{tot.}}$	$27.0 \pm 8.6_{\text{tot.}}$	$47.7 \pm 10.5_{\text{tot.}}$
Method SS	$13.1 \pm 5.5_{\text{tot.}}$	$29.6 \pm 9.0_{\text{tot.}}$	$41.5 \pm 10.3_{\text{tot.}}$
Combined	16.2	28.2	44.5
2017	4e	4μ	$2e2\mu$
Method OS	$16.1 \pm 4.9_{\text{tot.}}$	$33.1 \pm 10.4_{\text{tot.}}$	$45.7 \pm 10.2_{\text{tot.}}$
Method SS	$10.9 \pm 4.1_{\text{tot.}}$	$33.4 \pm 10.3_{\text{tot.}}$	$40.8 \pm 9.7_{\text{tot.}}$
Combined	13.0	33.3	43.2
2018	4e	4μ	$2e2\mu$
Method OS	$25.4 \pm 7.7_{\text{tot.}}$	$50.1 \pm 15.5_{\text{tot.}}$	$67.7 \pm 14.8_{\text{tot.}}$
Method SS	$16.1 \pm 5.9_{\text{tot.}}$	$51.9 \pm 15.8_{\text{tot.}}$	$60.7 \pm 14.2_{\text{tot.}}$
Combined	19.4	51.3	64.1

TABLE 6.6: Summary of the results given by the two methods for the prediction of the contribution of reducible background processes in the signal region for all three years. The weighted mean value of the results of the two methods is taken as the final estimate of this contribution, while the uncertainty of the result covers the uncertainties of both methods. The table shows symmetric individual uncertainties for the two methods, while these are treated as asymmetric in the combination and statistical analysis. Results are given for an integrated luminosity of 35.9, 41.5 and 59.7 fb^{-1} in 2016, 2017 and 2018 data, respectively.

The limited size of the samples in the control regions where FRS are measured and where those are appleid is the source of the statistical uncertainties of the method. The dominating statistical uncertainty is driven by the number of events in the control region and is typically in the range of 1-10%.

Compositions of reducible background processes (DY, $t\bar{t}$, WZ, $Z\gamma^{(*)}$) in the region where FRS are measured and where those are appleid are typically not the same. This is the main source of the systematic uncertainty of the fake ratio method.

This uncertainty can be estimated by measuring the FRs for individual background processes in the $Z+1L$ region in simulation. The weighted average of these individual fake ratios is the fake ratio which is measured in this sample (in simulation). The exact composition of the background processes in the 2P+2F region where FRS are to be applied can be determined from simulation, and one can reweigh the individual FRs according to the 2P+2F composition. The difference between the reweighed fake ratio and the average one can be used as a measure of the uncertainty on the measurement of the fake ratios. The effect of this systematic uncertainty is propagated to the final estimates, and it amounts to about 32% for $4e$, 33% for $2e2\mu$ and 35% for 4μ final state.

In order to estimate the uncertainty on the m_{4l} shape, the differences between the shapes of predicted background distributions for all three channels, and between both of the two methods are used. The envelope of differences between these shapes of distributions is used as an estimate of the shape uncertainty. The uncertainty is estimated to be roughly in the range of 5% - 15%. Since the difference of the shapes slowly varies with m_{4l}, it is taken as a constant versus m_{4l} and is absorbed in the much larger uncertainty on the predicted yield of backgrounds.

For differentiall measurements in $\mathrm{H} \to 4\ell$, shape of reducible background is attained from templates built from the control regions in the data for each differential bin of the observable considered for measurement. This procedure is similar to what was used for spin-parity studies as described in the Ref. [69]. More details on this procedure of differential measurements of $\mathrm{H} \to 4\ell$ will be given in Section 6.4.8.

The number of estimated signal and the background events, along with observed events in data passing all selections in mass range $m_{4\ell} > 70$ GeV are listed in Table 6.7 for full Run 2 datasets.

Reconstructed four-leptons invariant mass distribution for whole Run 2 datasets and MC samples in full and high mass range is shownn in the Figure 6.18.

6.4.7 Statistical Procedure

After the procedure of events seletions, signal and background modeling as needed ingredients, next step is to perform final measurements. An asymptotic approach as described in Ref. [85] with test statistics that is based on profile likelihood ratio [86] is used for this purpose. Maximization of likelihood is performed by fitting simultaneously background and signal parametrization to the observed $m_{4\ell}$ distribution, at the fiducial level with all reconstruction bins and all three final states. The parameter of interest (σ_{fid}) is directly extracted from the fit.

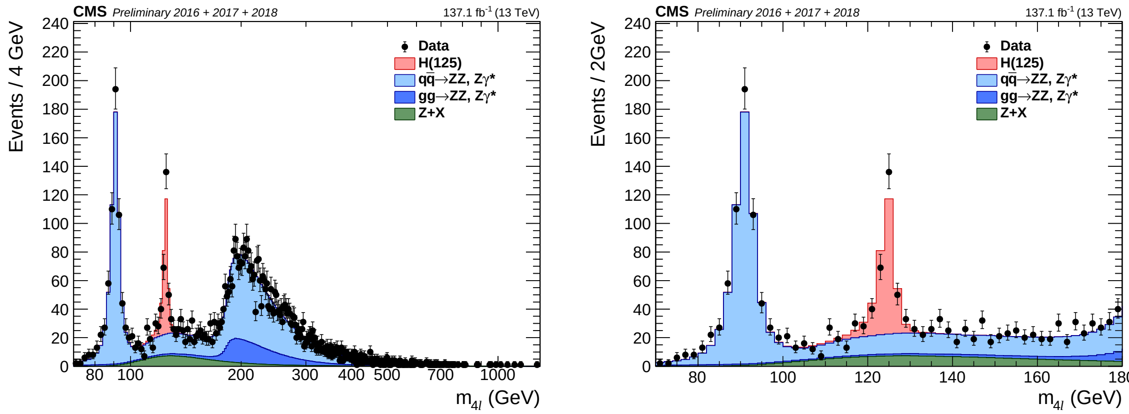


FIGURE 6.18: Invariant mass distributions of invariant mass of four-leptons using full Run 2 datasets in the full (left) and the low (right) mass range, along with predictions for Standard Model Higgs ($m_H = 125.0$ GeV), including $Z \to 4\ell$ resonance.

The estimated number events for a studied differential observable in a final state f and in the defined bin i are described in terms of $m_{4\ell}$ given by:

$$\begin{aligned} N^{\mathrm{f},i}_{\mathrm{obs}}(m_{4\ell}) &= N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell}) \\ &= \sum_j \epsilon^{\mathrm{f}}_{i,j} \cdot \left(1 + f^{\mathrm{f},i}_{\mathrm{nonfid}}\right) \cdot \sigma^{\mathrm{f},j}_{\mathrm{fid}} \cdot \mathcal{L} \cdot \mathcal{P}_{\mathrm{res}}(m_{4\ell}) \\ &\quad + N^{\mathrm{f},i}_{\mathrm{comb}} \cdot \mathcal{P}_{\mathrm{comb}}(m_{4\ell}) + N^{\mathrm{f},i}_{\mathrm{bkg}} \cdot \mathcal{P}_{\mathrm{bkg}}(m_{4\ell}). \end{aligned} \tag{6.7}$$

Where $N^{\mathrm{f},i}_{\mathrm{fid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{nonfid}}(m_{4\ell})$, $N^{\mathrm{f},i}_{\mathrm{comb}}(m_{4\ell})$ and $N^{\mathrm{f},i}_{\mathrm{bkg}}(m_{4\ell})$ denote the resonant fiducial signal, the nonfiducial resonant, the combinatorial contribution of fiducial signal, and background contributions in a defined bin i as functions of $m_{4\ell}$ (four lepton invariant mass), respectively. Then, probability density functions corresponding to the resonant (fiducial and the nonfiducial) signal, the combinatorial contributions of fiducial signal, and contribution from the background are $\mathcal{P}_{\mathrm{res}}(m_{4\ell})$, $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$ and $\mathcal{P}_{\mathrm{bkg}}(m_{4\ell})$ respectively. $\mathcal{P}_{\mathrm{res}}(m_{4\ell})$ is described by double sided Crystal Ball function details of which are described in Section 6.4.5.

$\epsilon^{\mathrm{f}}_{i,j}$ is the response matrix that connects reconstructed level events in a defined bin i with fiducial events in the bin j for studied differential observable. It is the probability of an event that was generated in ith bin at the fiducial level and is reconstructed in j-th bin. This is estimated through *unfolding procedure* for observables' bins at the fiducial level, corrected for the detection

efficiency and resolution effects. The procedure is discussed in Ref. [87] and similar to what is adopted for $H \rightarrow \gamma\gamma$ decay channel [88].
Figure 6.21 shows the efficiency matrix for $p_T(H)$ in $2e2\mu$ final state measured using different MC samples for SM production of Higgs. The efficiency matrix is dominated by diagonal elements. In case of jet related observables, off-diagonal contribution, however, is non-negligible. It is due to poor resolution offered by the detector to jets. In case of differential variable, jet multiplicity, neighbour elements of diagonal varies from 3% to 21% as shown in

TABLE 6.7: Number of observed candidates, expected background and the signal events passing all selections in mass range $m_{4\ell} > 70$ GeV corresponding to three different years. Z+X is estimated by control regions in the data while ZZ and signal events are estimated using MC simulation.

Channel	4e	4μ	2e2μ	4ℓ
2016 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	$192.7^{+18.6}_{-20.1}$	$360.2^{+24.9}_{-27.3}$	$471.0^{+32.6}_{-35.7}$	$1023.9^{+68.9}_{-76.0}$
$gg \rightarrow ZZ$	$41.2^{+6.3}_{-6.1}$	$69.0^{+9.5}_{-9.0}$	$101.7^{+14.0}_{-13.3}$	$211.8^{+28.9}_{-27.5}$
Z + X	$21.1^{+8.5}_{-10.4}$	$34.4^{+14.5}_{-13.2}$	$59.9^{+27.1}_{-25.0}$	$115.4^{+31.9}_{-30.1}$
Sum of backgrounds	$255.0^{+23.9}_{-25.1}$	$463.5^{+31.9}_{-33.7}$	$632.6^{+44.2}_{-46.1}$	$1351.1^{+85.8}_{-91.2}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$12.1^{+1.3}_{-1.4}$	23.8 ± 2.1	30.3 ± 2.6	66.3 ± 5.6
Total expected	$267.1^{+24.9}_{-26.1}$	$487.4^{+33.1}_{-34.9}$	$662.9^{+45.7}_{-47.5}$	$1417.4^{+89.1}_{-94.3}$
Observed	293	505	681	1479
2017 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	233.86^{+32}_{-36}	441.08^{+35}_{-40}	569.51^{+50}_{-54}	1244.45^{+104}_{-114}
$gg \rightarrow ZZ$	$49.11^{+8.7}_{-8.8}$	$81.50^{+11.1}_{-10.7}$	$121.05^{+17.0}_{-16.0}$	$251.66^{+35}_{-33.4}$
Z + X	$17.1^{+6.4}_{-6.1}$	$33.3^{+11.9}_{-11.4}$	$46.9^{+16.1}_{-15.6}$	$97.3^{+21.0}_{-20.2}$
Sum of backgrounds	300^{+39}_{-43}	556^{+43}_{-47}	738^{+62}_{-65}	1594^{+125}_{-134}
Signal ($m_{\mathrm{H}} = 125$ GeV)	$13.85^{+1.9}_{-2.1}$	$28.64^{+2.5}_{-2.4}$	$35.55^{+3.3}_{-3.3}$	$78.1^{+6.9}_{-7.0}$
Total expected	314^{+41}_{-45}	585^{+45}_{-49}	773^{+64}_{-67}	1672^{+130}_{-139}
Observed	307	602	797	1706
2018 (13 TeV)				
$q\bar{q} \rightarrow ZZ$	333^{+57}_{-53}	622^{+31}_{-44}	815 ± 73	1770^{+98}_{-101}
$gg \rightarrow ZZ$	$75.1^{+14.3}_{-13.5}$	$116.6^{+11.7}_{-12.8}$	176.9 ± 23.0	$368.5^{+29.5}_{-29.6}$
Z + X	19.3 ± 7.2	50.8 ± 15.2	64.6 ± 15.6	134.7 ± 22.9
Sum of backgrounds	$428^{+59.2}_{55.2}$	$790^{+36.4}_{-48.3}$	1057 ± 78.1	$2274^{+104.9}_{-107.7}$
Signal ($m_{\mathrm{H}} = 125$ GeV)	$19.6^{+3.3}_{-3.1}$	$40.8^{+2.5}_{-2.9}$	50.7 ± 5.6	$111.1^{+6.9}_{-7.0}$
Total expected	$447^{+59.3}_{55.2}$	$830^{+36.5}_{-48.4}$	1108 ± 78.3	$2385^{+105.1}_{-107.9}$
Observed	462	850	1130	2442

Figures **??-??** in the Appendix **??**. Regardless of particular differential bin, for jet multiplicity observable, the off-diagonal contributions are 3-8%, 3-20% and 2-14%; however, for a high resolution observable, e.g. $p_T(H)$, these are 1-5%, 1-5% and 1-4% for 2e2mu, 4mu and 4e channels respectively in the gluon gluon fusion production mode.

The f^{i}_{nonfid} fraction represents the ratio of nonfiducial signal to the fiducial signal contributions in a defined bin i at reconstruction level. It is the ratio of reconstructed events which are from outside the fiducial volume and reconstructed events which are from within the fiducial volume. Such events arise from imperfect detector resolution which causes the differences for quantities calculated at generated and reconstructed levels. Fraction of these events are also treated as background and will be referred "non-fiducial signal" in the disseration. It has been demonstrated with MC samples that these events can be described using same shape as resonant component. The values of such fraction are extracted from simulation repeatedly for SM. This fraction varies among different production modes of Higgs e.g. it is 4% for the $\mathrm{gg} \to \mathrm{H}$ and 3% for ttH production mode. The values for each model are listed in the Tables 6.9 2018.

$\sigma^{\mathrm{f},j}_{\mathrm{fid}}$ is the parameter of interest that represents signal cross section for a final state f in a defined bin j of fiducial region.

For associate production of Higgs, there is quite a possibility that one reconstructed lepton is not from $\mathrm{H} \to 4\ell$ decay process and orginates associated particle. Fraction of these events, denoted by $\mathcal{P}_{\mathrm{comb}}(m_{4\ell})$, vary with the production modes (i.e. WH, ZH and $t\bar{t}\mathrm{H}$) and are treated as background. They make broad invariant $m_{4\ell}$ mass distribution whose shape is described by a Landau distribution with parameters extracted from simulations.

Main goal is to measure fiducial cross sections after corrections of detector effects, detector resolution and systematic biases. These corrections are extracted with the help of simulated samples and then encoded in the parametrization of $m_{4\ell}$ spectra at the reconstruction level. All the systematic uncertainties in the measurements are encoded as nuisance parameters, which are profited properly in fit procedure.

6.4.8 Extraction of $\mathrm{H} \to 4\ell$ Fiducial Cross Section

Inclusive or Integrated Fiducial Cross section

From measurement point of view, the integrated or inclusive fiducial cross section is a special case of the general procedure adopted for fiducial cross section measurements as explained in previous Section 6.4.7 i.e. the number of considered bins is one.

Informations of different ingredient in Equation 6.7 for different production modes of Higgs are included in the form of tables. Reconstruction efficiency (ϵ) for the events reconstructed in fiducial volume is given in the Table 6.8 for each final state and as well as inclusive for all final states. It is obvious from the table that reconstruction efficiency varies with the production model. Also, in 4μ final state, efficiency is pretty higher that other states which is due to high detector resolution of CMS for muons. Table 6.9 shows the ratio (f_{nonfid}) of events reconstructed from out of fiducial region and the events reconstructed within fiducial volume, per final state. It is evident from the table, f_{nonfid} is higher for $t\bar{t}$Hproduction mode which is due to the contamination from hadronic activities. Source of the "nonfiducial signal" contributions i.e. f_{nonfid} is the imperfect correspondence of isolation at particle and reconstruction levels, as discussed in previous Section and in Section 6.4. Fraction of the signal events in mass range 105.0–140.0 GeV where at least one selected lepton does not originate from the Higgs boson decay are shown in Table 6.10. Fraction of these events in higher in case of ZH and $t\bar{t}$H production modes.

TABLE 6.8: Reconstruction efficiency (ϵ) for fiducial events per final state for different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 MC samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.404 ± 0.002	0.799 ± 0.002	0.572 ± 0.002	0.593 ± 0.001
VBF (POWHEG)	0.431 ± 0.003	0.805 ± 0.002	0.586 ± 0.002	0.608 ± 0.002
WH (POWHEG+MINLO)	0.415 ± 0.004	0.780 ± 0.003	0.580 ± 0.003	0.593 ± 0.002
ZH (POWHEG+MINLO)	0.428 ± 0.006	0.780 ± 0.005	0.581 ± 0.005	0.597 ± 0.003
ttH (POWHEG)	0.429 ± 0.006	0.745 ± 0.005	0.573 ± 0.005	0.585 ± 0.003

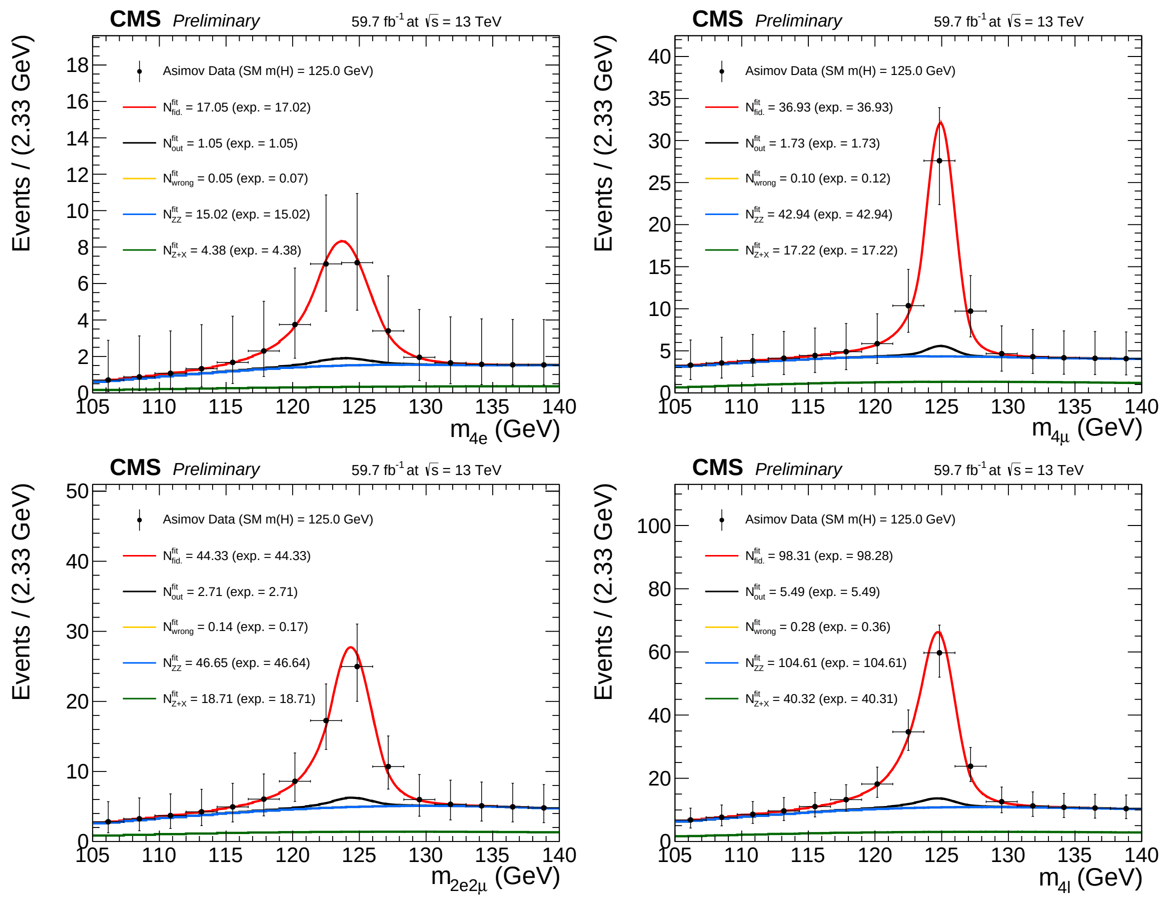


FIGURE 6.19: Inclusive four lepton mass distributions for an Asimov Dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of the signal and background pdfs in different final states at $\sqrt{s} = 13$ TeV with 2018 data.

TABLE 6.9: Ratio of events reconstructed from out of fiducial volume and the events reconstructed within the fiducial region (f_{nonfid}) per final state in different Standard Model signal models at $\sqrt{s} = 13$ TeV with 2018 MC samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.060 ± 0.002	0.046 ± 0.001	0.060 ± 0.001	0.055 ± 0.001
VBF (POWHEG)	0.050 ± 0.002	0.037 ± 0.001	0.046 ± 0.001	0.043 ± 0.001
WH (POWHEG+MINLO)	0.098 ± 0.004	0.058 ± 0.002	0.080 ± 0.002	0.075 ± 0.001
ZH (POWHEG+MINLO)	0.105 ± 0.007	0.071 ± 0.004	0.091 ± 0.004	0.087 ± 0.002
ttH (POWHEG)	0.278 ± 0.012	0.127 ± 0.005	0.193 ± 0.006	0.185 ± 0.004

TABLE 6.10: Signal event fraction in the mass range 105.0–140.0 GeV where minimum one lepton selected is originating from Higgs decay at $\sqrt{s} = 13$ TeV with 2018 MC samples.

Sample	$4e$	4μ	$2e2\mu$	4ℓ
gg→H (POWHEG)	0.002	0.002	0.002	0.002
VBF (POWHEG)	0.002	0.003	0.003	0.003
WH (POWHEG+MINLO)	0.036	0.031	0.042	0.037
ZH (POWHEG+MINLO)	0.199	0.207	0.212	0.208
ttH (POWHEG)	0.175	0.156	0.164	0.163

Differential Fiducial Cross section

In this dissertation, the kinematic properties of four-lepton system are exploited for Higgs differential cross section measurements and the results are reported for transverse momentum and rapidity of Higgs or the four lepton system($p_T(\mathrm{H})$ or $p_T(4\ell)$, $|y(\mathrm{H})|$ or $|y(4\ell)|$), N(jets), and $p_T(\mathrm{jet})$ which probe the production mechanism. For each kinematic or differential observable, selection procedure of the binning boundaries has been optimized in order to have a similar expected measured cross-section for every bin of the observable.

For the differential cross section measurements, the background is to be subtracted at the reconstruction level for each bin of the observable. The mass spectrum for the background in not same all differential bins because it does not have a resonance structure. The mass spectrum can also have non-negligible correlation with the observables.

For the irreducible backgrounds, the four-lepton mass spectra are modeled as the $m_{4\ell}$ template shapes through simulated events for each bin of the observable for differential measurements. However, for the reducible backgrounds the templates are built from the control regions in the data in the same way. The procedure is described in Ref. [69].

In each differential bin of $p_T(4\ell)$ fractions of background events are presented in the Table 6.11.

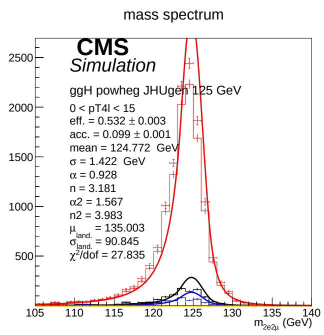
TABLE 6.11: Estimated fraction of background events in each bin of $p_T(4\ell)$, for each final state using 2018 samples.

Background	final state	$0-15$	$15-30$	$30-45$	$45-80$	$80-120$	$120-200$	$200-13000$
qqZZ	$2e2\mu$	0.483668	0.239196	0.130151	0.105276	0.0266332	0.011809	0.00326633
qqZZ	$4e$	0.459665	0.234399	0.136986	0.114916	0.0388128	0.00989346	0.00532725
qqZZ	4μ	0.492632	0.241203	0.110977	0.109173	0.0297744	0.0129323	0.00330827
ggZZ	$2e2\mu$	0.310446	0.309859	0.191901	0.168134	0.0187793	0.000880282	0.0
ggZZ	$4e$	0.3085	0.306332	0.19247	0.17433	0.0169994	0.00136908	0.0
ggZZ	4μ	0.321332	0.319311	0.194863	0.150987	0.0128703	0.000638196	0.0
Z+X (CR)	4l	0.0992521	0.211153	0.209307	0.305327	0.120303	0.0483797	0.00627828

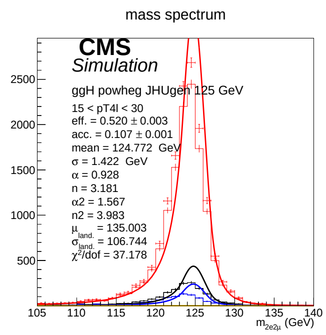
The fiducial differential cross section is extracted in the bins of kinematic observables at the fiducial level following the procedure outlined in Section 6.4.7. For signal yields, mass spectrum of signal is built in each cell of the response matrix using a Double Crytal Ball (DCB) function, similar to what was done in Ref. [69]. In practice, the same parameters of the CB function are used in each cell and the systematic uncertainties assigned on the energy scale and resolution generally cover discrepancy of signal mass spectrum among each cell.

Examples of the fitted signal line shapes for differential bins of $p_T(\mathrm{H})$ from gg→H (POWHEG+JHUGEN 125 GeV 2018 sample) in the individual final states are shown in Figure 6.20, and examples of the efficiency matrices for $p_T(\mathrm{H})$ for the $2e2\mu$ final state are shown in Fig 6.21. Many more examples can be seen in Appendix **??**.

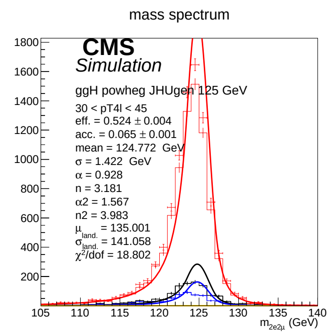
Figure 6.22 shows the four lepton mass distributions for an Asimov dataset generated using SM cross section values and efficiencies from the gg→H production mode from POWHEG+JHUGEN, and resulting fitted values of PDFs of signal and background for different bins of $p_T(\mathrm{H})$ (all final states combined).



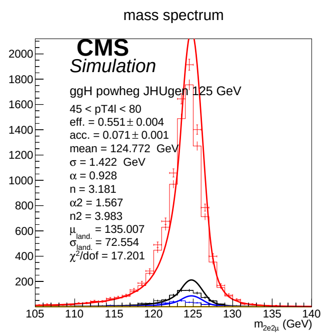
(a) $0.0 < p_T(\mathrm{H}) < 15.0$



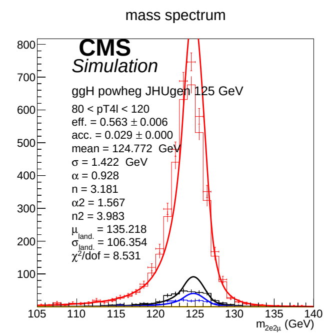
(b) $15.0 < p_T(\mathrm{H}) < 30.0$



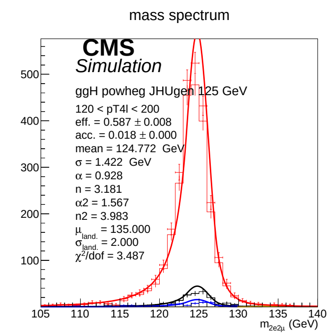
(c) $30.0 < p_T(\mathrm{H}) < 45.0$



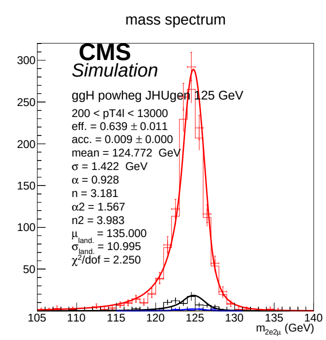
(d) $45.0 < p_T(\mathrm{H}) < 80.0$



(e) $80.0 < p_T(\mathrm{H}) < 120.0$

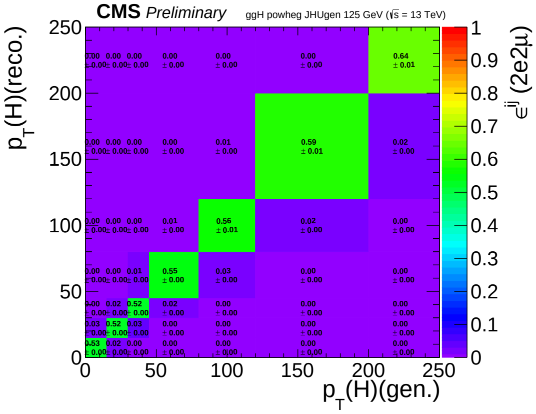


(f) $120.0 < p_T(\mathrm{H}) < 200.0$

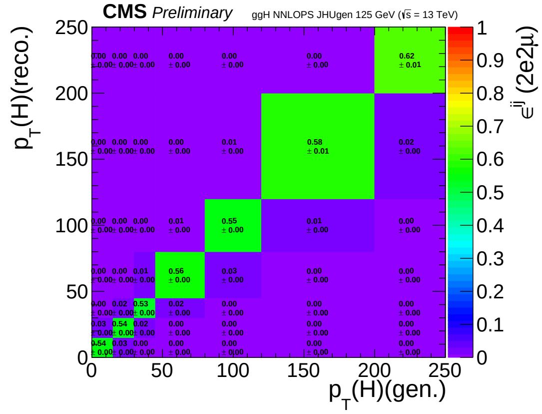


(g) $200.0 < p_T(\mathrm{H}) < 13000.0$

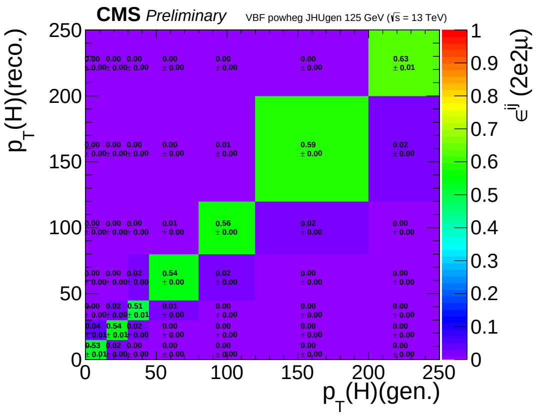
FIGURE 6.20: Example signal shapes at reconstruction level for a resonance of m(4ℓ) in $2e2\mu$ final state for the $gg \to$ H production mode from POWHEG+JHUGEN in different bins of $p_T(\mathrm{H})$. The black curve represents events which do not pass the fiducial volume selection. The curve has no effect on the result.



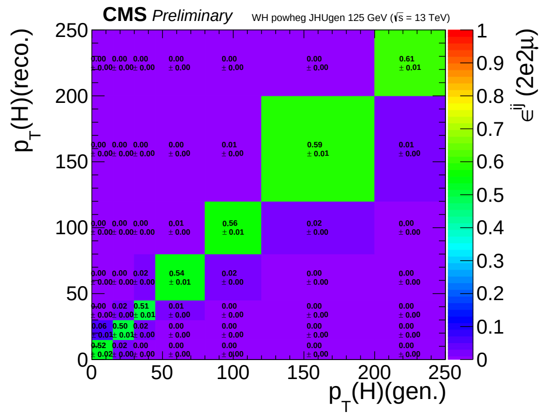
(a) $gg \to$H (POWHEG+JHUGEN)



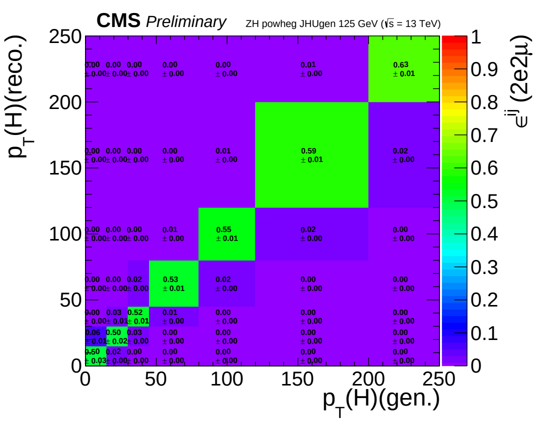
(b) $gg \to$H (MINLOHJJ)



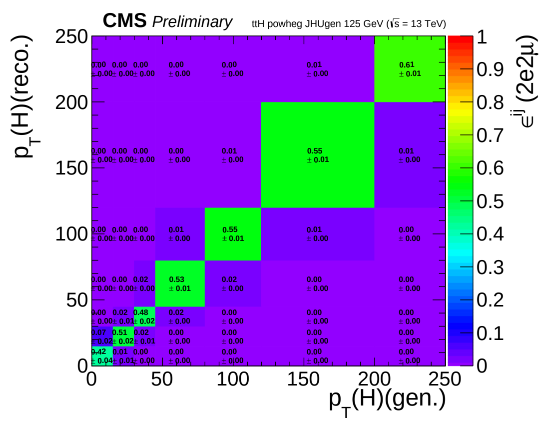
(c) VBF (POWHEG)



(d) WH (PYTHIA)



(e) ZH (PYTHIA)



(f) ttH (PYTHIA)

FIGURE 6.21: Efficiency response matrices for p_T(H) for different SM production modes in the $2e2\mu$ final state using 2018 samples.

6.5 Systematic Uncertainties

6.5.1 Experimental uncertainties souces

Main experimental uncertainties affecting signal and background are uncertainty on integrated luminosity (from 2.3% to 2.6%, depending on the year of data taking) and the uncertainty on efficiencies of lepton reconstruction and identification (ranging from 2.5% (4μ)–16.1% ($4e$)). Uncertainties on estimation of reducible backround, described in Section 6.4.6, originating from the background composition and misidentification rate uncertainty vary for different final states. Mismatch in composition of backgrounds between different samples where the fake rate is derived and where it is applied accounts between 24%(4μ) and 43%($4e$).

Lepton energy scale uncertainty is assessed by propagating down to the four-leptons invariant mass the uncertainty associated the lepton momentum corrections on each indiviudal lepton. It is estimated by exploiting $Z \to \ell\ell$ invariant mass distributions both in data and the simulation. Events are categorized on the basis p_T and η of either (randomly determined) of two leptons and integrating on the other lepton. The dilepton mass distributions are then fit by parameterization of a Breit-Wigner convoluted with a DCB function. The uncertainty is treated as uncorrelated (all up or all down). The new four-leptons invariant mass obtained this way is then fitted with only the mean value floating. Difference between new mean value and default one gives an estimation of impact of lepton energy scale. Values of the differences are reported in the Table 6.12, together with estimation of the impact of lepton momentum resolution, following a similar procedure.

TABLE 6.12: Difference (in %) between the nominal mean and the width of four-leptons or Higgs invariant mass distribution and those where the lepton momentum scale and resolution uncertainties have been propagated.

	4e	4μ	$2e2\mu$
Scale Difference (%)	0.087	0.049	0.069
Resolution Difference (%)	17	12	17

The results obtained this way suggest that the uncertainties from 2016 (20% on the resolution, 0.3% for the 4e channel for scale) are still valid.

6.5.2 Theoretical uncertainties

Estimation of the theoretical uncertainties affecting signal and the background include uncertainties from factorization and renormalization scale is done by varying the two scales 0.5-2 times the nominal values with their ratio kept in between 0.5-2. In the case of ggF production mode, the QCD scale uncertainty is broke down to 9 different sources. Other theoretical uncertainty source is PDF set choice which is determined RMS of the variation when different replias of default NNPDF set is used. Additional uncertainty on the background which is 10% on K factor applied on $ggZZ$ prediction is considered as described in Section 6.4.6. Systematic uncertainty of about 2% on the branching ratio of $\mathrm{H} \rightarrow 4\ell$ is only applicable to signal yield. Additional sources are the imprecise jet energy scale and resolution knowledge (from 2% to 22%), the efficiency on b-tagging and the light quarks ($i.e.$ u, d, s, c) and gluon jet mistag rate. While cross section measurement, signal cross section uncertainties arising from theoretical sources are removed.

TABLE 6.13: Experimental systematic uncertainties in $\mathrm{H} \rightarrow 4\ell$ measurements of 2016, 2017 and 2018 data.

Summary of relative systematic uncertainties			
Common experimental uncertainties			
	2016	2017	2018
Luminosity	2.6 %	2.3 %	2.5 %
Lepton selection efficiencies	2.5 – 9 %	3 – 12.5 %	0.7 – 11 %
Uncertainties related uncertainties			
Reducible background (Z+X)	36 – 43 %	6 – 32 %	31 –37 %
Uncertainties related to signal			
Lepton energy scale	0.04 – 0.3 %	0.04 – 0.3 %	0.04 – 0.3 %
Lepton energy resolution	20 %	20 %	20 %

6.5.3 Systematic uncertainties treatment when combining yearly measurements on data

In combination of three data taking periods, theoretical uncertainties as well as the experimental ones related to leptons or jets are treated as correlated while all other ones from experimental sources are taken as uncorrelated.

TABLE 6.14: Table theory uncertainties in the H $\to 4\ell$ Inclusive cross section measurements

Summary of inclusive theory uncertainties	
BR($H \to$ ZZ $\to 4\ell$)	2 %
QCD scale (qq $\to$ ZZ)	+3.2/-4.2 % %
PDF set (qq $\to$ ZZ)	+3.1/-3.4 %
Electroweak corrections (qq $\to$ ZZ)	$\pm$ 0.1 %

6.6 Results

6.6.1 Measurement of Integrated Fiducial Cross Section

Resulting fits to the inclusive m(4ℓ) distributions (extracted in $m_{4\ell}$ range viz 105 GeV to 140 GeV) for measurement of integrated H $\to 4\ell$ fiducial cross section in different final states, are shown in Figure 6.24. Extracted inclusive cross sections as function of colliding center of mass energy has been shown in the Figure 6.25(a) while the extracted integrated cross section results for different final states at 13 TeV corresponding to different years of data taking are shown in Figure 6.25(b).

Systematic on the measurement is represented by red bar, whereas black error bar represents total uncertainties, summed from the individuals in quadrature. Contributions of the sub-dominant component of the signal i.e. XH $=$ VBF $+$ VH $+$ $t\bar{t}$H is indicated in green separately. These results are also given in Tables 6.15 and 6.16.

The inclusive H $\to 4\ell$ measurements are compared with SM NNLO theoretical estimations where acceptance of dominant gg $\to$ H contribution is estimated using POWHEG at $\sqrt{s}$=13 TeVand HRES [82, 89] at $\sqrt{s}$=7 and 8 TeV. Total gluon fusion cross section and associated uncertainty has been taken from Ref. [90]. Definition of fiducial volume for $\sqrt{s}$=6–9 TeVuses the lepton isolation definition from Ref. [91], while for $\sqrt{s}$=12–14 TeVthe definition described in the text is used.

When extracting the integrated H$\to 4\ell$ fiducial cross section, the parameter of interest is total fiducial cross section and the fractions of $4e, 4\mu$, and $2e2\mu$ is floated in the fit. When extracting the fiducial cross section in the final states, three parameters of interest are the cross sections of three individual final states.

TABLE 6.15: Measured $\mathrm{H} \to 4\ell$ Inclusive fiducial cross section in $m_{4\ell}$ ranging from 105 GeV to 140 GeV in proton collisions at 13 TeV.

Fiducial cross section $\mathrm{H} \to 4\ell$ **at 13 TeV**	
Measured	$2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})^{+0.24}_{-0.19}(\mathrm{syst.})$ fb (model)
gg→H(POWHEG) + XH	$2.77^{+0.51}_{-0.39}$fb
gg→H(NNLOPS) + XH	$2.76^{+0.14}_{-0.14}$fb

Measured integrated $\mathrm{H} \to 4\ell$ fiducial cross section using 13 TeV data is in nice agreement with theoretical predictions within corresponding uncertainties. Dominant and sub-dominant uncertainties on the measurement are systematic and statistical uncertainties which are about 24% and 23% respectively. The theoretical uncertainty is about 11%.

TABLE 6.16: Measured H→ $4l$ integrated or inclusive fiducial cross section performed in $m_{4\ell}$ ranging from 105 GeV to 140 GeV.

Fiducial cross section (Inclusive) of H→ $4l$ (fb) at 13 TeV.				
	4l	$2e2\mu$	4μ	$4e$
Measured (stat. ⊕ sys.)	$2.73^{+0.23+0.24}_{-0.22-0.19}$	$1.30^{+0.16+0.10}_{-0.14-0.09}$	$0.75^{+0.10+0.04}_{-0.09-0.04}$	$0.69^{+0.14+0.13}_{-0.13-0.10}$
gg→H(POHWHEG) + XH	$2.77^{+0.51}_{-0.39}$	$1.30^{+0.xx}_{-0.xx}$	$0.78^{+0.xx}_{-0.xx}$	$0.70^{+0.xx}_{-0.xx}$
gg→H(NNLOPS) + XH	$2.76^{+0.14}_{-0.14}$	$1.30^{+0.07}_{-0.07}$	$0.76^{+0.04}_{-0.04}$	$0.70^{+0.04}_{-0.04}$

6.6.2 Measurement of Differential Fiducial Cross Sections

Observed data along with prediction of SM Higgs boson ($m_H = 125.09$ GeV) and expected background in each differential bin for chosen observables are shown in Figures 6.26 for production related observables. The differential results are only produced using 13 TeV data. The results of Higgs differential cross section measurements are produced for four-leptons transverse momentum and rapidity, transverse momentum of leading jet, jet multiplicity and are shown in the Figures 6.27. The uncertainty in the theoretical prediction for dominant process i.e. $\mathrm{gg} \to \mathrm{H}$ is estimated in the each differential bin of studied observable with POWHEG+JHUGEN which carries specific signal description. The uncertainty in the theoretical prediction for associated production mechanisms are assumed to be constant among differential bins and are obtained from Ref. [92]. The differential cross sections as a function of Higgs transverse momentum and rapidity are sensitive to pQCD calculations of $\mathrm{gg} \to \mathrm{H}$ production mode which is mediated by a loop, in which Higgs $p_T(H)$ is anticipated to be balanced with emission of the soft quarks and gluons. The differential distribution as a function rapidity of Higgs also probes PDFs of the colliding protons and representation of the gluon fusion production modes. The differential observables, transverse momentum p_T) of leading jet and jet multiplicity N(jets) probe hard quark and gluon radiations theoretical modelling in the process. Comparison of measurements with theoretical predictions from POWHEG and NNLOPS for the gg→H process (displayed in the blue and brown colors, respectively) together with predictions for XH = VBF (POWHEG) + VH + ttH(PYTHIA) (shown in green) is also shown. The error bars represents statistical and systematic uncertainty. The results are found to be compatible with SM based theoretical predictions within the uncertainties.

While extracting the differential measurements, the fractions of the $4e, 4\mu$ and $2e2\mu$ contributions to the fiducial cross sections in each bin is allowed to float. Results where the fractions are set to their SM predictions are included in Appendix **??**. Similarly, results where the model dependence uncertainty is assessed by taking into account the experimental constraints from the existing CMS measurements are included in Appendix **??**.

6.6.3 Comparison With ATLAS Results

ATLAS and CMS are two important general purpose experiments of the LHC. They are built with same goals with slightly different technology to obtain physics results. One experiment validates the results of the other experiments; one example is the announcement of Higgs discovery by both of experiments in the year 2012. Later, the properties of Higgs have also been studied using the data recorded with these individual experiments for run periods of LHC. There is a legacy of measurements from both of experiments as mentioned in Section 6.1.

Regarding Higgs cross section measurement in $H \to 4\ell$ decay channel, both experiments have published their results with full Run 2 data. Hence, it is important to show comparison of two recent independent measurements on the same subject from two different experiments [**CMS:2019chr**, **ATLAS:2020wny**]. Comparison is shown as under

- **Data and Triggers:** ATLAS used 139 $\mathrm{fb}^{-1}$and CMS used 137.1 $\mathrm{fb}^{-1}$of Full Run 2 LHC data. Triggers used by both experiments have little bit difference in p_T thresholds. For example, for Double Mu trigger ATLAS requires p_T threshold of leading and subleading muons to be 15 GeV and 15 GeV; however, CMS requirements are 17 GeVand 8 GeVfor leading and subleading muons respectively.

- **MC Samples (Signal):** Both experiments mainly use POWHEG with different PDFs to generate signal samples of different production modes of Higgs.

- **MC Samples (Background):** $\mathrm{q\bar{q}} \to \mathrm{ZZ}$ and $\mathrm{gg} \to \mathrm{ZZ}$ background samples are generated by using SHERPA by ATLAS experiment. However, CMS used POWHEG and MCFM for $\mathrm{q\bar{q}} \to \mathrm{ZZ}$ and $\mathrm{gg} \to \mathrm{ZZ}$ backgrounds respectively.

- **Lepton Selection:** For selection of leptons, requirements of p_T and η are almost same as per definitions of fiducial phase spaces by both experiments as shown in the Table 6.17. Lepton identification and impact parameter requirement criteria differs a bit. CMS requires $|SIP| < 4$ both for electron and muon; however, ATLAS requires $|SIP| < 3(5)$ for muon (electron). For lepton "dressing", ATLAS requires tracking based relative isolation; however, CMS requires particle-flow based relative isolation.

- **Jet Selection:** Both use $AK4$ jets with same p_T thresholds; however, η and $\Delta R(lepton, jet)$ requirements are slightly different. B-tagging algorithms are also different.

- **Fiducial phase space definition:** Both experiments perform measurement in a fiducial volume. Comparison of the definition of fiducial volumes are shown in Table 6.17. There is a slight difference in the definitions.

- **Assumed Higgs mass:** Assumed Higgs boson mass for measurements is 125.0 GeV and 125.09 GeV for ATLAS and CMS respectively.

- **Binned likelihood fit:** Both experiments perform binned likelihood fit to 4 lepton invariant mass to extract the cross section.

- **Criteria to define the bin boundaries of differential observables:** ATLAS ensures the statistical significance to be more than 2σ and the migrations between the bins are supposed to be minimum. In general, bin width is more than twice the experimental resolution. On the other hand, CMS aims for similar expected measured cross section in the differential bins.

- **Theory predictions for differential measurements:** ATLAS uses MADGRAPH5 and NNLOPS (normalized to N^3LO cross sections) as common to all observables. Additional predictions from NNLOJET, RADISH (normalized to their own respective predicted cross sections), PROPHECY and HTO4L are also used depending upon the differential abservables.

- **Overall inclusive and differential results:** Both experiments show the measurements in good agreement with SM predictions within the uncertainties of measurements.

The distribution of the reconstructed four-lepton invariant mass $m_{4\ell}$ is given in the Figure 6.28.

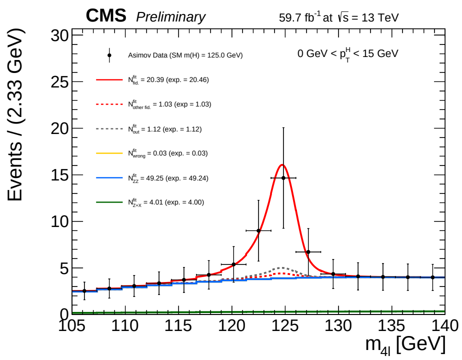
The inclusive or integrated cross section as measured from ATLAS is given as $\sigma_{fid} = 3.28 \pm 0.32 fb$ and with CMS is given as $\sigma_{fid} = 2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\text{stat.})^{+0.24}_{-0.19}(\text{syst.})$ fb. Inclusive cross section in different final state can be seen the Figure 6.29.

From CMS, current published results on differential cross section measurements in $H \to 4\ell$ decay channel, as described in the previous section, are based on production related observables which are four-leptons transverse

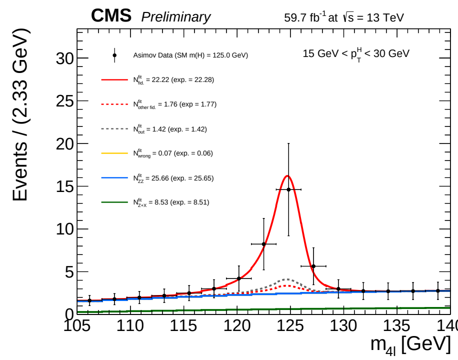
Parameter	CMS	ATLAS
p_T of the leading lepton	$p_T > 20$ GeV	$p_T > 20$ GeV
p_T of sub-leading lepton	$p_T > 10$ GeV	$p_T > 15$ GeV
p_T of next-to-sub-leading lepton	Not required	$p_T > 10$ GeV
Additional electrons (muons)	$p_T > 7$ (5) GeV and $\|\eta\| < 2.5$ (2.4)	$p_T > 5$ GeV and $\|\eta\| < 2.7$
Scalar p_T sum of all stable objects within $\Delta R < 0.3$ from the lepton	$< 0.35 p_T$	Not required
Leading pair Z_1 (from SFOS lepton pairs)	with smallest $\|m_Z - m_{ll}\|$	same
Sub-leading pair Z_2 (from remaining SFOS lepton pairs)	with higher $p_T$sum	with smallest $\|m_Z - m_{ll}\|$
Invarivant mass of Z_1 candidate	$40 < m(Z_1) <$ 120 GeV	$50 < m(Z_1) <$ 106 GeV
Invarivant mass of Z_2 candidate	$12 < m(Z_2) <$ 120 GeV	$12 < m(Z_2) <$ 115 GeV
Separation between selected four leptons	$\Delta R(\ell_i \ell_j) > 0.02$	$\Delta R(\ell_i \ell_j) > 0.1$
Invariant mass of any OS lepton pair(J/ψ veto)	$m(\ell_i^+ \ell_j^-) > 4$ GeV	$m(\ell_i^+ \ell_j^-) > 5$ GeV
Invariant mass of selected four leptons	$105 < m_{4\ell} < 140$ GeV	$105 < m_{4\ell} < 140$ GeV
Leading pair Z_1	SFOS lepton pair with smallest $\|mZ - m\text{ll}\|$	same

TABLE 6.17: Comparison of fiducial phase space selection from CMS and ATLAS experiments

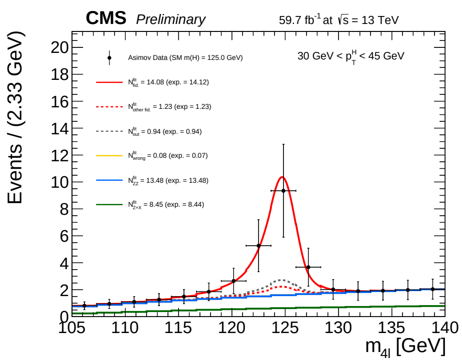
momentum and rapidity, transverse momentum of leading jet, jet multiplicity. Therefore, the relevant bservables are compared from ATLAS measurements. In the following, differential measurements from ATLAS and CMS are shown side-by-side. (Figures 6.30, 6.31, 6.32 and 6.33)



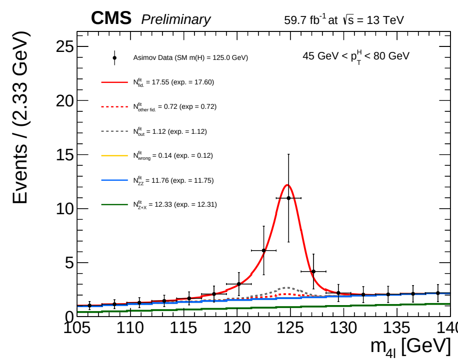
(a) 0.0 GeV $< p_T(\mathrm{H}) <$ 15.0 GeV



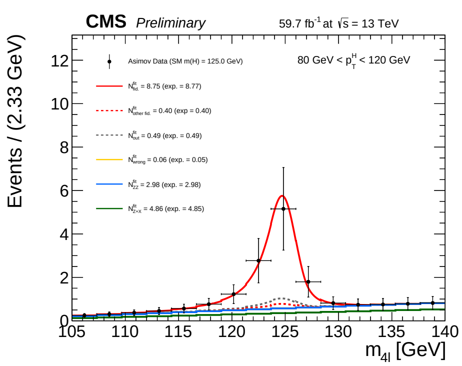
(b) 15.0 GeV $< p_T(\mathrm{H}) <$ 30.0 GeV



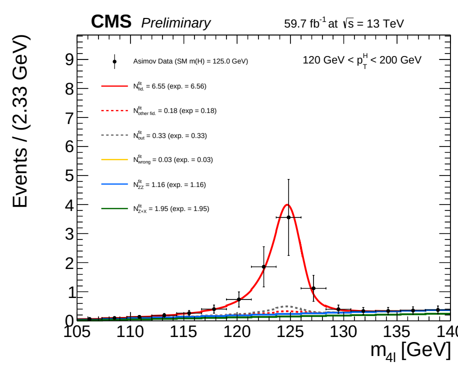
(c) 30.0 GeV $< p_T(\mathrm{H}) <$ 45.0 GeV



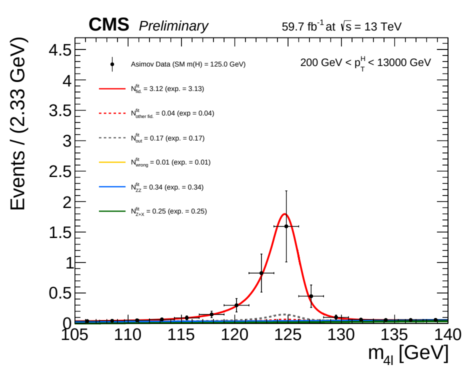
(d) 45.0 GeV $< p_T(\mathrm{H}) <$ 80.0 GeV



(e) 80.0 GeV $< p_T(\mathrm{H}) <$ 120.0 GeV



(f) 120.0 GeV $< p_T(\mathrm{H}) <$ 200.0 GeV



(g) 200.0 GeV $< p_T(\mathrm{H}) <$ 13000.0 GeV

FIGURE 6.22: Four lepton mass distributions for an Asimov dataset generated using SM cross section values and SM efficiencies where the gg→H prediction is from POWHEG+JHUGEN and resulting fitted values of PDFs of signal and background for different bins of $p_T(\mathrm{H})$ (all final states combined).

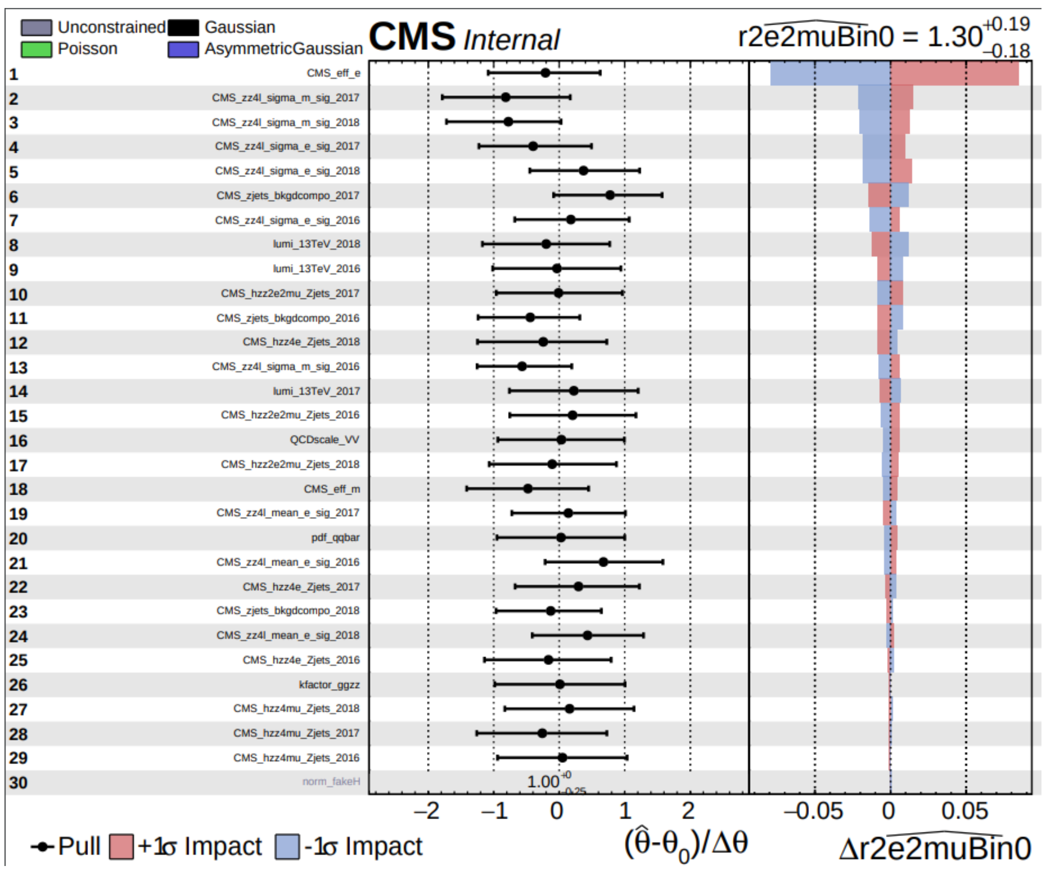
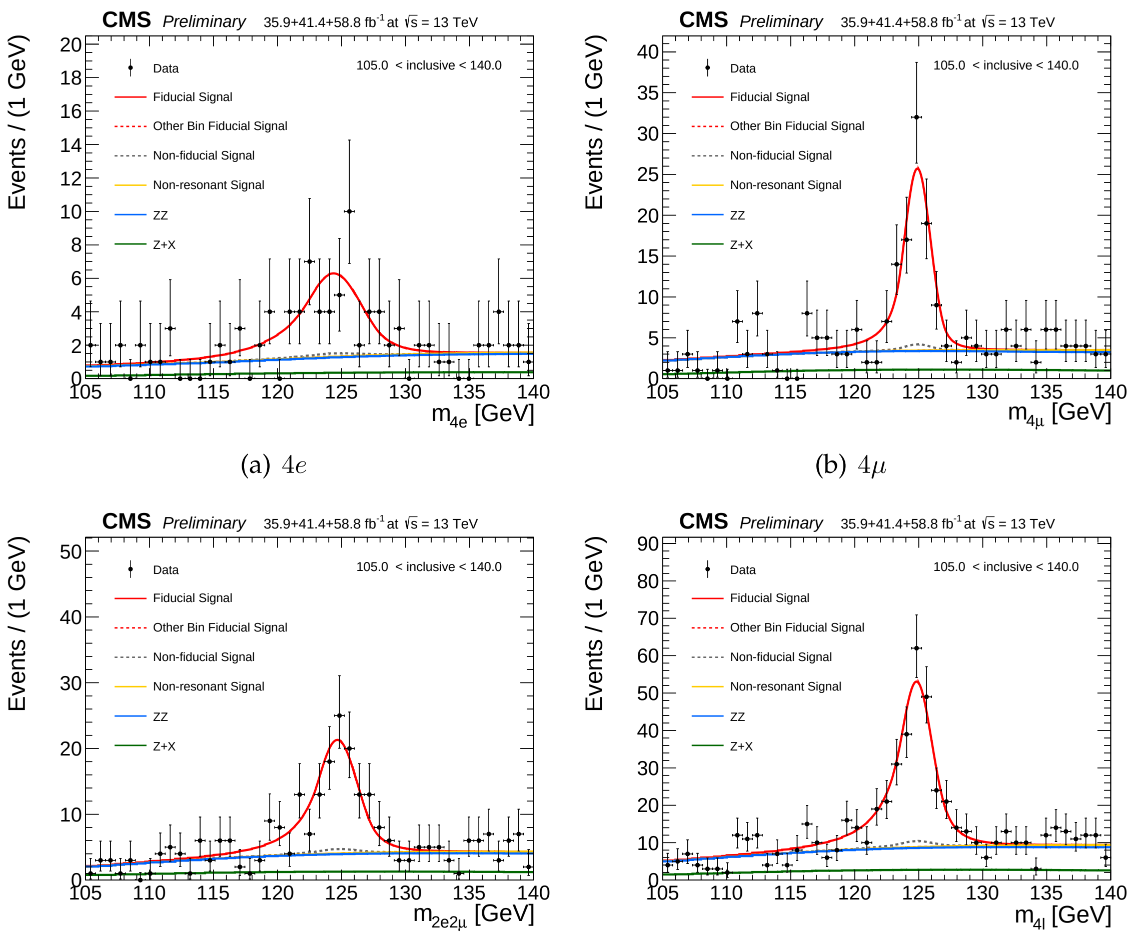


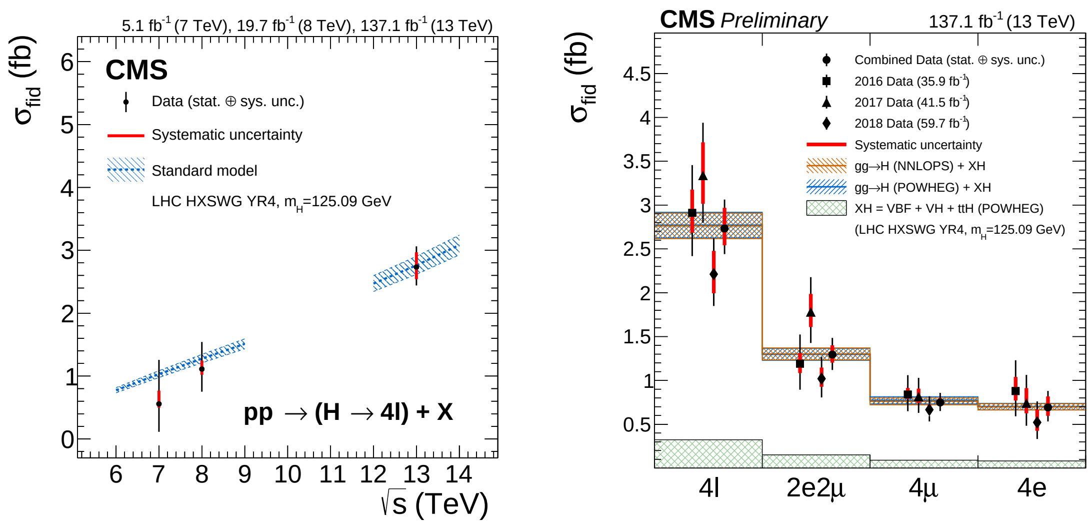
FIGURE 6.23: Pulls. Overall impact of different uncertainties sources on the inclusive measurement ($2e2\mu$ final state)



(a) $4e$ (b) 4μ

(c) $2e2\mu$ (d) 4ℓ

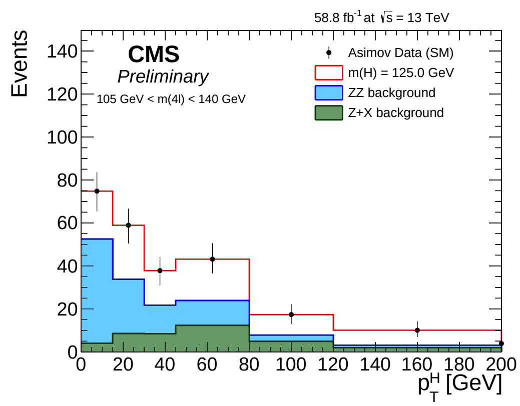
FIGURE 6.24: Inclusive four lepton invariant mass distributions in data and resulting fitted values of signal and the background PDFs in different final states.



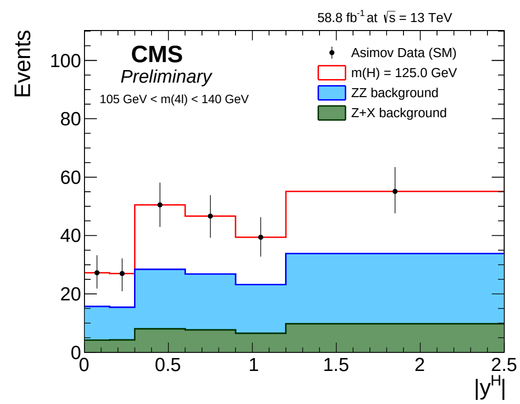
(a) $\mathrm{H} \to 4\ell$ (7, 8 and 13 TeV)

(b) $\mathrm{H} \to 4\ell$ (13 TeV)

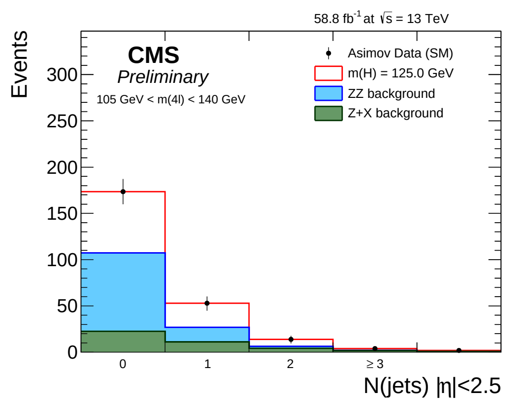
FIGURE 6.25: The measured fiducial cross section as a function of $\sqrt{s}$ (left). The measured inclusive fiducial cross section in different final states (right). The acceptance is calculated using at $\sqrt{s}$=13 TeV and HRES [82, 89] at $\sqrt{s}$=7 and 8 TeV and the total gluon fusion cross section and uncertainty are taken from Ref. [90]. The fiducial volume for $\sqrt{s}$=6–9 TeV uses the lepton isolation definition from Ref. [91], while for $\sqrt{s}$=12–14 TeV the definition described in the text is used.



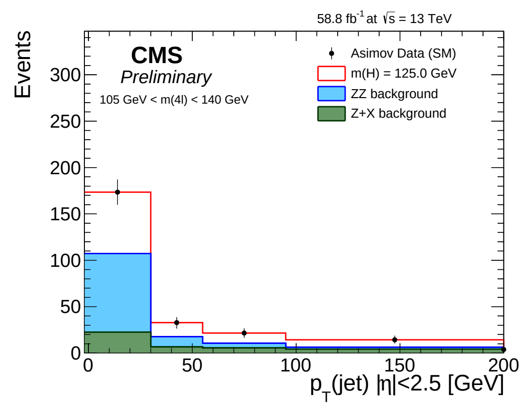
(a) $p_T(\mathrm{H})$



(b) $|y(\mathrm{H})|$

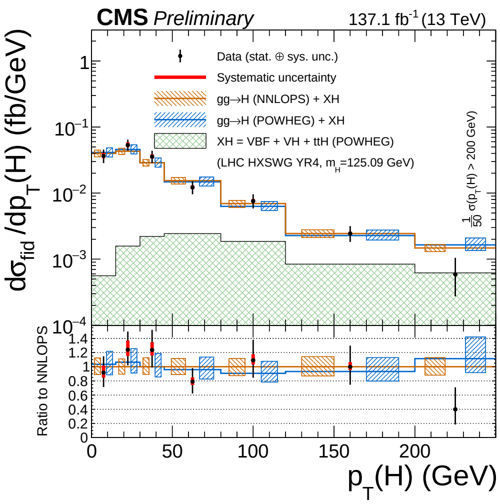


(c) N(jets), $|\eta| < 2.5$

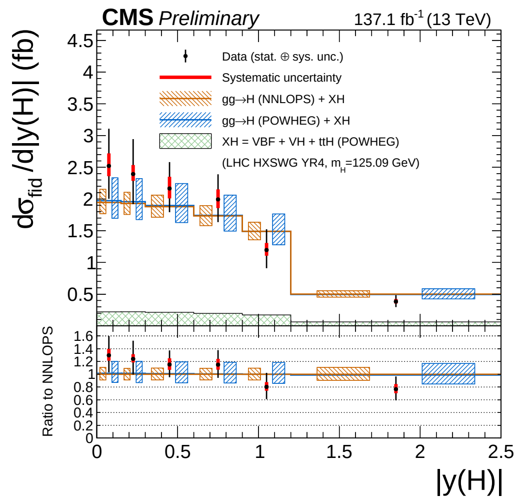


(d) $p_T(\mathrm{jet}), |\eta| < 2.5$

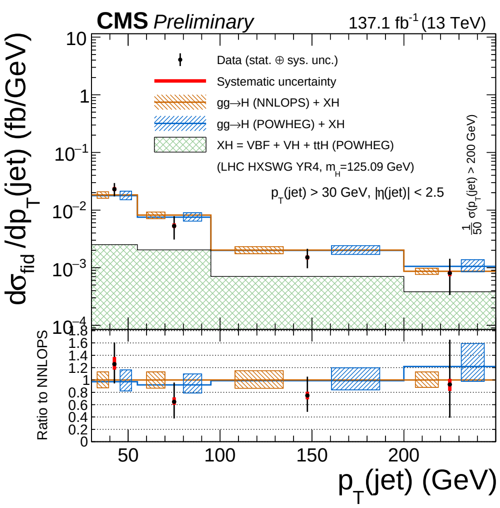
FIGURE 6.26: Expected yield using 2018 Asimov data along with SM signal ($m_H = 125.09$ GeV) and background expectations for different observables and for all final states combined, in mass window $105 < m_{4\ell} < 140$ GeV which is used forH $\rightarrow 4\ell$ measurements. The extraction of the signal is performed by exploiting the differences in the expected $m_{4\ell}$ spectra of the signal and the background contributions in each differential bin of an observable.



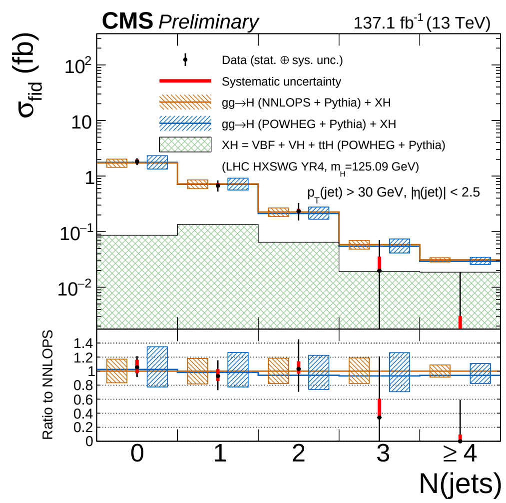
(a) $p_T(\mathrm{H})$



(b) $|y(\mathrm{H})|$



(c) $p_T(\text{leading jet}), |\eta| < 2.5$



(d) N(jets), $|\eta| < 2.5$

FIGURE 6.27: Differential fiducial cross section results for $p_T(\mathrm{H})$ (top left), $|y(\mathrm{H})|$ (top right), p_T of leading jet (bottom left) and N(jets) (bottom right) in $\mathrm{H} \to 4\ell$ using full Run 2 CMS data and comparison with the theoretical predictions. Systematic uncertainty is indicated by red bars and black bars show total statistical and the systematic uncertainties, combined in quadrature. The blue and brown colors show the theoretical predictions. The acceptances and the theoretical uncertainties in differential bins are computed using POWHEG. Subdominant production contributions $\mathrm{XH} = \mathrm{VBF} + \mathrm{VH} + t\bar{t}\mathrm{H}$ are shown in green separately. The inclusive cross section is normalized to SM estimates calculated at NNLL+NNLO level of accuracy for all theoretical predictions. The systematic uncertainties correspond to the generators accuracy are also taken in to account for differential predictions. The fraction of 4μ, $4e$ and $2e2\mu$ in each differential bin is allowed to float in fit.

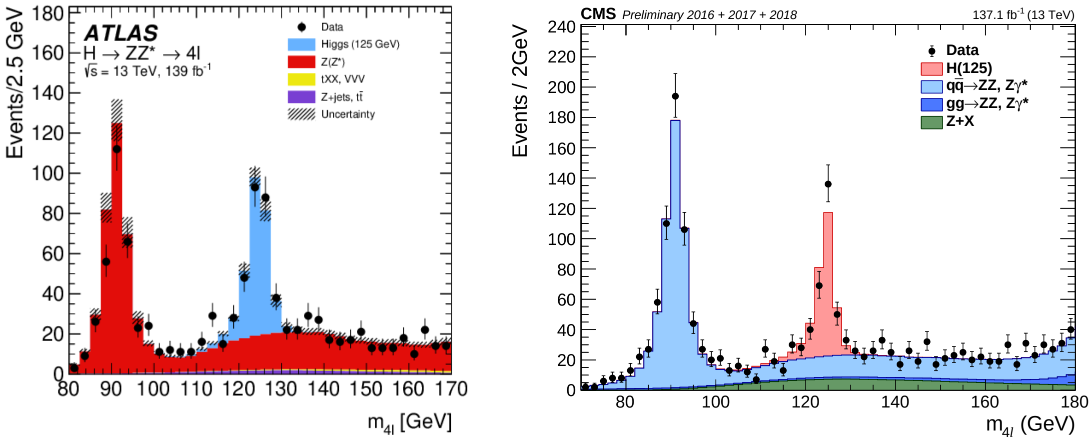


FIGURE 6.28: Left(Right): Reconstructed four-lepton invariant mass $m_{4\ell}$ at ATLAS (CMS) using an integrated luminosity of 139 fb^{-1} (137.1 fb^{-1}).

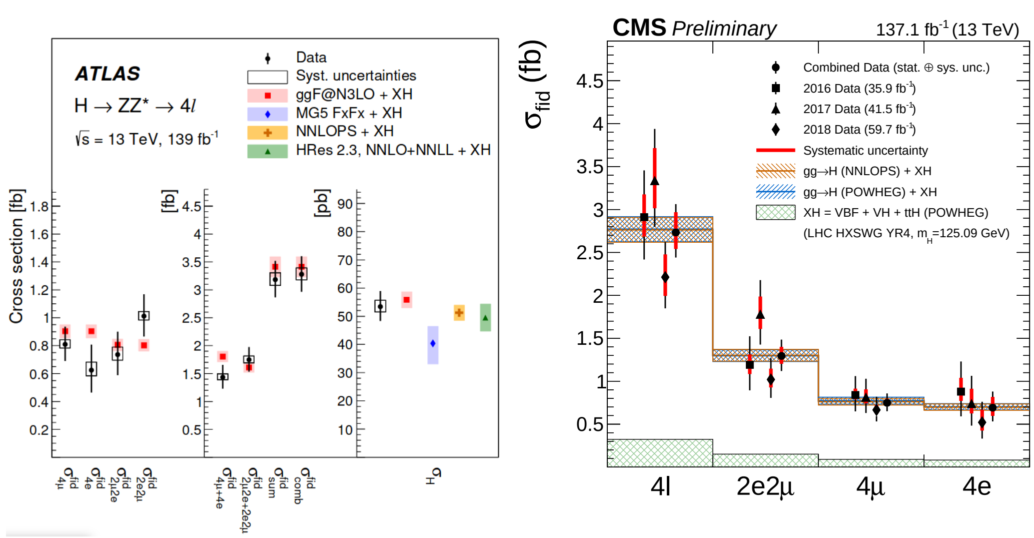


FIGURE 6.29: Left(Right): Inclusive cross section in differential final states at ATLAS (CMS) using an integrated luminosity of 139 fb^{-1} (137.1 fb^{-1}).

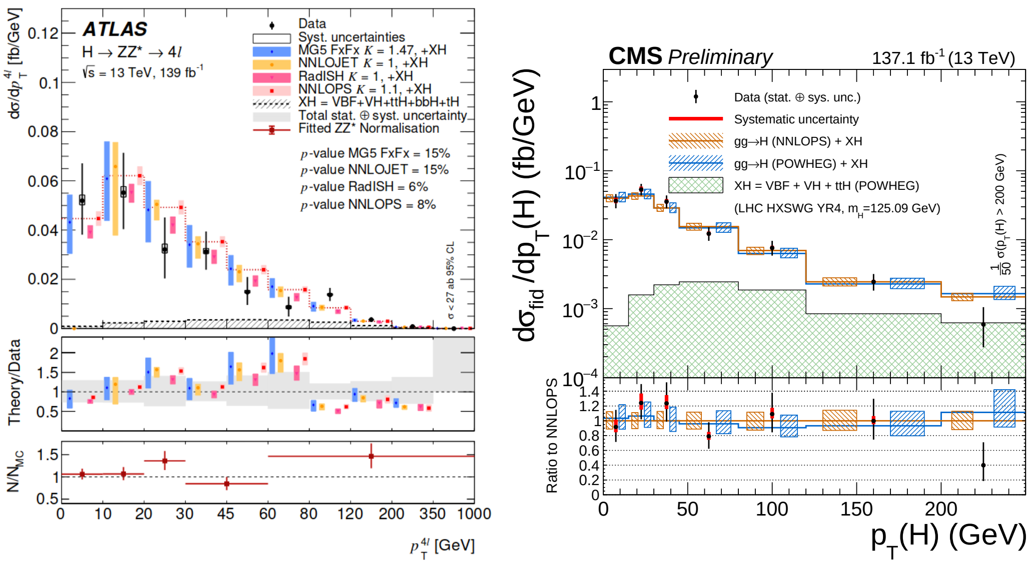


FIGURE 6.30: Left(Right): Higgs differential cross section as a function of transverse momentum of four lepton system with ATLAS (CMS) using an integrated luminosity of 139 fb^{-1} (137.1 fb^{-1}). Results are shown with theory prediction in each experiment as described in 6.6.3.

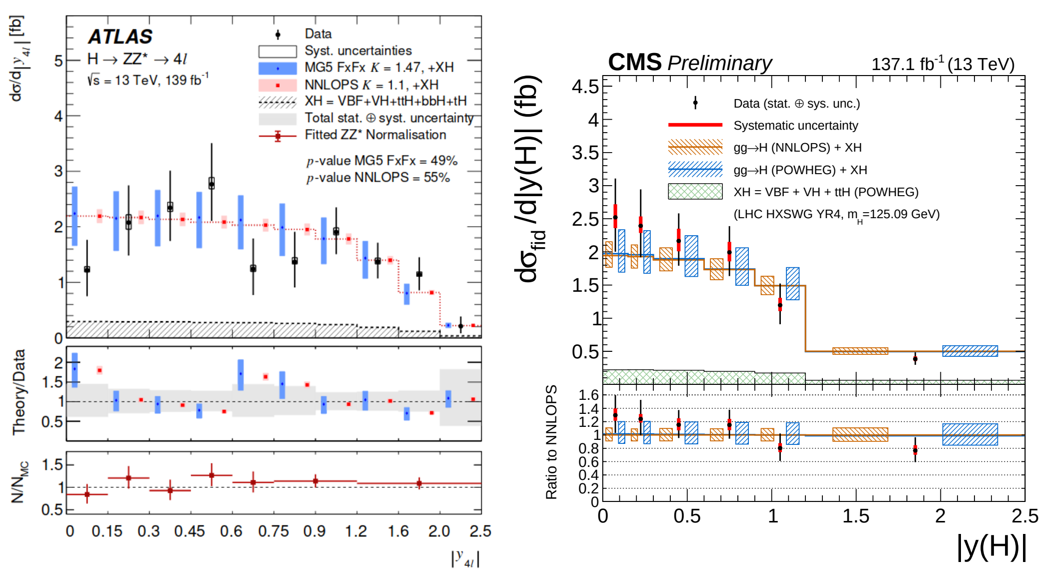


FIGURE 6.31: Left(Right): Higgs differential cross section as a function of rapidity of four lepton system with ATLAS (CMS) using an integrated luminosity of 139 fb^{-1} (137.1 fb^{-1}). Results are shown with theory prediction in each experiment as described in 6.6.3.

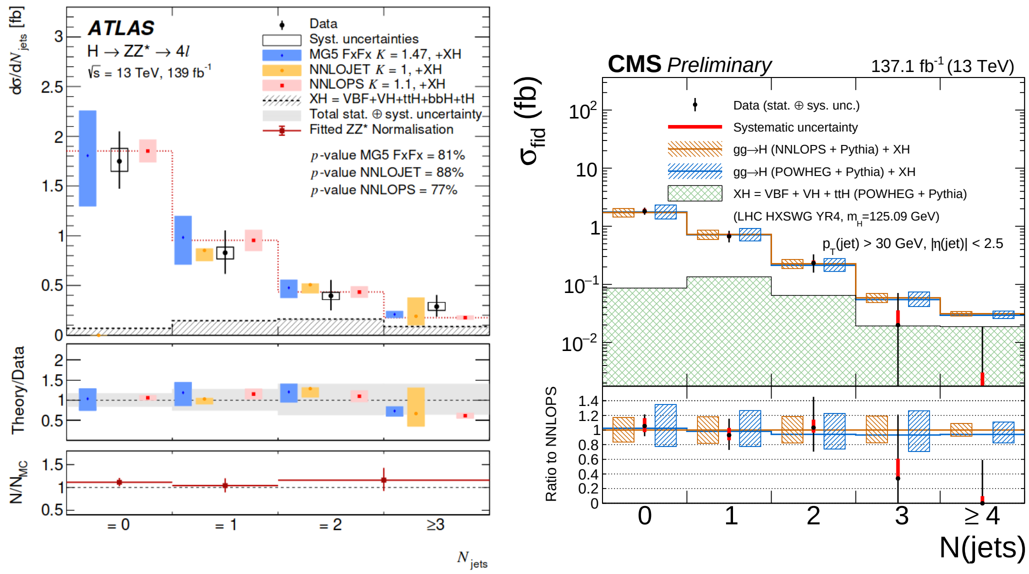


FIGURE 6.32: Left(Right): Higgs differential cross section as a function of jet multiplicity with ATLAS (CMS) using an integrated luminosity of 139 fb^{-1} (137.1 fb^{-1}). Results are shown with theory prediction in each experiment as described in 6.6.3.

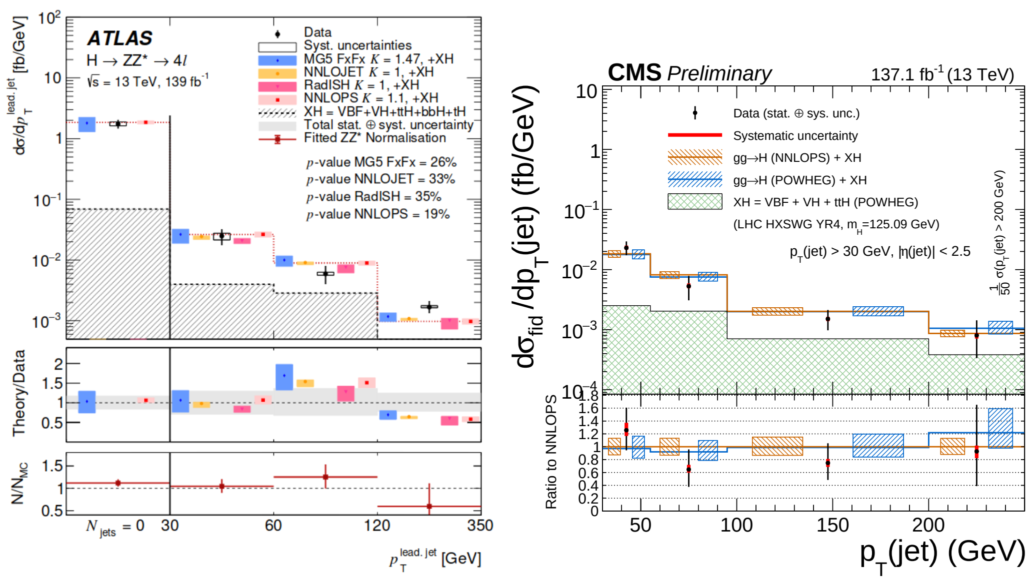


FIGURE 6.33: Left(Right): Higgs differential cross section as a function of transverse momentum of the leading jet with ATLAS (CMS) using an integrated luminosity of 139 fb^{-1} (137.1 fb^{-1}). Results are shown with theory prediction in each experiment as described in 6.6.3.

Chapter 7

Summary and Outlook

7.1 Summary of the dissertation

After many years of search, ATLAS and CMS collaborations of CERN announced the discovery much awaited Higgs boson in early July 2012 verifying the underlying concept of electroweak symmetry breaking (EWSB) in the SM of particle physics. This triggered essential precise studies of this particle to confirm its nature and to discover some new physics in case of some deviation. With availble LHC data, measured properties of Higgs are consistent with SM. Inclusive and differential cross section measurements of Higgs are its very important properties which help to probe its structure and its nature of interaction with decaying particles. These measurements provide a direct probe into SM Higgs sector perturbative QCD calculations at the TeV energy scale.

Measurement of inclusive and differential cross section of the Higgs boson production in proton-proton collisions at $\sqrt{s} = 13$ TeV corresponding to an integrated luminosity of 137.1 $\mathrm{fb}^{-1}$is presented in this dissertation. Measurements are performed by exploiting $H \to 4\ell$ decay channel ($\ell = e, \mu$), which is known to be a golden channel for such measurements. To minimize the dependence of the measurements on the underlying model of the Higgs boson production mechanism and its properties, the cross sections are extracted in a fiducial volume that closely matches with experimental acceptance. Remaining model dependence is extracted by repeating the measurement for SM and other exotic model and reported as a distinct systematic uncertainty. All measurements include the correction for the finite detection efficiency and resolution effects to allow for direct comparison with the SM predictions. As

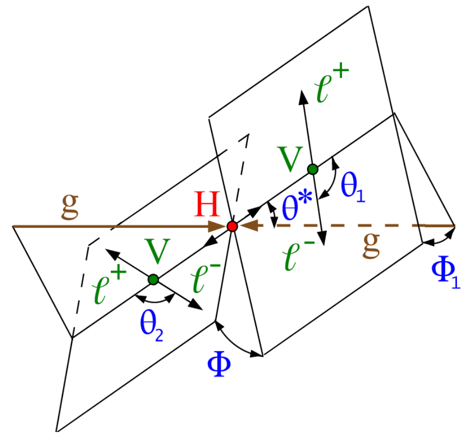


FIGURE 7.1: Higgs event signature

$\sigma \times BR$ strongly depend upon Higgs mass, the measurements have been extracted assuming $m_H = 125.09$ GeV, as reported by CMS experiment [51] from combined measurement in the $H \to 2\gamma$ and $H \to 4\ell$ decay channels.

The differential fiducial cross sections are presented as a function of kinematic observables which are sensitive to Higgs production mechanism: rapidity and transverse momentum of Higgs, associated jet multiplicity and transverse momentum of the leading jet. Full Run 2 data (137.1 fb^{-1}) enabled us to explore the differential observables in more fine bins than presented in previous measurements on this topic.

The integrated fiducial cross section is measured to be $2.73^{+0.30}_{-0.29} = 2.73^{+0.23}_{-0.22}(\mathrm{stat.})$ $^{+0.24}_{-0.19}(\mathrm{syst.})\ \mathrm{fb}$ at 13 TeVand SM predicts this to be $2.76^{+0.14}_{-0.14}$. The agreement between SM predictions and measured $H \to 4\ell$ cross section measurement at $\sqrt{s}$ =13 TeV is found be good. These model independent cross section measurements can be compared with any theoretical predictions in future with better accuracy.

7.2 Outlook

The presented measurements of Higgs differential cross sections are related to Higgs production. There are certain things that can be done which are beyond the scope of this dissertation:

– With more data, get more precise results

- Study more differential observables
- Include additional theoretical prediction(s) for comparison with the observed cross section in differential bins
- Include potential theory interpretations for differential observables

The current measurements are emerged with significant statistical uncertainties. The studies based on simulations show that the uncertainties on Higgs cross section measurements with 300 fb^{-1}pp collisions data (as expected from LHC Run 3) would be significantly less than the current theoretical uncertainties. More data will help to affirm the SM Higgs boson if measurements be in agreement with the SM expectations, elsewise any deviation from SM predictions will open the door to some new physics.

Studying more differential observables are also important from different aspects. There are more *production* and as well as *decay* related observables ($\theta^*, \theta_1, \theta_2, \Phi, \Phi_1$ as shown in Figure. 7.1) which can be studied with currently available data from CMS. With respect to earlier iterations, current data would allow to exploit more differential bins for a particular observable. Moreover, with more statistics there is opportunity to study the double differential cross section (as function of two observables). In $H \to 4\ell$, there are ongoing efforts for additional differential measurements alongwith potential theory interpretations; however, it is yet to accomplish to be made public. Preliminary version of some of the new differential measurements have been shown in Appendix. In the reported measurements, the observed cross sections are compared with respective theoretical predictions based on POWHEG and NNLOPS; however, including additional theoretical prediction(s) e.g. MADGRAPH5 are also underway to be part of next manuscript.

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